Elliptic Curves

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0 Introduction

§0.1 To Begin..

These are my notes for the course Elliptic Curves taught by Professor Yanki Lekili for the 2025-2026 academic year. These notes are a copy of [4] combined with any in-lecture anecdotes or extra exercises/ points that help me; any mistakes are mine.

§0.2 Content

The main objective of this course is to learn about solving polynomial equations over $\mathbb{Q}$ or $\mathbb{Z}$. k will often denote a field, which will generally be one of $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_p, \mathbb{Q}_p$. We may occasionally consider number fields such as $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{3})$. These polynomial equations will generally be in $k[x, y]$ or $k[x, y, z]$ and will usually be of highest degree 2 or 3.

We start with conics, which are equations of degree 2:

$$ax^2 + bxy + cy^2 + dx + ey + f = 0, \quad a, b, c, d, e, d \in k.$$

Later, we study elliptic curves, which are equations of the form

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_1, a_2, a_3, a_4, a_6 \in k.$$

To be specific, this is said to be in Weierstrass form. If $\mathrm{char}(k)$ is not equal to 2 or 3, then we can do a change of variables to bring it into

$$y^2 = x^3 + ax + b, \quad a, b \in k.$$

When the discriminant $4a^3 + 27b^2 \neq 0$, a result of Mordell says that the set of rational solution (or any number field), which we denote as $E(\mathbb{Q})$, is a finitely generated abelian group. By the basic structure theorem of finitely generated abelian groups, this means we must have

$$E(\mathbb{Q}) = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \colon y^2 = x^3 + ax + b\} \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{tors}.$$

Here, $E(\mathbb{Q})_{tors}$ is the finite group corresponding to points of finite order, and r is the rank, which counts how many independent points of infinite order are required to generate all other solutions. The main goal of these lectures is to prove Mordell's theorem and learn techniques how to prove $E(\mathbb{Q})$. There are three main ideas that will be prevalent

Geometric Method of Constructing Solutions

There is the elementary example of finding rational solutions on the circle $x^2 + y^2 = 1$. First take the point $P = (-1, 0)$ which is a rational solution. Then, any line with a rational slop t passing through P has the equation $y = t(x + 1)$. This will intersect the circle at exactly one other point, (x, y). If we substitute y into the circle equation and solve for x, this reveals that the second point must also have rational coordinates, giving the formulas

$$x = \frac{1 - t^2}{1 + t^2}, \quad y = \frac{2t}{1 + t^2},$$

for $t \in \mathbb{Q} \cup \{\infty\}$.

We can look at a newer example. Consider the elliptic curve given by

$$E\colon y^2 + y = x^3 - x.$$

There is a clear solution, $P = (0,0)$. To produce other solutions, we consider tangent lines!! First, we get the line ℓ_P, which is the tangent line to E at the point P. Recall that the tangent line is given by

$$\frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b) = 0.$$

We have that $\ell_P\colon y = -x$. Intersecting ℓ_P with E, we get another point $Q = (1,-1)$ on E. We now repeat this. The tangent at Q is given by $\ell_Q\colon y = -2x+1$, and intersecting this with E gives the point $R = (2,-3)$. The tangent line at R is given by $\ell_R\colon y = \frac{-11}{5}x + \frac{7}{5}$, and intersecting this with E, we get $S = \left(\frac{21}{25}, \frac{-56}{125}\right)$. Now, we ask the following questions:

- Can we continue this procedure infinitely?
- Do we get all rational solutions?

These are questions that will be answered through the course.

Descent

Theorem 0.2.1 (Fermat)
There is no right angled triangle with rational side lengths whose area is a square.

Proof. Note that we can always scale a given triangle, which changes its area by the square of that scale factor. Thus, it suffices to show that there is no right angled triangle with integer side lengths, whose sides may be assumed to be mutually prime and that the area is a square. We shall write then the side lengths as

$$p^2 - q^2, \quad 2pq, \quad p^2 + q^2,$$

where p, q are mutually prime and $p > q$ and $p - q$ is odd. The area is then given by $pq(p-q)(p+q)$. Let us suppose that this is a square. We note that by coprimality of p and q and since these have opposing parity, we see that $p, q, p+q, p-q$ are pairwise coprime. Thus, we may write

$$p = x^2, \quad q = y^2, \quad p+q = u^2, \quad p-q = v^2,$$

where u, v are odd and coprime. We now begin the decent. Consider the right triangle with sides $\left(\frac{u+v}{2}, \frac{u-v}{2}, x\right)$. The side lengths are pairwise coprime, and the area is $\frac{u^2-v^2}{8} = \frac{q}{4} = \left(\frac{y}{2}\right)^2$. To see this is the square of an integer, we must show that y is even. We know $p = x^2$ and $q = y^2$ have opposite parity. Since $u^2 = p+q$ and $v^2 = p-q$ are both odd, their sum $(p+q)+(p-q) = 2p$ must be congruent to 2 modulo 4. This implies that p is odd. Thus, $q = y^2$ must be even, so y is even. This new triangle thus has an integer area which is a perfect square. However, $4\left(\frac{y}{2}\right)^2 = q < pq(p-q)(p+q)$. Thus, we've attained a new-right-angled triangles with integer side lengths whose area is strictly smaller. We can repeated this argument, and by infinite descent, we reach the contradiction. $\square$

Exercise 0.2.2. Let $p, q \in k[t]$ be relatively prime polynomials $(\text{char}(k) \neq 2)$. Suppose that tehre exist linearly independent vectors $(a_i, b_i) \in k^2$ for $i = 1, 2, 3, 4$ such that $a_i p + b_i p$ is a square in $k[t]$ for all i. Using infinite descent, show that $p, q \in k$; that is, that they are constant polynomials.

Local to Global

Suppose we have an equation over $\mathbb{Q}$. It's clear that we can then multiply through by a common denominator so that we can write the equation over $\mathbb{Z}$, then reduce modulo p to get an equation over $\mathbb{F}_p$. Thus, it's clear that if the original equation had a solution in $\mathbb{Q}$, then there is a solution modulo p for every p. More generally, we can reduce this to any field $\mathbb{F}_q$, where $q = p^k$ for some k. Sometimes, we may let $p = \infty$ to mean that we consider the equation over $\mathbb{R}$.

The Hasse principle is concerned with the converse: given a solution to an equation in $\mathbb{F}_q$, where $q = p^k$, for all k and all p (or simply in $\mathbb{Q}_p$), and over $\mathbb{R}$, can we conclude that there is a solution over $\mathbb{Q}$? If so, the Hasse principle is satisfied for this equation.

We now given an example where there exists no solution over $\mathbb{Q}$, for there are no solutions in $\mathbb{Q}_2$.

Example 0.2.3

Consider $x^2 + y^2 = 2(x + y)z + z^2$. We claim that there are no non-trivial rational solutions to this equation. Putting $t = x + y + z$, we can rewrite this equation as

$$2x^2 + 2xy + 2y^2 = t^2.$$

Since this equation is homogeneous,it suffices to prove that there are no integer solutions; otherwise, we can scale out the denominators to get integer solutions. Similarly, we may assume that $\gcd(x, y, t) = 1$.

Since the LHS is divisible by 2, it follows that $2 \mid t$, so $2 \mid (x^2 + xy + y^2)$. Now, note that $x^2 + xy + y^2 = 0 \pmod 2$ if and only if $x = y = 0 \pmod 2$, but then we see that $2 \mid \gcd(x, y, t)$, which is a contradiction. It follows that the equation doesn't have a solution modulo 4; thus, not in $\mathbb{Q}_2$, and so in $\mathbb{Q}$.

The Hasse principle is the statement that quadratics equations such as the above have a solution over $\mathbb{Q}$ if and only if they can be solved modulo p-powers, for any p. Given a proof of this result for plane conics is our first goal in the course.

Example 0.2.4 (Non-Example)

Selmer's example given by $E\colon 3x^3 + 4y^3 + 5z^3 = 0$ shows that the Hasse principle fails for cubics. We shall prove this if time permits.

1 The p-adic Numbers

§1.1 Defining $\mathbb{Z}_p$ and $\mathbb{Q}_p$

Here is a quick and easy way of formulating the p-adic integers. A **p-adic integer** $x \in \mathbb{Z}_p$ is a formal solution to the system of congruences

$$x \equiv x_n \pmod{p^n}, \quad n = 1, 2, , \ldots$$

The consistency condition is that $x_n \equiv x_{n+1} \pmod{p^n}$. Two integers sequences (x_n) and (y_n) define the same p-adic integer if and only if $x_n \equiv y_n \pmod{p^n}$, for all n.

Remark 1.1.1 (Defining elements in $\mathbb{Q}_p$) — We can define the p-adic rational $\mathbb{Q}_p$ in the same way where the numbers x_n might be an arbitrary rational. That is, if $x_n = \frac{r}{s}$, then we think of $x \equiv x_n \pmod{p^n}$ as a solution to the congruence relation $sx \equiv r \pmod{p^n}$.

§1.2 Norms

Definition 1.2.1. A **norm** on a field k is a function $|\cdot|\colon k \to \mathbb{R}_{\geq 0}$ such that:

- $|x| = 0 \iff x = 0$
- $|xy| = |x||y|$
- $|x + y| \leq |x| + |y|$

Furthermore, if the norm satisfies the following **ultrametric inequality**

$$|x + y| \leq \max\{|x|, |y|\},$$

the norm is said to be **non-archimedean**.

Remark 1.2.2 — It is immediate that $|-x| = |x|$, for all $x \in k$.

Remark 1.2.3 — Given a norm on a field, it induces a metric defined by $d(x, y) = |x - y|$, for $x, y \in k$.

Remark 1.2.4 — It is clear that the ultrametric inequality implies the usual triangle inequality.

Notation 1.2.5. We shall denote the *usual absolute value* on $\mathbb{Q}$ (induced by the usual norm on $\mathbb{R}$ and $\mathbb{C}$) as $|\cdot|_\infty$.

Definition 1.2.6. The **p-adic norm** $|\cdot|_p\colon \mathbb{Q} \to \mathbb{R}_{\geq 0}$ is defined as follows: given a rational $r \neq 0$, write it as $r = p^\rho \frac{a}{b}$, where $a, b \in \mathbb{Z}$ and $p \nmid a, b$. Then, we let

$$|r|_p = p^{-\rho}.$$

We also define $|0|_p = 0$. There, we say that a number is is **p-adically small** if it is divisible by a high power of p.

Proposition 1.2.7

$|\cdot|_p\colon \mathbb{Q} \to \mathbb{R}_{\geq 0}$ is indeed a norm.

Proof. The only condition that is not clear is the triangle inequality. Let $r = \frac{p^\rho a}{b}$, $s = \frac{p^\sigma c}{d}$, where $\rho, \sigma, a, b, c, d \in \mathbb{Z}$ and $p \nmid a, b, c, d$. It follows that $|r|_p = p^{-\rho}$ and $|s|_p = p^{-\sigma}$, and WLOG, we shall assume that $\sigma \geq \rho$. It then follows that

$$r + s = p^\rho \frac{(ad + p^{\sigma-\rho}cd)}{bd},$$

and $p \nmid bd$. The numerator is an integer, but for $\sigma = \rho$, it can be divisible by p. It then follows that $|r+s|_p \leq p^{-\rho}$. In other words,

$$|r + s|_p \leq \max\{|r|_p, |s|_p\}.$$

□

Lemma 1.2.8 (Isosceles Property of Triangles)

Let $|\cdot|\colon k \to \mathbb{R}_{\geq 0}$ be a non-archimedean norm. If $|x| < |y|$, then $|x - y| = |y|$.

Proof. We have $|x - y| \leq \max\{|x|, |y|\} = |y|$, since $|x| < |y|$. On the other hand, $y = (y - x) + x$, so $|y| \leq \max\{|x - y|, |x|\} = |x - y|$, as $|x| < |y|$. □

Remark 1.2.9 — This is actually true for the converse as well; suppose $|x - y| = |y|$, so that

$$|y| = |x - y| \leq \max\{|x|, |y|\}$$

Suppose, for the sake of contradiction, that $|x| \geq |y|$. Then,

$$|x| = |(x - y) + y| \leq \max\{|x - y|, |y|\} = \max\{|y|, |y|\} = |y|,$$

which is a contradiction.

Exercise 1.2.10. Let $(k, |\cdot|)$ be a field with a non-archimedean norm. Let $a \in k$, and consider the unit disk $D = \{x\colon |x - a| < 1\}$ with center a. Let $b \in D$ be any point, then show that $D = \{x\colon |x - b| < 1\}$; that is, every point of the disk can be taken to be the center of the disk.

Solution: Let $x \in D = \{x\colon |x - a| < 1\}$. Then,

$$|x - b| \leq \max\{|x - a|, |a - b|\} < 1,$$

It follows that every $x \in \{x\colon |x - a| < 1\}$ is also contained in $\{x\colon |x - b| < 1\}$; vice versa.
□

Exercise 1.2.11. Let $n \in \mathbb{Z}$ be a composite number. For $r \in \mathbb{Q}$, write $r = \frac{n^\rho a}{b}$, where $n \in \mathbb{Z}$, $a, b \in \mathbb{Z}$, and $n \nmid a, b$, and define $|r|_n = n^{-\rho}$. Is $|\cdot|_n$ a norm on $\mathbb{Q}$.

Solution: No; take $n = 6$ for instance. Then,

$$6^{-1} = |6|_6 \neq |2|_6|3|_6 = 1$$

□

Definition 1.2.12. Given a field k with a norm $|\cdot|$, we say that a sequence $(x_n)_{n\geq 1}$ with $x_i \in k$ is a **Cauchy sequence** if $\forall \epsilon > 0$, $\exists n_0 \geq 1$ such that $|x_n - x_m| < \epsilon$ for all $n, m \geq n_0$.

Definition 1.2.13. Given a field k with a norm $|\cdot|$, we say that a sequence $(x_n)_{n\geq 1}$ **converges to a limit** x if $\forall \epsilon > 0$, $\exists n_0 \geq 0$ such that $|x_n - x| < \epsilon$ for all $n \geq n_0$.

Example 1.2.14

Let $p = 5$, and consider the sequence $x_1 = 4$, $x_2 = 34$, $x_3 = 334$, $\dots$, $x_n = 33\dots34$, $\dots$ of integers. Then, $(x_n)_{n\geq 1}$ is a Cauchy sequence for $|\cdot|_5$ (since there will be many 0s if we subtract one term from another). moreover, it converges to $\frac{2}{3}$, as can be seen by showing that $|3x_n - 2|_5 \to 0$ as $n \to \infty$.

Remark 1.2.15 — Given the example above, one could ask if the p-adic expansion of a rational number is eventually periodic or not. In fact, this is true! There is a theorem which states that in $\mathbb{Q}_p$, the numbers with eventually periodic p-adic expansions are exactly the rational numbers $\mathbb{Q}$ [2].

Definition 1.2.16. A field k is **complete** with respect to a norm if every Cauchy sequence has a limit.

Remark 1.2.17 — $\mathbb{Q}$, equipped with $|\cdot|_p$ is not complete for any p, as we can see from the following example.

Example 1.2.18

For instance, we can construct a sequence $\{x_n\}$ of integers such that

$$x_n^2 + 1 \equiv 0 \pmod{5^n}$$
$$x_{n+1} \equiv x_n \pmod{5^n}$$

We can start with $x_1 = 2$; supposing that we already have x_n, write $x_n^2 + 1 = 5^n c$. We aim to construct $x_{n+1} = x_n + b5^n$ such that $(x_n + b5^n)^2 + 1 \equiv 0 \pmod{5^{n+1}}$. Therefore, we need $2x_n b + c \equiv 0 \pmod 5$, but this can be solved since $5 \nmid x_n$. We've just constructed a 5-adic Cauchy sequence, as $|x_n - x_m|_5 \leq 5^{-n}$ for $m \geq n$. Suppose now that x_n converges to x, then $x_n^2 + 1$ converges to $x^2 + 1$. However, by our construction, $x_n^2 + 1 \to 0$, so it must be that $x^2 + 1 = 0$, but no such x exists in $\mathbb{Q}$.

Remark 1.2.19 — This tells us that $\mathbb{Q}$ is not complete with respect to the p-adic norm.

Exercise 1.2.20. For any prime p, find a $t \in \mathbb{Z}$ that is not a square and a Cauchy sequence $(x_n)_{n\geq 1}$ such that $x_n^2 \to t$ as $n \to \infty$.

Solution: I think this exercise should be done later, after seeing Hensel's lemma. Motivated by $x^2 = t$, we consider the sequence defined by

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{t}{x_n}\right).$$

If this sequence is Cauchy in $\mathbb{Q}_p$, then we are done, since then completeness implies the existence of a limit in $\mathbb{Q}_p$, so $\lim_{n\to\infty} x_n^2 = t$. The way we do this is rather constructive. We just let t be equal to some value and then show that the polynomial $x^2 - t$ has a root in $\mathbb{Q}_p$. We apply Hensel's lemma now with $x_0 = 1$. We get that $|f(1)|_p = |1-t|_p$ and $|f'(1)|^2 = |2|_p^2$. We want

$$|1-t|_p < |2|_p^2$$

at all p. Note that if $p = 2$, then the RHS is 2^{-2} and for odd p, the RHS is just 1. We also want t to be a nonsquare, which we can do just be consider t to be negative. Thus, we want t to be of the form $1 - kp$, for some positive integer k. For instance, $1 - 4p$ would work then. Further, $1 - 2^s p^r$ would definitely work for $s, r \in \mathbb{Z}_{>0}$. $\square$

§1.2.i Constructing $\mathbb{Q}_p$ and $\mathbb{Z}_p$

Just like the field $\mathbb{R}$ is the completion of the rationals with respect to the usual absolute value, the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$ is $\mathbb{Q}_p$; this is the p-adic numbers. Let us now give the construction of a completion of a normed field $(k, |\cdot|)$. We shall further assume that the norm is non-archimedean, but this changes very little.

The idea is to construct a bigger field by adding in the limit points of Cauchy sequences, but we need to be careful because the limit point may arise as a limit of many sequences. Define

$$\mathcal{C} := \{(x_n)_{n\geq 1} \colon x_n \in k, (x_n) \text{ a Cauchy sequence } \},$$

be the set of Cauchy sequences in k. We call two Cauchy sequences (x_n) and (y_n) equivalent if $|x_n - y_n| \to 0$ as $n \to \infty$. We define $\hat{k}$ to be the set of equivalence classes of Cauchy sequences. In other words,

$$\mathcal{I} := \{(x_n)_{n\geq 1} \in \mathcal{C} \colon \lim_{n\to\infty} x_n = 0\}.$$

Then, we define $\hat{k} := \mathcal{C}/\mathcal{I}$.

Remark 1.2.21 — We note that $\mathcal{C}$ is a commutative ring with addition and multiplication defined as

$$(x_n) \pm (y_n) = (x_n \pm y_n)$$
$$(x_n) \cdot (y_n) = (x_n y_n)$$

and $0 = (0)_{n\geq 1}$ and $1 = (1)_{n\geq 1}$.

By the remark above, to show that $\hat{k}$ is actually a field, we just need to show that $\mathcal{I}$ is a maximal ideal.

Lemma 1.2.22

Suppose $(x_n)_{n\geq 1} \in \mathcal{C} \setminus \mathcal{I}$. There exists an $n_0 \geq 1$ such that $|x_n| = |x_{n_0}|$ for all $n \geq n_0$.

Proof. Since $(x_n) \notin \mathcal{I}$, there exists $\epsilon > 0$ and for all $N \geq 1$, there exists $n(N) > N$ with $|x_{n(N)}| > \epsilon$. On the other hand, (x_n) is Cauchy, so there exists $M \geq 1$ such that $|x_n - x_m| < \epsilon$ for all $n, m \geq M$. Fixing some $\epsilon > 0$, find M and let $n_0 = n(M) > M$. then, for all $n \geq n_0$,

$$|x_{n_0}| > \epsilon > |x_n - x_{n_0}|.$$

It then follows by Lemma 1.2.8 that $|x_{n_0}| = |x_n|$ for all $n \geq n_0$. □

Remark 1.2.23 — The crux of this proof is the isosceles property of non-archimedean norms.

Claim 1.2.24. $\mathcal{I}$ is a maximal ideal.

Proof. We will show maximality by showing that an element in the complement has an inverse. It follows that $x_n \neq 0$ for all $n > n_0$. Hence, we can define a new sequence (y_n) with $y_n = \frac{1}{x_n}$ for $n \geq n_0$, and 0 otherwise. Then, $x_n y_n = 1$ for $n \geq n_0$, hence $(x_n)(y_n) - 1 \in \mathcal{I}$. Thus, $\mathcal{I}$ is maximal. □

We therefore have that $\hat{k} = \mathcal{C}/\mathcal{I}$ is a field. Furthermore, the norm on k extends to a norm on $\hat{k}$; that is, the map $|\cdot|\colon \hat{k} \to \mathbb{R}_{\geq 0}$, by letting $|(x_n)| = \lim_{n\to\infty} |x_n| = |x_{n_0}|$ is a norm on $\hat{k}$.

Remark 1.2.25 — When completing $\mathbb{Q}$ to $\mathbb{R}$, the possible values of $|\cdot|$ is enlarged, but when completing $\mathbb{Q}$ to $\mathbb{Q}_p$, the possible values remain the same: $\{p^n\}_{n\in\mathbb{Z}} \cup \{0\}$.

Lemma 1.2.26

The field embedding $k \to \hat{k}$ given by $x \to (x)_{n\geq 1}$ is dense.

Proof. Let $(x_n) \in \hat{k}$ and $\epsilon > 0$. Choose N such that if $m, n \geq N$, then $|x_m - x_n| < \epsilon$. Consider $y = (x_N)_{n\geq 1} \in k$. Then, $|x - y| = \lim_{n\to\infty} |x_n - x_N| < \epsilon$. □

It follows from the lemma above that we can identify elements of k with constant sequences in $\hat{k}$.

Lemma 1.2.27

$\hat{k}$ is complete.

Proof. Let (x_n) be a Cauchy sequence in $\hat{k}$, so x_n is itself an equivalence class of Cauchy sequences in k. By Lemma 1.2.26, for each $n \geq 1$, there exists $y_n \in k$ such that $|x_n - y_n| < \frac{1}{n}$. We claim that $y = (y_n)$ is a Cauchy sequence, and $x_n \to y$ as $n \to \infty$. Since

$$|y_m - y_n| \leq |y_m - x_m| + |x_m - x_n| + |x_n - y_n| < \frac{1}{m} + |x_m - x_n| + \frac{1}{n}$$

and (x_n) is Cauchy, it follows that (y_n) is Cauchy as well. Then,

$$|x_n - y| \leq |x_n - y_n| + |y_n - y| < \frac{1}{n} + |y_n - y|.$$

Now, $|y_n - y| \to 0$ as $n \to \infty$ by how we defined y, so $x_n \to y$ as $n \to \infty$. □

Example 1.2.28

Given $k = \mathbb{Q}$, $|\cdot|_p \colon \mathbb{Q} \to \mathbb{R}_{>0}$, we have $\hat{k} = \mathbb{Q}_p$ the p-adic numbers.

Definition 1.2.29. We define the ***p*-adic integers** $\mathbb{Z}_p := \{x \in \mathbb{Q}_p \colon |x|_p \leq 1\} \subset \mathbb{Q}_p$.

Remark 1.2.30 — $\mathbb{Z}_p$ is indeed a ring; we have that $|\alpha|_p, |\beta|_p \leq 1$ implies that $|\alpha\beta|_p, |\alpha + \beta|_p \leq 1$. Note that this relies on the ultrametric property.

Remark 1.2.31 — A rational number b is in $\mathbb{Q}_p$ when it takes the form $b = \frac{u}{v}$, where $u, v \in \mathbb{Z}$ and $p \nmid v$. In other words, $\mathbb{Z}_p \cap \mathbb{Q} = \mathbb{Z}_{(p)}$ is the localization of $\mathbb{Z}$ at the prime ideal (p), and $\mathbb{Z}_{(p)} = \{\frac{u}{v} \in \mathbb{Q} \colon p \nmid v\}$.

Definition 1.2.32. $\epsilon \in \mathbb{Z}_p$ with $|\epsilon|_p = 1$ are called the ***p*-adic units**.

Remark 1.2.33 — Every $\beta \neq 0$ in $\mathbb{Q}_p$ is of the form $\beta = p^n\epsilon$ for some $n \in \mathbb{Z}$ and ϵ is a unit. Note that for any nonzero $x \in \mathbb{Q}_p$, either $x \in \mathbb{Z}_p$ or $x^{-1} \in \mathbb{Z}_p$. The units are precisely the elements of $\mathbb{Q}_p$ such that both $x \in \mathbb{Z}_p$ and $x^{-1} \in \mathbb{Z}_p$.

Remark 1.2.34 — $\mathbb{Z}_p$ has a unique maximal ideal $(p) = \{x \in \mathbb{Z}_p \colon |x|_p < 1\}$. It's maximal because $\mathbb{Z}_p/(p) = \mathbb{F}_p$ is a field.

Remark 1.2.35 — We have that $\mathbb{Z} \subset \mathbb{Z}_p$. Note that $\frac{1}{p} \notin \mathbb{Z}_p$, since $|\frac{1}{p}|_p = p > 1$. In fact, $\mathbb{Q}_p = \mathbb{Z}_p[\frac{1}{p}]$, the field of fractions of $\mathbb{Z}_p$.

§1.2.ii Series in p-adic Numbers

In $\mathbb{Q}_p$ we define a series $\sum_{n=0}^{\infty} \beta_n$ in the usual way as the limit of partial sums $\sum_{n=0}^{N} \beta_n$. The convergence test is much easier in non-archimedean analysis.

Lemma 1.2.36

The series $\sum_{n=0}^{\infty} \beta_n$ converges if and only if $\beta_n \to 0$.

Proof. The proof of the forward direction is the same as that in analysis. For the converse, we want to show that the partial sums form a Cauchy sequence. Observe that

$$\left|\sum_{n=0}^{N} \beta_n - \sum_{n=0}^{M} \beta_n\right| = \left|\sum_{n=M+1} \beta_n\right|_p \leq \max_{M<n\leq N} |\beta_n|_p.$$

If M, N are large, then the LHS is small. Thus, the terms $\sum_{n=0}^{N} \beta_n$ form a Cauchy sequence, so converges by completeness of $\mathbb{Q}_p$. □

Remark 1.2.37 — There is no analogue of the harmonic series of real numbers in $\mathbb{Q}_p$, whose terms approach zero yet the sum diverges.

Lemma 1.2.38

The elements of $\mathbb{Z}_p$ are precisely the sums

$$\alpha = \sum_{n=0}^{\infty} a_n p^n,$$

where $a_n \in \{0, 1, \dots, p-1\}$ for all n.

Proof. By the previous lemma, the provided sum converges to an element in $\mathbb{Z}_p$. Conversely, suppose $\alpha \in \mathbb{Z}_p \subseteq \mathbb{Q}_p$. $\mathbb{Q}$ is dense in $\mathbb{Q}_p$, so there exists $b \in \mathbb{Q}$ such that $|\alpha - b|_p < 1$. $|\alpha|_p \leq 1$, so we have that $|b|_p \leq 1$ as well, implying that $b \in \mathbb{Z}_p$. Writing $b = \frac{r}{s}$, with $(r, s) = 1$ and $p \nmid s$, there exists a unique solution $a_0 \in \{0, \dots, p-1\}$ to the congruence $sa_0 - r \equiv 0 \pmod{p}$. In other words, $|b - a_0|_p < 1$. Then, since $|b - a_0|_p, |\alpha - b|_p < 1$, we get that $|\alpha - a_0|_p < 1$, so we can write

$$\alpha = a_0 + p\alpha_1,$$

where $|\alpha_1|_p \leq 1$; that is, $\alpha_1 \in \mathbb{Z}_p$. We can then do this inductively to get

$$\alpha = a_0 + a_1 p + \cdots + a_n p^n + \alpha_{n+1} p^{n+1},$$

with $\alpha_n \in \mathbb{Z}_p$. Note that this process goes forever, but $a_n \to 0$ as $n \to \infty$. Therefore, $\alpha = \sum_{n=0}^{\infty} a_n p^n$. $\square$

Corollary 1.2.39

Any $\alpha \in \mathbb{Q}_p$ can be uniquely written as

$$\alpha = \sum_{n \geq -T} a_n p^n,$$

where $a_{-T} \neq 0$ and $a_n \in \{0, 1, \dots, p-1\}$.

Proof. This follows from the observation that if $|\alpha|_P = p^T$, then $|p^T \alpha|_p = 1$, so $p^T \alpha \in \mathbb{Z}_p$. $\square$

This motivates us to think of $\mathbb{Z}_p$ to be analogous to $k[[X]]$ and $\mathbb{Q}_p$ as being analogous to $k((X))$, the ring of formal power series and formal Laurent series over a field k.

Remark 1.2.40 — Truncating the expansion above, we can approximate any element in $\mathbb{Z}_p$ using integers, giving the following corollary.

Corollary 1.2.41

$\mathbb{Z}$ is dense in $\mathbb{Z}_p$.

Proof. Let $\alpha = \sum_{n=0}^{\infty} a_n p^n \in \mathbb{Z}_p$ and fix some ϵ. Fix some N such that $p^{-N-1} < \epsilon$. Then, consider the sequence defined by $\alpha_N = \sum_{n=0}^{N} a_n p^n$. This is a finite sum, so is in $\mathbb{Z}$. Then, we have that

$$|\alpha - \alpha_N|_p = |\sum_{N+1}^{\infty} a_n p^n|_p \leq p^{-N-1} < \epsilon.$$

Therefore, we have shown that $\mathbb{Z}$ is dense in $\mathbb{Z}_p$. $\square$

Remark 1.2.42 — Note that we have a map of rings $\mathbb{Z}_p \to \mathbb{Z}/p^n\mathbb{Z}$ for any n, defined by the reduction modulo p^n map. That is, given $\alpha = \sum_{i=0}^{\infty} a_i p^i \in \mathbb{Z}_p$,

$$\alpha \mapsto \alpha \pmod{p^n} = \sum_{i=0}^{n-1} a_i p^i \pmod{p^n}.$$

We can show that these maps can be assembled to show that

$$\mathbb{Z}_p \cong \varprojlim_n (\mathbb{Z}/p^n\mathbb{Z}).$$

Exercise 1.2.43. Prove that a p-adic number $\alpha \in \mathbb{Q}_p$ has a finite expansion if and only if α is a positive rational number whose denominator is a positive power of p.

Exercise 1.2.44. Prove that a p-adic number $\alpha \in \mathbb{Q}_p$ is in $\mathbb{Q}$ if and only if it has an eventually periodic expansion; that is, there exists some N and s such that $a_i = a_{i+s}$ for all $i > N$.

Exercise 1.2.45. Show that every infinite sequence (x_n) with $x_n \in \mathbb{Z}_p$ has a convergent subsequence.

§1.3 Hensel's Lemma

We now turn to ask about solutions to polynomial equations in $\mathbb{Q}_p$, or whether numbers in $\mathbb{Q}_p$ admit square roots. In other words, we're interested in the algebraic closure of $\mathbb{Q}_p$. This is actually an infinite degree extension! We note that when we take the algebraic closure of $\mathbb{R}$ (which is $\mathbb{C}$), this is a finite extension so we get completeness in $\mathbb{C}$ as well. However, because the algebraic closure of $\mathbb{Q}_p$ is an infinite extension, we don't get completeness here. We have to take the completeness of the algebraic closure of $\mathbb{Q}_p$ to fully get completeness and algebraic closure (which is where analysis happens); this is called $\mathbb{C}_p$, but we won;t study this.

Going back to the original problem, we attack arithmetic problems which arise from polynomial equations by *Hensel's lemma*, which allows one to lift solutions over $\mathbb{F}_p$ to $\mathbb{Q}_p$. Here are some preliminary examples/ exercises.

Example 1.3.1

Let $\epsilon \in \mathbb{Z}_p$ be a unit. For $p \neq 2$, a necessary and sufficient condition for $\epsilon = \alpha^2$ for some $\alpha \in \mathbb{Q}_p$ is that ϵ is a square modulo p.

Proof. Let $p \neq 2$, and take $x \in \mathbb{F}_p$ such that $x^2 = \epsilon \pmod p$. Then, we inductively construct $\alpha_1 = x$, α_2, ..., that is subject to the conditions

$$|\alpha_n^2 - \epsilon|_p \leq p^{-n}$$
$$|\alpha_{n+1} - \alpha_n|_p \leq p^{-n}$$

Supposing that we already have constructed α_n, we set $\alpha_{n+1} = \alpha_n + p^n\beta$. Since $\alpha_{n+1}^2 \equiv \alpha_n^2 + 2p^n\alpha_n\beta \pmod{p^{n+1}}$ and $p^n \mid \alpha_n^2 - \epsilon$, what we need to arrange is that $2\alpha_n\beta \equiv \frac{\epsilon - \alpha_n^2}{p^n} \pmod p$. We can arrange this since $p \nmid 2$ and $p \nmid \epsilon$. Given this, we let $\alpha = \lim\limits_{n\to\infty} \alpha_n$. □

Example 1.3.2

Let $p > 0$ be a prime, and $p \equiv 2 \pmod 3$. For any integer a such that $p \nmid a$, there exists an $x \in \mathbb{Z}_p$ with $x^3 = a$.

Proof. Consider the group homomorphism $x \mapsto x^3$ from $\mathbb{F}_p^\times$ to itself. $3 \nmid p - 1$, so there are no element of order 3 in $\mathbb{F}_p^\times$. Thus, this map is injective and also surjective. This means we can find x_1 with $x_1^3 \equiv a \pmod p$. next, suppose that we have $x_n^3 \equiv a \pmod{p^n}$ and pose $x_{n+1} = x_n + p^n y$. We then seek to solve $x_{n+1}^3 \equiv a \pmod{p^{n+1}}$. We can compute that

$$x_{n+1}^3 \equiv (x_n + p^n y)^3 \equiv x_n^3 + 3p^n x_n^2 y \pmod{p^{n+1}}.$$

By assumption, $p^n \mid x_n^3 - a$, so if we let y be such that $3x_n^2 y = \frac{a - x_n^3}{p} \pmod p$, which we can do since $p \nmid 3x_n^2$ as $p \nmid a$ and $p \neq 3$, then $p^{n+1} \mid x_{n+1}^3 - a$ as required. □

Exercise 1.3.3. Show that there is no 7-adic number $x = 2 + y$ with $|y|_7 \leq 7^{-1}$ such that $x^3 + x^2 - 2x - 1 = 0$.

Solution: $|y|_7 \leq 7^{-1}$ implies that $7 \mid y$. $x = 2 + y$, so in other words, $x \equiv 2 \pmod 7$. Further, $x = 2 + y$, so x is a 7-adic unit: $|x|_7 = 1$. Then, $x = 2 + 7a$, for some $a \in \mathbb{Z}_p$. Taking modulo 7 is not enough, as we see below:

$$x^3 + x^2 - 2x - 1 \equiv 8 + 4 - 4 - 1 \equiv 0 \pmod 7$$

Therefore, we take modulo 7^2 next.

$$\begin{aligned} x^3 + x^2 - 2x - 1 &= (2 + 7a)^3 + (2 + 7a)^2 - 2(2 + 7a) - 1 \\ &\equiv 8 + 6 \cdot 7a + 4 + 4 \cdot 7a - 4 + (-2) \cdot 7a - 1 \pmod{7^2} \\ &\equiv 8 \cdot 7a \equiv 7(8a) \pmod{7^2} \end{aligned}$$

We require $8a \equiv 0 \pmod 7$, so $a \equiv 0 \pmod 7$ then. □

Lemma 1.3.4 (Hensel's Lemma)

Let $f = \sum_{i=0}^n a_i X^i \in \mathbb{Z}_p[X]$, and suppose that there exists $x_0 \in \mathbb{Z}_p$ such that

$$|f(x_0)| < |f'(x_0)|^2, \tag{1}$$

where $f'(X)$ denotes the (formal) derivative. Then, there exists a unique root of f in $\mathbb{Z}_p$ satisfying $|x - x_0| < |f'(x_0)|$. Further, $|x - x_0| = \frac{|f(x_0)|}{|f'(x_0)|} < |f'(x_0)|$ and $|f'(x)| = |f'(x_0)|$.

Proof. We can rewrite (1) as $|\frac{f(x_0)}{f'(x_0)}| < |f'(x_0)| \leq 1$, since $f'(x_0) \in \mathbb{Z}_p$. This implies that there is some y_0 with $|y_0| < 1$ such that $f(x_0) + y_0 f'(x_0) = 0$. Let us define (finitely many) polynomials $f_1, f_2, \ldots, \in \mathbb{Z}_p[X]$ via the identity

$$f(X+Y) = f(X) + f_1(X)Y + f_2(X)Y^2 + \ldots,$$

which is a finite sum. Then, $f_1(X) = f'(X)$. We then have

$$|f(x_0 + y_0)| \leq \max_{j \geq 2} |f_j(x_0) y_0^j|.$$

Here, note that $|f_j(x_0)| \leq 1$ since $f_j(X) \in \mathbb{Z}_p[X]$ and $x_0 \in \mathbb{Z}_p$. Thus,

$$|f(x_0 + y_0)| \leq |y_0|^2 = \frac{|f(x_0)|^2}{|f'(x_0)|^2} < |f(x_0)|.$$

Similarly,

$$|f'(x_0 + y_0) - f'(x_0)| \leq |y_0| < |f'(x_0)|,$$

so the isosceles property (Lemma 1.2.8) tells us that $|f'(x_0 + y_0)| = |f'(x_0)|$. Now set $x_1 = x_0 + y_0$, and note that $|f(x_1)| < |f'(x_1)|^2$, so we can repeat this process. In this fashion, we obtain a sequence $(x_n)_{n \geq 0}$ using Newton's formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ with the properties

(i) $|x_n| \leq 1$, so $x_n \in \mathbb{Z}_p$.

(ii) $|f'(x_n)| = |f'(x_0)|$

(iii) $|f(x_{n+1})| \leq \frac{|f(x_n)|^2}{|f'(x_n)|^2} = \frac{|f(x_n)|^2}{|f'(x_0)|^2} < |f(x_n)|$, so $f(x_n) \to 0$ as $n \to \infty$.

(iv) $|x_{n+1} - x_n| = |y_n| = \frac{|f(x_n)|}{|f'(x_n)|} = \frac{|f(x_n)|}{|f'(x_0)|} \to 0$ as $n \to \infty$.

It follows that $(x_n)_{n \geq 0}$ is Cauchy. Then, the element $x := \lim_{n \to \infty} x_n \in \mathbb{Z}_p$ exists and $f(x) = 0$. To see that $|x - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$, we claim that $|x_n - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$. We prove this by induction. For $n = 1$, this is clear, so assume true for up to some n. Now, observe that

$$|x_{n+1} - x_n| = \frac{|f(x_n)|}{|f'(x_n)|} < \frac{|f(x_0)|}{|f'(x_0)|}$$

for all $n \geq 1$. Now, note that $|x_1 - x_0| = \frac{|f(x_0)}{|f'(x_0)|}$, and if $|x_n - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$ for some n, then it follows that $|x_{n+1} - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$ since we have $|x_{n+1} - x_0| = |x_n - x_0|$ by the isosceles property again.

Finally, we prove uniqueness. Suppose $\tilde{x}$ is another solution with $|\tilde{x} - x_0| < |f'(x_0)|$ and $x \neq \tilde{x}$. Then, $\tilde{x} = x + \tilde{y}$. We have

$$0 = f(x + \tilde{y}) - f(x) = f'(x)\tilde{y} + f_2(x)\tilde{y}^2 + \ldots.$$

However, $|\tilde{y}| = |x - x_0 + x_0 - \tilde{x}| \leq \max\{|x - x_0|, |\tilde{x} - x_0|\} < |f'(x_0)| = |f'(x)|$ and $|f_i(x)| \leq 1$. The first term on the RHS of the above equation has strictly greater norm than the others, so the sum can't be 0, which is a contradiction. $\square$

Remark 1.3.5 (Alternate Forms of Hensel's Lemma) — Let $: \mathbb{Z}_p \to \mathbb{Z}_p/(p) = \mathbb{F}_p$ denote the reduction homomorphism to the residue field, and let $\overline{f} = \sum_{i=0}^{n} \overline{a_i} X^i \in \mathbb{F}_p[X]$. Instead of (1), some other may state this result by assuming the existence of a simple root of $\overline{f}$ over $\mathbb{F}_p$ (which is easier to check but a stronger assumption). Indeed, assuming that there exists $\overline{x_0} \in \mathbb{F}_p$ such that

$$\overline{f}(\overline{x_0}) = 0 \text{ and } \overline{f'}(\overline{x_0}) \neq 0, \tag{2}$$

this implies (1). If we let $x_0 \in \mathbb{Z}_p$ be any lift of $\overline{x_0} \in \mathbb{F}_p$, then we have that $p \mid f(x_0)$ but $p \nmid f'(x_0)$, so $|f(x_0)|_p < 1$ and $|f'(x_0)| = 1$.

Remark 1.3.6 — Hensel's lemma holds over complete DVRs. There is a nice treatment in [1].

Example 1.3.7

Consider $f(x) = x^2 - 11 \in \mathbb{Z}_7[X]$. Note that $x_0 = 2$ gives a solution over $\mathbb{F}_7 \cong \mathbb{Z}_7/(7)$. Moreover, $f'(x) = 4 \neq 0 \in \mathbb{F}_7$, so the assumptions are satisfied. Then, we let $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 + \frac{7}{4} = \frac{15}{4}$. We have $f(x_1) = \frac{225}{16} - 11 = \frac{7^2}{2^4}$, so this gives a solution over $\mathbb{Z}_7/(7^2)$. Moreover,$f'(x_1) = \frac{15}{2} \neq 0 \in \mathbb{F} - 7$ as it should. Thus, we can continue to define $x_2 = x_2 - \frac{f(x_1)}{f'(x_1)} = \frac{401}{120}$. We can see that $f(x_2) = \frac{7^4}{2^6 3^2 5^2}$, so we get a solution over $\mathbb{Z}_7/(7^4)$. Note that we were looking for a solution modulo 7^3, but got lucky and found one modulo 7^4 instead. As proven above, we can continue this to attain a solution in $\mathbb{Q}_7$. We can write out the 7-adic expansion as follows. First, we observe the identities:

$$\begin{aligned} 401 &= 2 + 1.7 = 1.7^2 + 1.7^3 \\ \frac{1}{2} &= 4 + 3.7 + 3.7^2 + 3.7^3 + O(7^4) + \dots \\ \frac{1}{3} &= 5 + 4.7 + 4.7^2 + 4.7^3 + O(7^4) + \dots \\ \frac{1}{5} &= 3 + 1.7 + 1.7^2 + 1.17^3 + O(7^4) + \dots \end{aligned}$$

Then, a solution in $\mathbb{Z}_7/(7^4)$ is given by

$$\begin{aligned} &(2+1.7 = 1.7^2+1.7^3)(4+3.7+3.7^2+3.7^3)^3(5+4.7+4.7^2+4.7^3)(3+1.7+1.7^2+1.17^3) \\ &\qquad = 2 + 2.7 + 4.7^2 + 4.7^3 + O(7^4). \end{aligned}$$

Remark 1.3.8 — The method used in Hensel's lemma is an adaptation of Newton's method of approximations to the roots of a real polynomial, where successive approximations are defined by the formula $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$ However, in the non-archimedean case, it's guaranteed that the sequence converges, whereas in the real case, it *usually* converges, but not always. For instance, taking $f(X) = X^3 - X$ and choosing $x_0 = \frac{1}{\sqrt{5}}$, then we get $x_i = \frac{(-1)^i}{\sqrt{5}}$, for all $i \geq 0$.

Example 1.3.9

Consider $f(x) = x^2 - 11$ in $\mathbb{Z}_7$. $x_0 = 2$ works in $\mathbb{F}_p$, so $7^{-1} = |f(x_0)|_7 < |f'(x_0)|_7^2 = 1$. Then, by Hensel's lemma, there is a unique solution. We can define

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 + \frac{7}{4} = \frac{15}{4}.$$

This gives

$$f(x_1) = f(\frac{15}{4}) = \frac{225}{16} - 11 = \frac{7^2}{16},$$

so our solution indeed improved. Indeed, $f'(x_1) = \frac{15}{2}$, so $|f'(x_1)|_7 = 1$. Next,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{401}{2^3 \cdot 3 \cdot 5},$$

so

$$f(x_2) = \frac{7^4}{2^6 3^2 5^2},$$

and so we actually got a better solution than expected.

Corollary 1.3.10

Consider $x^{p-1} - 1 \in \mathbb{Z}_p[X]$. Then, for every $a \in \mathbb{F}_p^\times$, there exists a unique $\alpha \in \mathbb{Z}_p$ such that $f(\alpha) = 0$ and $\alpha = a \in \mathbb{Z}_p/(p) = \mathbb{F}_p$.

Proof. Let $x_0 \in \mathbb{Z}_p$ be any element that lifts a; that is $\overline{x_0} = a \in \mathbb{Z}_p/(p)$. We have $f'(x_0) = (p-1)x_0^{p-2} \neq 0 \in \mathbb{F}_p$, so $|f'(x_0)|_p = 1$. On the other hand, over $\mathbb{F}_p$, we have $x^{p-1} - 1 = \prod_{0 \neq a \in \mathbb{F}_p}(x - a)$ by Fermat's little theorem. Thus, $|f(x_0)|_p < 1$. Therefore, Hensel's Lemma implies that f has a root in $\mathbb{Z}_p$ lifting a. $\square$

Example 1.3.11

Take $p = 5$ for instance. Here are the four solution solutions to the equation $x^4 = 1$ over $\mathbb{Z}_5$, up to $O(5^{11})$.

$$[1] = 1$$
$$[2] = 2 + 1.5 + 2.5^2 + 1.5^3 + 3.5^4 + 4.5^5 + 2.5^6 + 3.5^7 + 3.5^9 + 2.5^{10} + O(5^{11})$$
$$[3] = 3 + 3.5 + 2.5^2 + 3.5^3 + 1.5^4 + 2.5^6 + 1.5^7 + 4.5^8 + 1.5^9 + 2.5^{10} + O(5^{11})$$
$$[4] = 4 + 4.5 + 4.5^2 + 4.5^3 + 4.5^4 + 4.5^5 + 4.5^6 + 4.5^7 + 4.5^8 + 4.5^9 + 4.5^{10} + \dots$$

Example 1.3.12

Let $\epsilon \in \mathbb{Z}_2$ be a unit. Then, $\epsilon = \alpha^2$ for some $\alpha \in \mathbb{Q}_2$ if and only if $\epsilon \equiv 1 \pmod 8$.

Proof. We apply Hensel's lemma. If $\varepsilon = \alpha^2$, then $|\alpha|_2^2 = 1$; hence, α is a unit. In $\mathbb{Z}_2/8\mathbb{Z}_2 = \mathbb{Z}/8\mathbb{Z}$, the units are $1, 3, 5$ and 7 which square to $1 \pmod 8$. Conversely, suppose $\epsilon \equiv 1 \pmod 8$. Let $f(X) = X^2 - \epsilon \in \mathbb{Z}_2[X]$. We have $|f(1)|_2 = |1 - \epsilon|_2 \leq 1/8$ and $|f'(1)|_2 = |2|_2 = 1/2$, so $|f(1)|_2 < |f'(1)|_2^2$. Therefore, by Hensel's lemma, $f(X)$ has a root in $\mathbb{Z}_2$. $\square$

§1.4 Exercises

Exercise 1.4.1. For each of the sets of p, m, r given, either find an $x \in \mathbb{Z}$ such that

$$|r - x|_p \le p^{-m},$$

or show that no such x exists.

(i) $p = 257, r = \frac{1}{2}, m = 1$

(ii) $p = 3, r = \frac{7}{8}, m = 2$

(iii) $p = 3, r = \frac{7}{8}, m = 7$

(iv) $p = 3, r = \frac{5}{6}, m = 9$

(v) $p = 5, r = \frac{1}{4}, m = 4$

Solution: (i) Since $|2|_p = 1$, this is equivalent to finding x such that

$$|1 - 2x|_{257} \le 257^{-1} \iff v_{257}(1 - 2x) \ge 1.$$

Then, $x = -128$ works.

(ii) Since $|8|_3 = 1$, this is equivalent to finding

$$|7 - 8x|_3 \le 3^{-2} \iff v_3(7 - 8x) \ge 2.$$

Taking $x = 2$ works.

(iii) This is equivalent to finding

$$|7 - 8x|_3 \le 3^{-7} \iff v_3(7 - 8x) \ge 7.$$

We can see that we can try to solve the equation $7 - 8x = 3^7 \cdot a$, for some integer pair x, a. Modulo 8, we see that $a \equiv 5 \pmod 8$. Thus, $x = \frac{7 - 3^7 \cdot 5}{8} = -1366$ works. Of course, if we want the solution by Professor Yanki Lekili, we can add 3^7 to this to get 821.

We can also do this by constructing a p-adic number, which I guess is the desired route. We find solutions to $8x - 7 \equiv 0 \pmod{3^i}$, for each $i = 2, \dots, 7$ to construct such a sequence. From the previous part, we know that

$$8 \cdot 2 - 7 \equiv 0 \pmod{3^2}.$$

Next, we search for a_2 such that $x = 2 + 3^2 a_2$ satisfies $8x - 7 \equiv 0 \pmod{3^3}$. $a_2 = 1$ works, and actually $8(2 + 9) - 7 = 81$, so we've solved the congruence for $i = 3$ as well. Next, we search for a_4 such that $x = 2 + 3^2 + 3^4 a_4 \equiv 0 \pmod{3^5}$. $a_4 = 1$ works, and $8(2 + 9 + 81) - 7 = 729 = 3^6$, so this works for $i = 6$ as well. Finally, we want to search for a_6 such that $x = 2 + 3^2 + 3^4 + 3^6 a_6$ satisfies $8x - 7 \equiv 0 \pmod{3^7}$, and $a_6 = 1$ works: $8(2 + 9 + 81 + 729) - 7 = 6561 = 3^8$, so this x works:

$$x = 2 + 3^2 + 3^4 + 3^6 = 821.$$

(iv). This is equivalent to finding x such that

$$|5 - 6x|_3 \le 3^{-10} \iff v_3(5 - 6x) \ge 10.$$

This is not possible because $5-6x \equiv 2 \pmod 2$, so it can't be divisible by any power of 3.

(v). This is equivalent to finding

$$|1-4x|_5 \leq 5^{-4} \iff v_5(1-4x) \geq 4.$$

We search for the constants a_0, a_1, a_2, a_3 in the solution $x = a_0 + a_1 5 + a_2 5^2 + a_3 5^3$. Clearly, $a_0 = 4$ satisfies $1-4x \equiv 0 \pmod 5$. Next, $a_1 = 3$ satisfies $1-4x \equiv 0 \pmod{5^2}$. Next, $a_2 = 3$ satisfies $1-4x \equiv 0 \pmod{5^3}$. Finally, $a_3 = 3$ works. Therefore,

$$x = 4 + 3\cdot 5 + 3 \cdot 5^2 + 3 \cdot 5^3 = 469$$

satisfies the inequality. □

Exercise 1.4.2. For given p, m, r, either find an $x \in \mathbb{Z}$ such that

$$|r - x^2|_p \leq p^{-m}$$

or show that no such x exists.

(i) $p = 5, r = -1, m = 4$

(ii) $p = 5, r = 10, m = 3$

(iii) $p = 13, r = -4, m = 3$

(iv) $p = 2, r = -7, m = 6$

(v) $p = 7, r = -14, m = 4$

(vi) $p = 7, r = 6, m = 3$

(vii) $p = 7, r = \frac{1}{2}, m = 3$

Solution: (i) We want to find an x such that

$$|-1-x^2|_5 \leq 5^{-4} \iff v_5(x^2+1) \geq 4.$$

We can see that $\left(\frac{-1}{5}\right) = (-1)^{\frac{5-1}{2}} = 1$, so -1 is a quadratic residue and so we can find an x. Let $x = a_0 + a_1 5 + a_2 5^2 + a_3 5^3$. We have two choices for a_0: 2 and 3. So write $x = 2 + a_1 5 + a_2 5^2 + a_3 5^3$ and $x' = 3 + b_1 5 + b_2 5^2 + b_3 5^3$. Then, $a_1 = 1$ and $b_1 = 3$, so

$$\begin{aligned} x &= 2 + 1 \cdot 5 + a_2 5^2 + a_3 5^3 \\ x' &= 3 + 3 \cdot 5 + b_2 5^2 + b_3 5^3. \end{aligned}$$

At the next step, we get $a_2 = 2$ and $b_2 = 2$, so

$$\begin{aligned} x &= 2 + 1 \cdot 5 + 2 \cdot 5^2 + a_3 5^3 \\ x' &= 3 + 3 \cdot 5 + 2 \cdot 5^2 + b_3 5^3. \end{aligned}$$

Finally, we get $a_3 = 1$ and $b_3 = 3$. Thus, we've constructed our solutions:

$$\begin{aligned} x &= 2 + 1 \cdot 5 + 2 \cdot 5^2 + 1 \cdot 5^3 \\ x' &= 3 + 3 \cdot 5 + 2 \cdot 5^2 + 3 \cdot 5^3. \end{aligned}$$

(ii). We want to check

$$|10 - x^2|_5 \leq 5^{-3} \iff v_5(10 - x^2) \geq 3.$$

In other words, we want $10 - x^2 \equiv 0 \pmod{5^3}$. We see that $5 \mid x^2$, so $25 \mid x^2$, but $25 \nmid 10$, so no solution.

(iii). We want to find a solution to

$$|-4 - x^2|_{13} \leq 13^{-3} \iff v_{13}(4 + x^2) \geq 3.$$

Write $x = a_0 + a_1 13 + a_2 13^2$. Then, a_0 can be 3 or 10. Thus, let $x = 3 + a_1 13 + a_2 13^2$ and $x' = 10 + b_1 13 + b_2 13^2$. Then, $a_1 = 2$ and $b_1 = 10$ works. Then, we solve we finally get $a_2 = 10$ and $b_2 = 2$. This gives the final solution of:

$$\begin{aligned} x &= 3 + 2 \cdot 13 + 10 \cdot 13^2 \\ x' &= 10 + 10 \cdot 13 + 2 \cdot 13^2 \end{aligned}$$

(iv). We want

$$|-7 - x^2|_2 \leq 2^{-6} \iff v_2(7 + x^2) \geq 6.$$

This is pretty easy. Note that taking $x = 5$ gives $7 + 5^2 = 32$, which is exactly 2^5. Therefore, we just need a bit more:

$$x = 1 + 1 \cdot 2^2 + a_4 2^4 + a_5 2^5.$$

In fact, $a_5 = 0$ since if we square this now, the 2^5 term has coefficient $2a_5$. Thus, we just need to solve $32 + 160a_4 \equiv 0 \pmod{2^6}$, so $a_4 = 1$ works. Thus, we've found a solution:

$$x = 1 + 1 \cdot 2^2 + 1 \cdot 2^4.$$

(v). We want to find

$$|-14 - x^2|_7 \leq 7^{-4} \iff v_7(14 + x^2) \geq 4.$$

In particular, $7 \mid x$, so $49 \mid x^2$. Particularly, $7^2 \mid x^2$, but not 14, so there is no solution.

(vi). We want

$$|6 - x^2|_7 \leq 7^{-3} \iff v_7(6 - x^2) \geq 3.$$

However, we note that $\left(\frac{-1}{7}\right) = -1$, so there is no solution.

(vii). We want

$$|1 - 2x^2|_7 \leq 7^{-3} \iff v_7(1 - 2x^2) \geq 3.$$

First in $\mathbb{F}_7$, we see the solutions 2 and 5. Thus, write $x = 2 + a_1 7 + a_2 7^2$ and $x' = 5 + b_1 7 + b_2 7^2$. $a_1 = 6$ and $b_1 = 0$ work, so we currently have $x = 2 + 6 \cdot 7 + a_2 7^2$ and $x' = 5 + a_2 7^2$. Finally, we can get $a_2 = 5$ and $b_2 = 1$, which give:

$$\begin{aligned} x &= 2 + 6 \cdot 7 + 5 \cdot 7^2 \\ x' &= 5 + 1 \cdot 7^2 \end{aligned}$$

$\square$

Remark 1.4.3 — These types of questions eventually boils down to seeing if solutions exist in each modulo-power; if they do, we can just iterate.

Exercise 1.4.4. Let $p \equiv 2 \pmod 3$. For any integer a such that $p \nmid a$, show that there is a $x \in \mathbb{Z}_p$ with $x^3 = a$.

Solution: Consider the group homomorphism $x \mapsto x^3$ from $\mathbb{F}_p$ to $\mathbb{F}_p$. We note that there are no elements of order 3 in $\mathbb{F}_p$ because $3 \nmid p-1$, so this map is injective and therefore surjective. This implies that we can find a solution to $x_0^3 = a$ in $\mathbb{F}_p$. Now suppose, by induction, that we've found a solution modulo p^n; we call this x_{n-1}. Then, we verify that we can lift this to a solution modulo p^{n+1}. Let $x_n = x_{n-1} + cp^n$ be our candidate. Then,

$$(x_n)^3 = (x_{n-1} + cp^n)^3 \equiv x_{n-1}^3 + 3x_{n-1}^2 cp^n \pmod{p^{n+1}}.$$

Since $x_{n-1}^3 = p^n d$, for some $p \nmid d$ we just want to find a c such that

$$d + 3x_{n-1}^2 c \equiv 0 \pmod p.$$

Indeed, we can, since $d, x_{n-1}, 3$ all don't divide p. Thus, by induction, this is true. $\square$

Exercise 1.4.5. Find the number of solutions to $X^3 - 5X + 10 = 0$ in $\mathbb{Q}_p$ for $p = 2, 3, 5$.

Solution: We first note that we can assume that $|x|_p \leq 1$, for otherwise, the norm of $x^3, -5x, 10$ would differ so they cannot cancel. Now, let

$$f(X) = X^3 - 5X + 10, \quad f'(X) = 3X^2 - 5.$$

Let us first try $p = 2$. modulo 2, we find that $\bar{f}(X) = X^3 + X$, so $0, 1 \in \mathbb{F}_2$ are both solutions. By Hensel's lemma, we see that $|f'(0)|_2 = 1$, so lifts to a solution in $\mathbb{Z}_2$. Let us see if we can find a solution of the form $1 + 2x$ now, where $x \in \mathbb{Z}_2$.

$$f(1+2x) = 1 + 6x + 12x^2 + 8x^3 - 5 - 10x + 10 = 6 - 4x + 12x^2 + 8x^3 \equiv 2 \pmod 4,$$

so we cannot find a solution which lifts $1 \in \mathbb{F}_2$. Thus, we have one solution here.

Next, we try $p = 3$. $\bar{f}(X) = X^3 + X + 1$, so $1 \in \mathbb{F}_3$ is a solution. We can check that $|f'(1)|_3 = 1$, so this lifts uniquely via Hensel. Thus, we have one solution here.

Finally, $p = 5$. Here, $\bar{f}(X) = X^3$. We have that $0 \in \mathbb{F}_5$ is a root, but $f'(0) = -5$. Thus, let's check for solutions $5x$, where $x \in \mathbb{Z}_5$. We have that

$$f(5x) = 125x^3 - 25x + 10 \equiv 10 \pmod{25},$$

so we don't have any solutions in this case. $\square$

Exercise 1.4.6. Find the number of solutions to $X^3 + 2X + 16 = 0$ in $\mathbb{Q}_2$.

Solution: We can assume that $|x|_2 \leq 1$, because otherwise, the norms would differ and so there would be no way cancellation occurs. Let

$$f(X) = X^3 + 2X + 16, \quad f'(X) = 3X^2 + 2.$$

We have that $\bar{f}(X)) = X^3$. This has roots at $0 \in \mathbb{F}_2$, but $F'(0) = 2$. Therefore, $|f(0)|_2 = 2^{-4} < 2^{-2} = |f'(0)|_2^2$. $\square$

Exercise 1.4.7. Find the number of solutions to $X^3 + X + 2 = 0$ in $\mathbb{Q}_2$.

Solution: We can assume that $|x|_2 \leq 1$, because otherwise, the norms would differ and so there would be no way cancellation occurs. Let

$$f(X) = X^3 + X + 2, \quad f'(X) = 3X^2 + 1.$$

We have $\bar{f}(X) = X^3 + X$, so $0, 1 \in \mathbb{F}_2$ are roots. 0 uniquely lifts via Hensel. As for 1, not yet. Thus, check if $2x + 1$, where $x \in \mathbb{Z}_2$, can be a root.

$$f(2x+1) \equiv 6x + 1 + 2X + 1 + 2 \equiv 0 \pmod 4,$$

so any x works. Take $x = 1$. Then,

$$|f(3)|_2 = 2^{-5} < 2^{-4} = |f'(3)|_2^2.$$

Therefore, $3 \in \mathbb{F}_4$ lifts uniquely via Hensel's lemma. $\square$

2 Algebraic Curves

§2.1 The Basics

Now, here we won't go into detail but rather give a crash course on the material. Let $0 \neq f \in k[x, y]$ and $k \subset K$ be a field extension. Note that we shall usually either take $K = k$ or $K = \overline{k}$. The point of K/k is that k is where the coefficients of f reside in, while K is where the solutions to f reside in.

Notation 2.1.1. Given a field k, we denote by $\overline{k}$ to be the algebraic closure of k.

Definition 2.1.2. A field k is **algebraically closed** if any non-constant polynomial $f \in k[x]$ has a root.

> **Remark 2.1.3** — It follows from the definition that f can be factored as
>
> $$f(x) = c\prod(x - r_i)^{e_i}, \quad c, r_i \in k,$$
>
> where the r_i are the distinct roots of f. This tells us that a polynomial of degree d has d roots, counted with multiplicity.

> **Example 2.1.4**
>
> Some algebraically closed fields are $\mathbb{C}$ or $\overline{\mathbb{Q}}$.

Definition 2.1.5. An **affine algebraic curve** over K is the subset of K^2 of the form

$$C_f(K) = \{(x, y) \in K^2 : f(x, y) = 0\}.$$

Definition 2.1.6. The **degree** of an affine algebraic curve C_f, denoted $\deg C_f = \deg f \in \mathbb{N}_{>0}$, is the total degree, so if

$$f(x, y) = \sum_{i,j=0}^{n} a_{ij}x^i y^j, \quad \text{then} \quad \deg f = \max\{i + j : a_{ij} \neq 0\}.$$

Definition 2.1.7. We call algebraic curves of degree one, two and three as **lines**, **conics** and **cubics**, respectively.

> **Remark 2.1.8** — Nearly everything that we do is independent under an *affine linear change of co-ordinates*; that is, we will consider two curves **equivalent** if they are related by a change of co-ordinates:
>
> $$\begin{aligned} \tilde{x} &= px + qy + u \\ \tilde{y} &= rx + sy + v \end{aligned}$$
>
> with $p, q, r, s, u, v \in k$ and $\left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ invertible over k.

Definition 2.1.9. Let C be an algebraic curve defined by a polynomial $f(x,y) \in k[x,y]$. A point (a,b) is called a **singular point** of C if

$$f(a,b) = \frac{\partial f}{\partial x}(a,b) = \frac{\partial f}{\partial y}(a,b) = 0.$$

We say that C is **non-singular** or **smooth** if C has no singular points over $K = \bar{k}$.

Remark 2.1.10 — In the definition for C smooth, it's important to signify that C has no singular points over the algebraic closure.

Exercise 2.1.11. Show that the curve $f(x,y) = y^2 + xy - x^3$ is not smooth. This singularity is called a **node**.

Solution: We have that $\frac{\partial f}{\partial x} = y - 3x^2$ and $\frac{\partial f}{\partial y} = 2y + x$. We see that these are both zero when $(x,y) = (0,0)$. Thus, not smooth. $\square$

Exercise 2.1.12 (Hyperelliptic Curve)**.** Show that the algebraic curve defined by $f(x,y) = y^2 - p(x)$ where $p \in k[x]$ a polynomial, is smooth if and only if $p(x)$ has no multiple roots.

Solution: First suppose f is smooth and suppose for contradiction that $p(x)$ has a root with multiplicity, say at a. Then, $p(a) = 0$ and $p'(a) = 0$. Since $p(a) = 0 = b^2$, it follows that $b = 0$, but then $\frac{\partial f}{\partial y}(a,b) = 0$. Thus, (a,b) is a singular point, but f is smooth; a contradiction. For the converse, suppose that $p(x)$ has no roots with multiplicity. If f is not smooth, then there exists some point (a,b) such that $f(a,b) = \frac{\partial f}{\partial x}(a,b) = \frac{\partial f}{\partial y}(a,b) = 0$. However, this second condition tells us that $b = 0$, so $\frac{\partial f}{\partial x}(a,b) = p'(a) = 0$. Therefore, $p(a) = 0 = p'(a)$, which is a contradiciton to $p(x)$ having no roots with multiplicty. $\square$

Remark 2.1.13 — In order to see the singular points of a polynomial, one might need to look for them in an algebraic extension of k. For example, consider $f(x,y) = y^2 - (x^4 - 4x^2 + 4) \in \mathbb{Q}[X,Y]$. Then, partial derivatives give $x^3 - 2x = 0$ and $2y = 0$. We see that $(x,y) = (0,0)$ doesn't lie on the curve, so we consider $(x,y) = (\pm\sqrt{2}, 0)$. This is a singular point which is not in $\mathbb{Q}$, highlighting the significance of taking the algebraic closure.

Definition 2.1.14. A polynomial $f(x,y) \in k[X,Y]$ which has no nonconstant polynomial factors other than scalar multiples of itself is called **irreducible**.

Remark 2.1.15 — Recall that $k[x,y]$ is a UFD, so every polynomial can be factored into irreducible factors.

Definition 2.1.16. We say that a curve defined by f is **irreducible**, if f is irreducible over $\bar{k}[X,Y]$.

Remark 2.1.17 — The notion of irreducibility of a polynomial really depends on the coefficient field k. For instance, $x^2 + y^2$ is irreducible in $\mathbb{Q}[X,Y]$, but is reducible in $\mathbb{C}[X,Y]$.

Definition 2.1.18. The **tangent line** to an algebraic curve C_f at a smooth point (a,b) is given by

$$\frac{\partial F}{\partial x}(a,b)(x-a)+\frac{\partial F}{\partial y}(a,b)(y-b)=0.$$

Of main interest to us is the intersection $C\cap D$ of two algebraic curves. Particularly given two such curves, we would like to understand how many points of intersection there are. By trial and error, one might get to the conclusion that the equality

$$\#(C\cap D)=\deg C\cdot\deg D$$

should be true, but there are some annoying technical issues. Let us point these out now.

Field of definition:

Intersections may not be defined over the field K. Two curves $y=f(x)$ and $y=0$ for $f(x)\in k[x]$ intersect at the zeroes of $f(x)$. We would then expect that there are $\deg(f)$ zeroes, but this fact is only guaranteed if k is algebraically closed.

Example 2.1.19

x^2-1 has 2 zeroes, but x^2+1 has no zeroes over $\mathbb{Q}$.

The solution to this is to work over the algebraic closure of k. Alternatively, we could be more conservative and work in a finite algebraic extension (that depends on the curves) which guarantees that the intersection points are defined over k.

Intersections at infinity:

If we take the polynomials f and g sufficiently generically, we can ensure that all the intersections of C_f and C_g lie in k^2, so we will get $\deg f\cdot\deg g$ intersections. We've only consider the most generic case, so it's possible that C_f and C_g are in a *special* position with respect to each other, which pushes some of the intersection points to 'infinity'. This phenomena is illustrated in the following example:

Example 2.1.20

Consider the lines $y=x$ and $y=tx+1$ for some $t\in k$. For almost all values of t, these two lines intersect at one point, namely at $(x,y)=(\frac{1}{1-t},\frac{1}{1-t})$. However, we see that as $t\to 1$, the point of intersection goes to 'infinity', and as a matter of fact, $y=x$ and $y=x+1$ do not intersect.

To address this issue, one introduces a *compactification* of the affine space k^2; that is, we consider the projective space $\mathbb{P}^2(k)$. The idea is to identify $(x,y)\in k^2$ with the one dimensional k-linear subspace of k^3 spanned by $(x,y,1)$. Every one-dimensional linear space in k^3 which is not in the plane $\{(x,y,z)\in k^3 : z=0\}$ contains a unique point of the form $(x,y,1)$. Therefore, the remaining subspaces of $\{(x,y,z)\in k^3 : z=0\}$ can be thought as 'points at infinity'. We shall denote study the projective space and how to go between an affine curve and its projectivization in more detail.

Multiplicities:

We need to count multiplicities. There is a definition given for multiplicity.

Definition 2.1.21. If $\underline{0} = (0,0)$ is the intersection point of two curves $f(x,y) = 0$ and $g(x,y) = 0$ for $f, g \in k[x,y]$, so $f(\underline{0}) = g(\underline{0}) = 0$, then the **intersection number** of f and g at 0 is

$$(C_f \cap C_g)_0 = \dim_k k[[x,y]]/(f,g) < \infty.$$

For $p \in C_f \cap C_g$, we write $(C_f \cdot C_g)_p$ for the **intersection number** or **intersection multiplicity** at p.

Example 2.1.22

Take $y = 0$ and $y = x^2$. Then, we calculate

$$\dim_k(k[x,y]/(y, y - x^2)) = 2,$$

since $k[x,y]/(y, y-x^2)$ is spanned by 1 and x.

Remark 2.1.23 — If g is linear, we can compute the intersection multiplicity of a curve C_f and a line $C_g = L$ in a straight-forward manner. By a suitable change of co-ordinates, we can assume that C_f and L intersects at the origin in k^2, so we can parametrize points of L as

$$x = at, \quad y = bt, \text{ for some } a, b \in k^\times.$$

If we substitute this into f, we get

$$p(t) = f(at, bt) = f_1(a,b)t + f_2(a,b)t^2 + \cdots + f_d(a,b)t^r,$$

where the $f_i(x,y)$ are homogeneous polynomials of degree i such that $f(x,y) = \sum_{i=1}^d f_i(x,y)$. Then, the multiplicity of the intersection is the order of vanishing of $p(t)$ at $t = 0$, and is equal to

$$\max_i \{f_1(a,b) = f_2(a,b) = \cdots = f_i(a,b) = 0\}.$$

Exercise 2.1.24. Let C_f be a smooth curve, and L be a tangent line on a point (x,y) to C_f. Show that the multiplicity of the intersection of L with C_f at the point (x,y) is greater than 1. (Generally, it will be 2 but it can be higher.)

Definition 2.1.25. A non-singular point P of a curve C_f is called a **flex** or an **inflection point** if the intersection multiplicity of the tangent line at P to C_f with C_f is ≥ 3.

Common factors:

We may have a common factor in the intersection, such as the following.

Example 2.1.26

$x^2 - y^2 = 0$ and $x^3 - y^3 = 0$ have more than 6 points in common. This is because they share a common factor $x - y$.

The solution in this situation is simple: remove the common factors.

With these precautions in mind, we get that if f and g are polynomials in $k[x, y]$ with k algebraically closed, C_f and C_g intersect at $\deg(f) \cdot \deg(g)$ points; this gives Bezout's theorem (Theorem 2.2.17). To be more precise about intersections at infinity, we will now study the projective space and projectivizations in a bit more detail.

§2.2 The Projective Space

Let k be any field. Then

$$\mathbb{P}_k^2 = k^3 \setminus \{0\} / \sim$$

is the set of equivalence classes (x_0, x_1, x_2) such that $x_i \in k$ are not all zero modulo $\sim$, where

$$\mathbf{x} \sim \mathbf{y} \iff \mathbf{x} = \lambda \cdot \mathbf{y},$$

where $\lambda \in k \setminus \{0\} = k^\times$.

Definition 2.2.1. The **projective n-space** is defined as

$$\mathbb{P}_k^n = k^{n+1} \setminus \{0\} / \sim .$$

Definition 2.2.2. The **homogeneous coordinates** $[x_0 : \cdots : x_n]$ are an equivalence class of non-zero vectors in k^{n+1} modulo $\sim$, so

$$\mathbb{P}_k^n = \{[x_0 : \cdots : x_n] \mid x_i \in k \text{ not all zero}\}.$$

Definition 2.2.3. The **affine n-space** is $\mathbb{A}_k^n = k^n$.

We have an embedding $\varphi \colon \mathbb{A}_k^n \to \mathbb{P}_k^n$ given by

$$(x_1, \dots, x_n) \mapsto [1 : x_1 : x_2 : \cdots : x_n].$$

Injectivity of this map is clear. The points of $\mathbb{P}_k^n$ in the complement of the image of φ are called **points at infinity**. They are given by the equivalence classes of non-zero vectors of the form $[0 : x_1 : \cdots : x_{n-1} : x_n]$, which can in turn be identified with the projective space of one lower dimension, which leads to the decomposition

$$\mathbb{P}_k^n = \mathbb{A}_k^n \cup \mathbb{P}_k^{n-1}.$$

Example 2.2.4

$\mathbb{P}_k^1$ consists of the points $[1 : y]$ and $[0 : y] = [0 : 1]$, where $y \in k$.

Example 2.2.5

We see that $\mathbb{P}_k^2$ is obtained by compactifying $\mathbb{A}_k^2$ by adding a line $\mathbb{P}_k^1$ at infinity.

Remark 2.2.6 — There is nothing special about the first coordinate x_0; we may construct embeddings $\varphi_i \colon \mathbb{A}_k^n \to \mathbb{P}_k^n$ by $(x_0, x_1, \dots, x_n) \mapsto [x_0 : \cdots : x_{i-1} : 1 : x_{i+1} : \cdots : x_n]$.

Let $\ell : k^{n+1} \to k$ be a non-trivial linear function given by

$$(x_0, \dots, x_n) \mapsto \alpha_0 x_0 + \dots + \alpha_n x_n,$$

with not all $\alpha_i \in k$ are zero. The image of

$$\ker \ell = \{(x_0, \dots, x_n) \in k^{n+1} \colon \alpha_0 x_0 + \dots + \alpha_n x_n = 0\} \subset \mathbb{A}^n_{k+1}$$

with respect to the quotient map $k^{n+1} \setminus \{0\} \to \mathbb{P}^n_k$ is a **linear hyperplane**. This can be generalized by taking homogeneous polynomials in general.

Definition 2.2.7. A polynomial $F(X_0, \dots, X_n) \in k[X_0, \dots, X_n]$ is **homogeneous** of degree $d \in \mathbb{N}$ if

$$F(X_0, \dots, X_n) = \sum_{i_0 + \dots + i_n = d} \alpha_{i_0, \dots, i_n} X_0^{i_0} \dots X_n^{i_n},$$

so you only have degree d terms.

Remark 2.2.8 (Homogenizing) — If f is a degree d polynomial in $k[x_1, \dots, x_n]$, then here is how to **homogenize** it. Change x_i to X_i and then introduce a new variable X_0 and multiply each term with a suitable power of X_0 such that the resulting polynomial is homogeneous of the smallest possible degree.

Remark 2.2.9 (Dehomogenizing) — If F is a degree d homogeneous polynomial in $k[X_0, \dots, X_n]$, then here is how to *dehomogenize* it. Choose i with $0 \leq i \leq n$, set $X_i = 1$ and change all the other X_j to x_j. If we chose $i = 0$ then this recovers the initial equation.

Remark 2.2.10 — If $f \in k[x_1, \dots, x_n]$ then the **points at infinity** of $f = 0$ are the zeroes of F, the homogenization of f, which are in $\mathbb{P}^n_k$ but not in $\mathbb{A}^n_k$.

Remark 2.2.11 — If $F \in k[X_0, \dots, X_n]$ is homogeneous of degree d and $k \subset K$ a field extension, then

$$C_F(K) = \{[x_0 : \dots : x_n] \in \mathbb{P}^n_K \colon F(x_0, \dots, x_n) = 0\}$$

is well-defined. Homogenization allows us to extend an algebraic subset in $\mathbb{A}^n_K$ to $\mathbb{P}^n_K$.

Definition 2.2.12. For $F \in k[X, Y, Z]$, we call this subset $C_F \subset \mathbb{P}^2_K$ the **projective curve over** K defined by F.

Example 2.2.13

$X^2 + YZ + Z^2 = 0$ is homogeneous of degree two and gives rise to a conic in $\mathbb{P}^2_k$. Taking $Y = 1$, this gives rise to $X^2 + Z + Z^2 = 0$, which represents the $\mathbb{A}^2_k$ part. Now we check the points at infinity. Set $Y = 0$. Then, we have $X^2 + Z^2 = 0$. If $k = \mathbb{C}$ gives the points $[1 : 0 : i], [1 : 0 : -i]$. Now, consider when $Z = 0$. Then, we just get $X^2 = 0$, giving $X = 0$, $Y = 1$, so we get a point $[0 : 1 : 0]$ with multiplicity two. This corresponds with Bézout's theorem.

Example 2.2.14

$x^2 + x^3 = y^2$ and $xy = 1$ homogenizes to $X^2Z + X^3 = Y^2Z$ and $XY = Z^2$.

Example 2.2.15

$X^2 + Y^2 = Z^2$ and $YZ = X^2$ dehomogenizes to $x^2 + y^2 = 1$ and $y = x^2$.

Remark 2.2.16 — Nearly everything that we do is independent under a *projective linear change of coordinates*; that is, we will consider two curves **equivalent** if they are related by a change of coordinates:

$$\begin{aligned}\tilde{X} &= c_{11}X + c_{12}Y + c_{13}Z\\ \tilde{Y} &= c_{21}X + c_{22}Y + c_{23}Z\\ \tilde{Z} &= c_{31}X + c_{32}Y + c_{33}Z\end{aligned}$$

where $c_{ij} \in K$ and the matrix (c_{ij}) is invertible over K. Note that if $p = [x : y : z] \in \mathbb{P}^2_k$ and we have $F \in k[X, Y, Z]$ such that $F(p) = 0$, then if we consider

$$\tilde{F}(X, Y, Z) = F(c_{11}X + c_{12}Y + c_{13}Z, c_{21}X + c_{22}Y + c_{23}Z, c_{31}X + c_{32}Y + c_{33}Z),$$

the point $P = [x : y : z]$ gets mapped to

$$\tilde{P} = [d_{11}x + d_{12}y + d_{13}z : d_{21}x + d_{22}y + d_{23}z : d_{31}x + d_{32}y + d_{33}z],$$

where $d = (c_{ij})^{-1}$. Then, $\tilde{F}(\tilde{P}) = 0$.

§2.2.i Bézout's Theorem

We can now state the main theorem on intersections of algebraic curves.

Theorem 2.2.17 (Bézout's theorem)

Let $K = \overline{k}$ be an algebraic closure of k. If $F, G \in k[X_0, X_1, X_2]$ be homogeneous non-zero polynomials without common factors, then

$$\sum_{p \in C_F(K) \cap C_G(K)} (C_F \cdot C_G)_p = \deg F \cdot \deg G.$$

We will not prove this result as we've previously covered it.

Corollary 2.2.18

If F and G are two homogeneous polynomials in $k[X, Y, Z]$, for k any field not necessarily algebraically closed, then either the curves C_F and C_G in $\mathbb{P}^2_k$ have at most $\deg F \cdot \deg G$ points in common, or F and G have a common factor.

Proof. Immediate from Bézout applied to $\overline{k}$. □

Definition 2.2.19. We say that a point $P \in \mathbb{P}^n_k$ is a **singular point** of $F = 0$ if $F(P) = 0$ and $(\frac{\partial F}{\partial X_i})(P) = 0$ for all $i = 0, \ldots, n$.

Definition 2.2.20. We say that $C_F = \{F = 0\}$ is **non-singular** or **smooth** if all of its points over $K = \overline{k}$ are non-singular, that is, for all points $P \in C_F$, there exists at least one i such that $(\frac{\partial F}{\partial X_i})(P) \neq 0$.

> **Exercise 2.2.21.** A point P is singular if and only if there exists some dehomogenization f of F, obtained by setting some $X_i = 1$, such that $f(P) = 0$ and $\frac{\partial f}{\partial X_i} = 0$ for all $i = 1, \dots, n$.

Definition 2.2.22. Let $f, g \in k[x, y]$ be non-zero polynomials with $f(P) = g(P) = 0$. We say that $f = 0$ and $g = 0$ **intersect transversely** at P if P is smooth for both f and g and the tangent lines of $f = 0$ and $g = 0$ at P are different.

> **Exercise 2.2.23.** If $f(P) = g(P) = 0$, then the multiplicity of intersection of C_f and C_g at P is one if and only if the intersection is transversal at P.

Solution: If the linear terms of f and g define distinct lines, then in the ring $k[[x, y]]$ we have $(f, g) = (x, y) + (x, y)^2$, since (x, y) is equal to the space generated by the linear terms of f and g. Hence, it follows from Nakayama's lemma that $(x, y) = (f, g)$. This gives $k[[x, y]]/(x, y) = k[[x, y]]/(f, g)$ which implies that the intersection number is 1. Conversely, suppose the intersection multiplicity is equal to 1. Then, if the linear parts were not linearly independent, then $(f, g) \subset \ell + (x, y)^2$, where ℓ is a linear subspace. Then, $\dim_k(k[[x, y]]/(f, g)) \geq \dim_k(k[[x, y]]/(\ell + (x, y)^2) = 2$, which is a contradiction. $\square$

> **Remark 2.2.24** — If we take a curve $C_F \subset \mathbb{P}^2_k$, and $F(P) = 0$, $P \in C_F$. Let $\ell_F(P)$ be the tangent line at P. Then, by the above,
>
> $$(C_F \cdot \ell_F)_P > 1.$$

> **Exercise 2.2.25.** Let $f(x, y) = y^2 - x^3$ and $g(x, y) = y^2 - x^3 - x^2$. Show that $(C_f \cdot C_g)_{(0,0)} = 4$.

Solution: We need to calculate $\dim_k(k[[x, y]]/(f, g))$. We have that $f - g = x^2$, so we have that $k[[x, y]]/(f, g) = k[[x, y]]/(y^2 - x^3, x^2)$. Therefore, a set of generators is $1, x, y, xy$, so the dimension is four. $\square$

§2.2.ii Conics in $\mathbb{P}^2_k$

Any conic in $\mathbb{P}^2_k$ is given by a quadratic form

$$Q(X_1, X_2, X_3) = \sum_{i,j=1}^{3} q_{ij} X_i X_j$$

where $q_{ij} = q_{ji} \in k$. Thus Q is completely described by a symmetric 3×3 matrix.

> **Remark 2.2.26** — We assume $\operatorname{char} k \neq 2$ when dealing with conics.

Claim 2.2.27. The conic is nonsingular if and only if

$$\det(q_{ij}) \neq 0.$$

Proof. Indeed, we can easily compute that Q is singular if and only if there exists a non-trivial solution to the system of linear equations:

$$\partial_{X_1}Q = 2(q_{11}X_1 + q_{12}X_2 + q_{13}X_3) = 0$$
$$\partial_{X_2}Q = 2(q_{21}X_1 + q_{22}X_2 + q_{23}X_3) = 0$$
$$\partial_{X_3}Q = 2(q_{31}X_1 + q_{32}X_2 + q_{33}X_3) = 0$$

This is equivalent $\det(q_{ij}) = 0$. □

Lemma 2.2.28

Let k be a field with $\operatorname{char} k \neq 2$. Let $C_Q \subset \mathbb{P}^2_k$ be a conic defined by $Q \in k[X_1, X_2, X_3]$. Then C_Q is singular if and only if Q is a product of two linear polynomials over the algebraic closure.

Proof. By diagonalization of quadratic forms for $Q \in k[X_1, X_2, X_3]$ of homogeneous degree 2, after rescaling by a non-zero scalar, and a permutation of variables, we can assume that

$$Q(X_1, X_2, X_3) = a_1X_1^2 + a_2X_2^2 + a_3X_3^2.$$

Then, by the above discussion, we see that C_Q is singular if and only if $a_1a_2a_3 = 0$.Now, say $a_3 = 0$, then we have

$$Q(X_1, X_2, X_3) = (\sqrt{a_1}X_1 + \sqrt{-a_2}X_2)(\sqrt{a_1}X_1 - \sqrt{-a_2}X_2).$$

Conversely, suppose $Q = L_1L_2$ where $L_i \in \overline{k}[X_1, X_2, X_3]$ are linear forms. If $L_1 = L_2$, then every point on Q is singular (we can arrange by a projective linear transformation that $L_1 = L_2 = X_1$). Suppose that L_1 and L_2 are distinct, then by Bézout's theorem, they intersect at a point P. Now, observe that

$$\partial Q_{X_i} = (\partial_{X_i}L_1)L_2(P) + L_1(P)\partial_{X_i}L_2(P) = 0$$

Hence, Q is singular at $P \in L_1 \cap L_2$. □

§2.2.iii Higher Degree Curves

Remark 2.2.29 — For curves of higher degree, singular does not imply reducible. For example, the cubic curve defined by $F(X, Y, Z) = X^3 - Y^2Z$ is singular at $[0 : 0 : 1]$ but F is irreducible over k.

Definition 2.2.30. A smooth point P of a curve C_F is called a **flex** (inflection point) if the intersection multiplicity of C_F and the tangent line ℓ_F at P is greater than 2; that is, if $(C_F \cdot \ell_F)_P > 2$.

Proposition 2.2.31

Let $C_F \subset \mathbb{P}^2_k$ be a smooth curve defined by a degree d homogeneous polynomial $F \in k[X_1, X_2, X_3]$. Assume $\operatorname{char} k \nmid 2(d-1)$. Then a point $P \in C$ is a flex point if and only if $H(P) = 0$, where $H(X_1, X_2, X_3)$ is $\det\left(\frac{\partial^2 F}{\partial X_i X_j}\right)_{1 \leq i,j \leq 3}$, the **Hessian**.

Proof. As $F(P) = 0$, the Taylor expansion of F at $P = (P_1, P_2, P_3)$ is of the form

$$F(P_1 + X_1, P_2 + X_2, P_3 + X_3) = \sum_{i=1}^{3} \frac{\partial F}{\partial X_i}(P)X_i + \frac{1}{2}\sum_{i,j=1}^{3} \frac{\partial^2 F}{\partial X_i X_j}(P)X_iX_j + \dots$$

Let $L = \sum_{i=1}^{3} \frac{\partial F}{\partial X_i}(P)X_i$ be the tangent line at P, and $Q = \sum_{i,j=1}^{3} \frac{\partial^2 F}{\partial X_i X_j}(P)X_iX_j$ be the **osculating conic**. F is homogeneous, so we have $F(\lambda X_1, \lambda X_2, \lambda X_3) = \lambda^d F(X_1, X_2, X_3)$. Differentiating with respect to λ and evaluating at $\lambda = 1$ gives the **Euler relation**:

$$\sum_i X_i \frac{\partial F}{\partial X_i} = dF$$

Differentiating again, we arrive at

$$\sum_{i,j} X_iX_j \frac{\partial^2 F}{X_iX_j} = d(d-1)F.$$

Hence, we see that $P \in Q$. Moreover, the tangent line to Q at P is given by

$$2\sum_i \left(\sum_j \frac{\partial^2 F}{X_iX_j}(P)P_j\right) X_i = 2(d-1)\sum_i \frac{\partial F}{\partial X_i}(P)X_i = 0$$

Hence, when $2(d-1) \neq 0$, it coincides with L. Therefore, we conclude that Q is smooth at P.

The point P is an inflection point if and only if $L \subset Q$ (parametrize L as $(P_1 + at, P_2 + bt, P_3 + ct)$ and compute the vanishing of F restricted to L). This implies that Q is singular at some other point $S = (S_1, S_2, S_3)$, hence $H(P) = 0$ as S is in the kernel of the matrix $\left(\frac{\partial^2 F}{X_iX_j}(P)\right)_{1\leq i,j\leq 3}$.

Conversely, if Q is singular at P (the Hessian vanishes at that point), then L is smooth at P, hence $L \subset Q$. Hence, P is an inflection point. □

Exercise 2.2.32. Check that a smooth point $P \subset C$ is an inflection point if and only if the tangent line L to C at P is contained in the osculating conic Q.

Solution: Do... □

Exercise 2.2.33. Consider the cubic curve $C\colon \{X^3 + Y^3 + Z^3 + 3XYZ = 0\}$ over k of characteristic not equal to 2 or 3, and $P = [0 : -1 : 1]$ on this curve. Show that the tangent line at P is $X - Y - Z = 0$ and the osculating conic is $(X - Y - Z)(Y - Z) = 0$. Hence, conclude that P is an inflection point (but observe that it is a smooth point of both C and Q).

Solution: Let $F(X, Y, Z) = X^3 + Y^3 + Z^3 + 3XYZ$. Then, $\frac{\partial F}{\partial X}(P) = -3$, $\frac{\partial F}{\partial Y}(P) = 3$, $\frac{\partial F}{\partial Z}(P) = 3$, so the tangent line at P is $X - Y - Z = 0$. For the osculating conic, we calculate all second orders:

$$F_{XX}(P) = 0, F_{XY} = 3, F_{XZ} - 3, F_{YY} = -6, F_{YZ} = 0, F_{ZZ} = 6.$$

Thus, the osculating conic is

$$6XY - 6XZ - 6Y^2 + 6Z^2.$$

The osculating conic can be factored as $6(Y - Z)(X - Y - Z)$. Since the tangent line is contained in the osculating conic, P is an inflection point. □

Exercise 2.2.34. Suppose $F \in k[X,Y,Z]$ is a homogeneous polynomial of degree 3. Let H_F be the Hessian of F computed with respect to X,Y,Z. Now, consider $C := (c_{ij})_{i,j=1}^3$ to be an invertible matrix and use it to change coordinates to $\tilde{X}, \tilde{Y}, \tilde{Z}$, and consider the Hessian $H_{\tilde{F}}$ of the cubic $\tilde{F}$ corresponding to F computed using $\tilde{X}, \tilde{Y}, \tilde{Z}$. How are H_F and $H_{\tilde{F}}$ related?

Corollary 2.2.35

Let $C \subset \mathbb{P}^2$ be a smooth projective cubic over a field k that is algebraically closed and of characteristic $\neq 2$. Then there exists a projective linear change of coordinates on $\mathbb{P}^2$ such that the equation of the curve C takes the form (called the **Weierstrass form**):

$$Y^2Z = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$$

and the roots of the polynomial $f(X) = X^3 + a_2X^2 + a_4X + a_6 \in k[X]$ are distinct.

Proof. Let $C = \{F(X,Y,Z) = 0\} \subset \mathbb{P}^2$. By the previous theorem, the inflection points on C are given by the intersection of C and the cubic curve $H_F = 0$. By Bézout's theorem, there are 9 inflection points (here we use that k is algebraically closed and $\operatorname{char} k \neq 2$). Thus, let $P \in C$ be an inflection point. Choose coordinates such that $P = [0 : 1 : 0]$ and the tangent line at P is given by $Z = 0$. As P is an inflection point, we see that $F(t,1,0) = ct^3$ for some $c \in k$, so F has no X^2Y, XY^2 or Y^3 terms. Therefore, we can write it as

$$\alpha Y^2Z + a_1XYZ + a_3YZ^2 = \beta X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

Moreover, as P is a smooth point, it is easy to check that $\alpha, \beta \neq 0$ by passing to the derivatives. Hence, by rescaling $(X \to \alpha\beta X, Y \to \alpha^2\beta Y)$, we can reduce to

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

Finally, replacing Y by $Y - \frac{1}{2}a_1X - \frac{1}{2}a_3Z$, we arrive at the desired form. The fact that the resulting $f(X)$ has distinct roots follows from the smoothness assumption. □

Remark 2.2.36 — Over a field of characteristic 2, we cannot do the last step, so the general Weierstrass form of an elliptic curve is given by

$$E\colon y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_1, a_2, a_3, a_4, a_6 \in k.$$

3 Plane Conics and the Hasse-Minkowski Theorem

Definition 3.0.1. Let $X \subseteq \mathbb{P}^n_{\mathbb{Q}}$ be a projective algebraic variety. The **local global principle** holds for X if $X(\mathbb{Q}) \neq \emptyset$ if and only if $X(\mathbb{Q}_p) \neq \emptyset$ for all p which are prime numbers and $X(\mathbb{R}) \neq \emptyset$.

Here are some historical results/ remarks

- Hasse-Minkowski implies that conics, and all dimensions for all quadratic hypersurfaces, satisfy the local-global principle.
- Selmer and Chaubauty showed that the cubic $3X^3 + 4Y^3 + 5Y^3 = 0$ doesn't satisfy the local-global principle.

Theorem 3.0.2 (Hasse-Minkowski)

Let Q be the projective curve defined by

$$aX^2 + bY^2 + cZ^2 = 0, \quad a, b, c \in \mathbb{Z}, \quad abc \neq 0, \quad (a, b) = (b, c) = (c, a) = 1,$$

where a, b, c are square-free. Let $\Sigma := \{p \mid 2abc\}$. Then the following are equivalent.

1. Q has infinitely many solutions in $\mathbb{P}^2_{\mathbb{Q}}$.
2. Q has a solution in $\mathbb{P}^2_{\mathbb{Q}}$.
3. Q has a solution in $\mathbb{P}^2_{\mathbb{Q}_p}$ for all p, and in $\mathbb{P}^2_{\mathbb{R}}$.
4. Q has a solution in $\mathbb{P}^2_{\mathbb{Q}_p}$ for all $p \in \Sigma$.

Remark 3.0.3 — As we will explain, this theorem reduces to problem of solving a general conic over $\mathbb{Q}$ to solving it over $\mathbb{Q}_p$ for finitely many primes p. Let's explain how to get rid of the infinitely many primes that are not in Σ first.

Proposition 3.0.4

Let Q be a conic $aX^2 + bY^2 + cZ^2 = 0$ with $a, b, c \in \mathbb{Z}_p$ and $p \nmid 2abc$. Then there is always a solution over $\mathbb{Q}_p$.

Proof. Since $p \nmid 2abc$, we have $|a|_p = |b|_p = |c|_p = 1$. We first reduce the equation modulo p. Pick a nonzero x, and let

$$A := \{\, ax^2 + by^2 : y \in \mathbb{F}_p \,\}, \quad B := \{\, -cz^2 : z \in \mathbb{F}_p \,\}.$$

Claim 3.0.5. $|A| = |B| = (p + 1)/2$.

Proof. Fix x, and consider the map $y \mapsto ax^2 + by^2$ in $\mathbb{F}_p$. If $ax^2 + by_1^2 = ax^2 + by_2^2$ in $\mathbb{F}_p$, then since $p \nmid b$, we have $y_1^2 = y_2^2$, so $y_1 = y_2$ or $y_1 = -y_2 \neq 0$. Hence the map is two-to-one except at 0, so the image has size $1 + \frac{p-1}{2} = \frac{p+1}{2}$. The argument is similar for B. $\square$

With this claim, we have

$$|A| + |B| = \frac{p+1}{2} + \frac{p+1}{2} = p + 1 > p = |\mathbb{F}_p|.$$

Therefore $A \cap B \neq \emptyset$, or equivalently $ax^2 + by^2 + cz^2 = 0$ has a nontrivial solution over $\mathbb{F}_p$. Call such a solution (x_0, y_0, z_0) with $x_0 \neq 0$. Here, y_0 and z_0 are arbitrary lifts of the solution in $\mathbb{F}_p$. Now, consider the polynomial $f(x) := ax^2 + by_0^2 + cz_0^2 \in \mathbb{Z}_p[x]$. This has a solution in $\mathbb{F}_p$, namely x_0, and we want to lift this. Since we want to apply Hensel's lemma, we check that $f'(x_0) = 2ax_0 \not\equiv 0 \pmod p$, so we attain a nonzero solution to $ax^2 + by^2 + cz^2 = 0$ over $\mathbb{Z}_p$. $\square$

Remark 3.0.6 — This gets rid of the local case for *big* primes.

Exercise 3.0.7.

1. Let $p > 2$ be prime and let $a, b, c \in \mathbb{Z}$, $p \nmid b$. Show that $bx^2 + c$ takes precisely $\frac{1}{2}(p+1)$ distinct values modulo p for $x \in \mathbb{Z}$.
2. Suppose that, further, $a \in \mathbb{Z}$, $p \nmid a$. Show that there are $x, y \in \mathbb{Z}$ such that $bx^2 + c \equiv ay^2 \pmod p$.

Solution: 1. View $bx^2 + c$ as a map from $\mathbb{F}_p$ to $\mathbb{F}_p$. Then,

$$bx^2 + c \equiv 0 \pmod p \iff x^2 \equiv -b^{-1}c \pmod p.$$

Then, restricting the map to the group $\mathbb{F}_p^\times$, we get $\frac{p-1}{2}$ solutions, plus 1, for zero.

2. We note that

$$|\{bx^2 + c \colon x \in \mathbb{F}_p\}| = |\{ax^2 \colon x \in \mathbb{F}_p\}| = \frac{p+1}{2},$$

so we can apply the pigeon hole principle. $\square$

Exercise 3.0.8. Let $p > 2$ be prime, $a_{ij} \in \mathbb{Z}$ $(1 \leq i, j \leq 3)$ with $a_{ij} = a_{ji}$, and let $d = \det(a_{ij})$. Suppose that $p \nmid d$. Show that there exist $x_1, x_2, x_3 \in \mathbb{Z}$, not all divisible by p, such that

$$\sum_{i,j} a_{ij} x_i x_j \equiv 0 \pmod p.$$

Solution: First, note that any quadratic form over a field of characteristic not 2 is diagonalizable. Therefore, the quadratic form is $ax_1^2 + bx_2^2 + cx_3^2$, where $a, b, c \in \mathbb{Z}$. Then, $d \nmid abc$. By setting any one of x_1, x_2, x_3 to some specific value (say 1), we can consider the equation modulo p and apply the previous exercise. $\square$

Exercise 3.0.9. Let $a, b, c \in \mathbb{Z}_p$, and $|a|_p = |b|_p = |c|_p = 1$, where $p > 2$ is prime. Show that there are $x, y \in \mathbb{Z}_p$ such that $bx^2 + c = ay^2$.

Solution: There exists a nontrivial solution (x_0, y_0) in $\mathbb{F}_p$ to the equation $bx^2+c-ay^2=0$. In particular, $x_0 \neq 0$. Then, we use Hensel's lemma on the polynomial $f(x) = bx^2+c-ay_0^2$. Then, $f'(x) = 2bx$, so $f'(x_0) \neq 0$. Thus, we can lift (x_0, y_0) to a solution in $\mathbb{Z}_p$ by Hensel's lemma. □

Now, let Q be a conic $aX^2 + bY^2 + cZ^2 = 0$ over $\mathbb{Q}$. We can assume the following:

- $a, b, c \in \mathbb{Z}$, by rescaling
- a, b, c are relatively prime, by rescaling
- a, b, c are square-free, since we can rescale individual variables by squares
- a, b, c are pairwise relatively prime, since if p is a prime number such that $p \mid a$ and $p \mid b$, then pa and pb are divisible by p^2, so we absorb the p^2 into X and Y.

To determine solutions over $\mathbb{R}$, consider a conic Q given by $aX^2+bY^2+cZ^2 = 0$ over $\mathbb{R}$. We rescale each of a, b, c by a nonzero square. Over $\mathbb{R}$ we can assume $\{a, b, c\} = \{1, -1\}$. There are two cases to consider:

- If $X^2 + Y^2 + Z^2 = 0$, then $Q(\mathbb{R}) = \emptyset$.
- If $X^2 + Y^2 - Z^2 = 0$, then $Q(\mathbb{R}) \neq \emptyset$.

Thus, just by looking at the signs you can tell whether Q has a solution or not over the reals.

It remains to analyze whether there are solutions over $\mathbb{Q}_p$ for $p \mid 2abc$. Suppose first p is odd and $|a|_p = 1/p$ and $|b|_p = |c|_p = 1$, then Q becomes the singular conic $bY^2 + cZ^2 = 0$ over $\mathbb{F}_p$. Running through the finitely many elements in $\mathbb{F}_p$, we can check if there is a nontrivial solution, and if there is, we can obtain a solution over $\mathbb{Q}_p$ by an application of Hensel's lemma as before. Finally, for $p = 2$, the analysis is a bit more tedious as usual, but we can reduce mod 8, and analyze if we have a solution over $\mathbb{Z}/8$, and then lift that to $\mathbb{Q}_2$. In fact, the following exercises show we can be a bit more precise.

> **Exercise 3.0.10.** Let $a, b, c \in \mathbb{Z}$, $2 \nmid abc$. Show that a necessary and sufficient condition that the only solution in $\mathbb{Q}_2$ of $ax^2 + by^2 + cz^2 = 0$ is the trivial one is that $a \equiv b \equiv c \pmod 4$.

Solution: Suppose $(a_1, a_2, a_3) \neq 0$ is a nontrivial solution in $\mathbb{Q}_2$ then we can assume that $\max |a_i|_2 = 1$ by multiplying with an element of $\mathbb{Q}_2$. This means that at least one of the a_i is a unit. Now, since $aa_1^2 + ba_2^2 + ca_3^2 = 0$ and $2 \nmid abc$, it follows that precisely two of the a_i are units. Because of the non-archimedean inequality, we must have two of the $|aa_1|_2^2, |ba_2|_2^2, |ca_3|_2^2$ must be equal and the other one is less than or equal to. Suppose, for instance, that $|a_2|^2 = |a_3|^2 = 1$, and $|a_1|^2 \leq 1$. By examining modulo 2, we see then that $|a_1| < 1$. Now, $2 \mid a_1$, hence it follows that $b + c \equiv 0 \pmod 4$ but b, c are odd, hence b is not equivalent to c modulo 4.

Conversely, suppose that $(a, b, c) \neq (1, 1, 3)$ or $(1, 3, 3)$ modulo 4, and we want to construct a solution in $\mathbb{Q}_2$. By multiplying the equation with -1, we can assume that we are in the case where $(a, b, c) = (1, 1, 3)$ modulo 4, or equivalently we are interested in the equation $ax^2 + by^2 = (-c)z^2$. Now, multiply both sides with $(-1/c)$ and redefine a, b to reduce to the case $ax^2 + by^2 = z^2$ where we still have $(a, b) = (1, 1)$ modulo 4.

Now, we use the fact that $ax^2 + by^2$ is a square in $\mathbb{Q}_2$ if and only if $ax^2 + by^2 \equiv 1 \pmod 8$. We may have that a, b are either 1 or 5 modulo 8. So it suffices to find solutions for the four equations:

$$x^2 + y^2 \equiv 1(8), \quad 5x^2 + y^2 \equiv 1(8), \quad x^2 + 5y^2 \equiv 1(8), \quad 5x^2 + 5y^2 \equiv 1(8).$$

It is very easy to solve these congruence equations. For instance $(3, 0)$, $(1, 2)$, $(2, 1)$, $(2, 1)$ are solutions in the respective cases. $\square$

Exercise 3.0.11. Let $a, b, c \in \mathbb{Z}$ such that $|a|_2 = \frac{1}{2}$ and $2 \nmid bc$. Then, $ax^2 + by^2 + cz^2 = 0$ has a nontrivial solution in $\mathbb{Q}_2$ if and only if either $b + c \equiv 0 \pmod 8$ or $a + b + c \equiv 0 \pmod 8$.

§3.1 Geometry of Numbers

First, we explain a continuous generalization of the Pigeonhole principle: If n papers are filed in k pigeon-holes, and $n > mk$ then one pigeon-hole must contain at least $m + 1$ papers.

Consider a measurable subset $S \subseteq \mathbb{R}^n$, we define its volume as the Lebesgue integral

$$\mathrm{vol}(S) = \int_S \mathbf{1}_S(x)\, d\mu$$

where $\mathbf{1}_S(x)$ is the characteristic function of S:

$$\mathbf{1}_S(x) = \begin{cases} 1 & x \in S, \\ 0 & x \notin S. \end{cases}$$

Lemma 3.1.1 (Blichfeld)

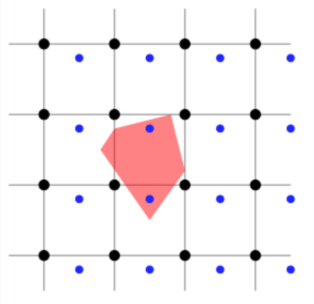
Let $\mathcal{S} \subseteq \mathbb{R}^n$ be a measurable set and assume that it has volume $\mathrm{vol}(\mathcal{S}) > m \in \mathbb{Z}_{>0}$. Then there exist $s_0, \ldots, s_m \in \mathcal{S}$ such that $s_i - s_j \in \mathbb{Z}^n \subset \mathbb{R}^n$ for all $0 \le i, j \le m$.

Proof. Let $U = [0, 1)^n \subseteq \mathbb{R}^n$. Then U is measurable and $\mathrm{vol}(U) = 1$, and $U + \mathbb{Z}^n = \mathbb{R}^n$. For any set $\mathcal{S} \subseteq \mathbb{R}^n$, we have

$$\begin{aligned} m < \mathrm{vol}(\mathcal{S}) &= \int_{\mathbb{R}^n} \mathbf{1}_{\mathcal{S}}(\mathbf{x}) d\mu \\ &= \sum_{\mathbf{t} \in \mathbb{Z}^n} \int_{U+\mathbf{t}} \mathbf{1}_{\mathcal{S}}(\mathbf{x}) d\mu \\ &= \sum_{\mathbf{t} \in \mathbb{Z}^n} \int_U \mathbf{1}_{\mathcal{S}}(\mathbf{x} + \mathbf{t}) d\mu \\ &= \int_U \sum_{t \in \mathbb{Z}^n} \mathbf{1}_{\mathcal{S}}(\mathbf{x} + \mathbf{t}) d\mu. \end{aligned}$$

Therefore, the function $\sum_{\mathbf{t} \in \mathbb{Z}^n} \mathbf{1}_{\mathcal{S}}(\mathbf{x} + \mathbf{t})$ must take a value greater than m at some point in U since $\mathrm{vol}(U) = 1$. In other words, there exists $\mathbf{x}_0 \in U$ such that $|(\mathbf{x}_0 + \mathbb{Z}^n) \cap \mathcal{S}| > m$. Now, let s_j be distinct elements from $(\mathbf{x}_0 + \mathbb{Z}^n) \cap \mathcal{S}$. $\square$

Remark 3.1.2 — The condition $s_i - s_j \in \mathbb{Z}^n$ is equivalent to $s_i - s_0 \in \mathbb{Z}^n$, for all i. Geometrically, this lemma then says that we can translate the integer points so that a set of volume greater than one will contain at least two integer points:



Remark 3.1.3 — We used the following properties of the Lebesgue integral.

If $\{f_j\}$ are nonnegative measurable functions, then:

$$\int_{\mathcal{S}} \sum_j f_j d\mu = \sum_j \int_{\mathcal{S}} f_j d\mu.$$

In other words, one can always interchange an infinite sum and the integral sign when dealing with nonnegative functions. This requires more care when dealing with Riemann integrals. However, if $\mathcal{S}$ is bounded, then we only have a finite sum, so this interchange of integration and summation is justified easily.

We also use the translation invariance of the Lebesgue measure:

$$\int_{U+\mathbf{t}} f(\mathbf{x})\, d\mu = \int_U f(\mathbf{x}+\mathbf{t})\, d\mu.$$

The measurable sets $\mathcal{S}$ that we will use in these lectures are all regular, bounded shapes, and the notion of $\operatorname{vol}(\mathcal{S})$ that we use coincides with any reasonable intuitive concept of volume.

Definition 3.1.4. $\mathcal{S} \subseteq \mathbb{R}^n$ is **symmetric** if for all $\mathbf{x} \in \mathcal{S}$, $-\mathbf{x} \in \mathcal{S}$.

Definition 3.1.5. $\mathcal{S} \subseteq \mathbb{R}^n$ is **convex** if for all $\mathbf{x}, \mathbf{y} \in \mathcal{S}$ and $t \in [0,1]$, one has $t\mathbf{x}+(1-t)\mathbf{y} \in \mathcal{S}$.

Corollary 3.1.6 (Minkowski's geometry of numbers)
Let $\Lambda \subseteq \mathbb{Z}^n$ be a subgroup of finite index m. Let $\mathcal{C} \subseteq \mathbb{R}^n$ be a convex, symmetric set such that

$$\operatorname{vol}(\mathcal{C}) > 2^n \cdot m.$$

Then $\Lambda \cap \mathcal{C} \neq \{\mathbf{0}\}$ (that is, Λ and $\mathcal{C}$ have a common point other than $\mathbf{0}$).

Proof. Let

$$\mathcal{S} = \tfrac{1}{2}\mathcal{C} = \left\{\tfrac{1}{2}\mathbf{x} \colon \mathbf{x} \in \mathcal{C}\right\}.$$

Then

$$\operatorname{vol}(\mathcal{S}) = 2^{-n} \cdot \operatorname{vol}(\mathcal{C}) > m.$$

So by Lemma 3.1.1, there exist $s_0, \ldots, s_m \in \mathcal{S}$ such that $s_i - s_j \in \mathbb{Z}^n$ for all $0 \le i, j \le m$.

Now,
$$|\{s_i - s_0 \colon i = 0, \ldots, m\}| = m + 1 > [\mathbb{Z}^n : \Lambda],$$
there exist $i \neq j$ such that $s_i - s_0 \equiv s_j - s_0 \pmod{\Lambda}$, so $s_i - s_j = (s_i - s_0) - (s_j - s_0) \in \Lambda$.

On the other hand, $2s_i \in \mathcal{C}$ and $-2s_i \in -\mathcal{C} = \mathcal{C}$ by symmetry of $\mathcal{C}$, so $s_i - s_j = \frac{1}{2}(2s_i) + \frac{1}{2}(-2s_j) \in \mathcal{C}$, as $\mathcal{C}$ is convex. □

Remark 3.1.7 — Note that $\mathrm{vol}((-1,1)^n) = 2^n$, and this is a convex, symmetric set such that the only lattice point it contains is $\mathbf{0}$.

Our main interest is to use these results to get the Hasse principle for conics. Here is a simple application before diving into that.

Theorem 3.1.8

If $m \in \mathbb{Z}_{>0}$ such that there exists t with $t^2 \equiv -1 \pmod{m}$, then m is a sum of two squares.

Proof. Define
$$\Lambda = \left\{(x, y) \in \mathbb{Z}^2 \colon y \equiv tx \pmod{m}\right\} \subseteq \mathbb{Z}^2.$$
This has index m, because it is the kernel of
$$\varphi \colon \mathbb{Z}^2 \to \mathbb{Z}/m\mathbb{Z}, \quad (x, y) \mapsto y - tx.$$
Let
$$V = \left\{(x, y) \in \mathbb{R}^2 \colon x^2 + y^2 < 2m\right\}.$$
Then
$$\mathrm{vol}(V) = 2m \cdot \pi > 4m = 2^2 \cdot m = 2^2 \cdot [\mathbb{Z}^2 : \Lambda].$$
Corollary 3.1.6 implies that there exists $(x, y) \in V \cap \Lambda$ such that $(x, y) \neq (0, 0)$. Then $(x, y) \in V$ implies that $x^2 + y^2 < 2m$, and $(x, y) \in \Lambda$ implies that $y \equiv tx \pmod{m}$, so
$$x^2 + y^2 \equiv x^2 + (tx)^2 = x^2(t^2 + 1) \equiv 0 \pmod{m}.$$
So $x^2 + y^2 = m$. □

Exercise 3.1.9. Let $m \in \mathbb{Z}$, $m > 1$ and suppose that there is some $f \in \mathbb{Z}$ such that $f^2 + f + 1 \equiv 0 \pmod{m}$. Show that $m = u^2 + uv + v^2$ for some $u, v \in \mathbb{Z}$.

Solution: We use a tactic similar to the Theorem 3.1.8. Consider the open ellipse $x^2 + xy + y^2 < 2m$. Its area is πab where a, b are the lengths of semi-major and semi-minor axes, which can be found by setting $x = y$ and $x = -y$. This gives $a = 2\sqrt{\frac{m}{3}}$ and $b = 2\sqrt{m}$, respectively. Hence, the area is equal to $\frac{4\pi m}{\sqrt{3}}$, which is greater than $4m$.

Now, consider the lattice $L \subseteq \mathbb{Z}^2$ given by $y \equiv fx \pmod{m}$. This is clearly an index m subgroup of $\mathbb{Z}^2$. Hence, by Corollary 3.1.6, there is a non-zero (u, v) in L and in the open ellipse above. This satisfies $0 < u^2 + uv + v^2 < 2m$ and $u^2 + uv + v^2 \equiv u^2(1 + f + f^2) \equiv 0 \pmod{m}$. Hence, $u^2 + uv + v^2 = m$, as required. □

Remark 3.1.10 — To find the major/ minor axes, we must find these principal directions in the first place. This is done by attaining eigenvector for the matrix defining the quadratic form. Particularly,

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x^2 + xy + y^2.$$

The eigenvector of the matrix are $(1, -1)$, $(1, 1)$. These are the direction vectors, so correspond to $y = x$ and $y = -x$, with respective eigenvalues $\frac{3}{2}$ and $\frac{1}{2}$. To find the major/ minor axes, we must solve the equation $t^2\lambda = 2m$. Thus, we have

$$\frac{3}{2} \cdot t^2 = 2m \implies t = 2\sqrt{\frac{m}{3}}$$

and

$$\frac{1}{2} \cdot t^2 = 2m \implies t = 2\sqrt{m}.$$

Therefore, we have the major/ minor axes. Note that more generally, if we have some ellipse bounded by c, and the quadratic form defining the ellipse has eigenvalues λ_1, λ_2, then $\sqrt{\frac{c}{\lambda_i}}$ determines the length of the major/ minor axes.

Exercise 3.1.11. Find a prime $p > 0$ for which there is an $f \in \mathbb{Z}$ such that $1 + 5f^2 \equiv 0 \pmod{p}$ but p is not of the shape $u^2 + 5v^2$ $(u, v \in \mathbb{Z})$.

Solution: Consider $p = 7$. Then $f = 2$ satisfies $1 + 5f^2 \equiv 0 \pmod{7}$. But $7 \neq u^2 + 5v^2$ for any $u, v \in \mathbb{Z}$, as can be verified directly, or by noting that 2 is not a quadratic residue modulo 5.

$p = 3$ is valid too. $f = 1$ satisfies $1 + 5 \cdot 1^2 \equiv 0 \pmod{3}$. However, clearly 3 is not of this form. □

Remark 3.1.12 — The point of this exercise is that the approach to this problem as in the previous problem fails. Indeed, the area of the ellipse $x^2 + 5y^2 < 2p$ is $2p\pi/\sqrt{5}$ which is not greater than $4p$.

Let us now return to the proof of the local to global principle for conics. We first establish the following result which states that the existence of one rational point on a smooth conic implies that it is isomorphic to the projective line.

Proposition 3.1.13

Let k be any field of characteristic $\mathrm{char}(k) \neq 2$, and let $Q \neq \emptyset$ be a conic $F = 0$ over k for a degree two homogeneous polynomial $F \in k[X, Y, Z]$ such that $F = 0$ is non-singular. If $O \in Q(k)$, then we can construct a bijection between points of $Q(k)$ and points on a projective line over k that does not contain O.

Proof. Let $\pi(P)$ be the projection of O through P onto a line ℓ over k not containing O. We claim that $P \mapsto \pi(P)$ is a well-defined map. Any line can intersect Q in at most two points, and it intersects Q in exactly one point if and only if it is a tangent.

- If $P \neq O$, then there is a unique line $\overrightarrow{OP}$ between O and P. Since ℓ does not contain O, $\overrightarrow{OP}$ and ℓ are distinct, so $\overrightarrow{OP} \cap \ell$ is a unique point $\pi(P)$, making $\pi(P)$ well-defined in this case.

- The tangent line of Q at O intersects Q at O with multiplicity at least two, so it does not intersect Q at any other point. However, it still has a unique intersection point with ℓ, so we define $\pi(O)$ to be this intersection point.

Note that $\pi(P)$ has coordinates in k, since we're intersecting a line with a line; in other words, everything that appears is linear.

Claim 3.1.14. $P \mapsto \pi(P)$ is a bijection.

Proof.

- The map is injective: if $\pi(P) = \pi(P')$, then $\overrightarrow{OP} = \overrightarrow{OP'}$, so P, P', O are collinear, implying $P = P'$.

- The map is surjective: if $O \neq L \in \ell$, then there is another intersection point on $\overline{OL} \cap Q$ over the algebraic closure of k. Although the line intersects Q in two points over the algebraic closure, neither of which may be a rational point, we claim that if $0 \neq h \in k[x]$ is a quadratic polynomial and it has one root over k, then its other root is also defined over k. The reason for this is the following. Let $h(x) = ax^2 + bx + c$. Suppose $\alpha \in k$ is a root of $h(x)$, so we can factor $h(x) = (x - \alpha)g(x)$ for some $g(x) \in k[x]$. Since $\deg(g) = 1$, we have $g(x) = (x - \beta)$ for some $\beta \in k$. Therefore, β is the other root. Hence, the other intersection point is a point $P \in Q$.

$\square$

$\square$

Remark 3.1.15 — When working with conics, we know that if one of the intersection points is in k, then the other is in k as well.

Corollary 3.1.16

If k is infinite, then either Q has no points, or it has infinitely many.

Remark 3.1.17 — Here is a more explicit proof of Proposition 3.1.13. Suppose the conic is given by $y^2 = ax^2 + b$ such that $ab \neq 0$ and $P = (p_0, q_0)$ is a rational point. Then, every other point (p, q) with $p \neq p_0$ is given by

$$p = \frac{2tq_0 - (t^2 + a)p_0}{a - t^2}, \qquad q = \frac{(t^2 + a)q_0 - 2atp_0}{a - t^2}$$

for $t \in k \setminus \pm\sqrt{a}$. To see this, consider the line $y = t(x - p_0) + q_0$ through (p_0, q_0) of slope $t \in k$ and find the other intersection point with the conic.

§3.2 Hasse-Minkowski Theorem

Let $X \subseteq \mathbb{P}^n(\mathbb{Q})$ be a projective algebraic variety, such as hypersurfaces or plane curves. The **local-global principle** holds for X if $X(\mathbb{Q}) \neq \emptyset$ if and only if $X(\mathbb{Q}_p) \neq \emptyset$ for all prime p and $X(\mathbb{R}) \neq \emptyset$. Hasse-Minkowski implies that conics, and all quadratic hypersurfaces of any dimension, satisfy the local-global principle. Selmer and Chabauty proved that $3X^3+4Y^3+5Z^3 = 0$ does not. We now come to the proof of Hasse–Minkowski for $Q = aX^2 + bY^2 + cZ^2 = 0$.

Theorem 3.2.1 (Hasse-Minkowski)
Let Q be the projective curve defined by

$$aX^2 + bY^2 + cZ^2 = 0, \quad a, b, c \in \mathbb{Z}, \quad abc \neq 0, \quad (a, b) = (b, c) = (c, a) = 1,$$

where a, b, c are square-free. Let $\Sigma := \{p \mid 2abc\}$. Then the following are equivalent.

(i) Q has infinitely many solutions in $\mathbb{P}^2_{\mathbb{Q}}$.

(ii) Q has a solution in $\mathbb{P}^2_{\mathbb{Q}}$.

(iii) Q has a solution in $\mathbb{P}^2_{\mathbb{Q}_p}$ for all p, and in $\mathbb{P}^2_{\mathbb{R}}$.

(iv) Q has a solution in $\mathbb{P}^2_{\mathbb{Q}_p}$ for all $p \in \Sigma$.

Proof. (i) implies (ii) implies (iii) implies (iv), and we proved that (ii) implies (i), so proving (iv) implies (ii) is sufficient.

The crux of this is to use the existence of local solutions to construct a subgroup $\Lambda \subset \mathbb{Z}^3$ of index $4|abc|$ with the property that

$$Q(x, y, z) \equiv 0 \pmod{4|abc|}$$

for $(x, y, z) \in \Lambda$. Once we achieve that, we consider the ellipsoid (a convex, symmetric set)

$$\mathcal{C} = \{|a|X^2 + |b|Y^2 + |c|Z^2 < 4|abc|\},$$

whose volume satisfies

$$\mathrm{vol}(\mathcal{C}) = \frac{4}{3}\pi\frac{\sqrt{|4abc|^3}}{\sqrt{|abc|}} = \frac{\pi}{3} \cdot 2^3 \cdot |4abc| > 2^3 \cdot [\mathbb{Z}^3 : \Lambda].$$

Hence, by Minkowski's theorem (Corollary 3.1.6), we get a non-zero point $(r, s, t) \in \Lambda \cap \mathcal{C}$. For such r, s, t, we have

$$|ar^2 + bs^2 + ct^2| \leq |a|r^2 + |b|s^2 + |c|t^2 < 4|abc|,$$

hence, it follows that

$$ar^2 + bs^2 + ct^2 = 0$$

as required.

Thus, it remains to construct Λ with the required properties. We will use local solutions to construct Λ by imposing congruence relations.

Claim 3.2.2. If $p \mid a$, then there is an r_p such that

$$b + cr_p^2 \equiv 0 \pmod{p}.$$

Proof. We are assuming (iv), so there exists $(x, y, z) \in \mathbb{P}^2(\mathbb{Q}_p)$ with $ax^2 + by^2 + cz^2 = 0$. Without loss of generality, assume $x, y, z \in \mathbb{Z}_p$, not all divisible by p. If $|y|_p < 1$, then $cz^2 = -(ax^2 + by^2) \equiv 0 \pmod{p}$, so $|z|_p < 1$, and then $|ax^2|_p \leq 1/p^2$, so $|x|_p < 1$, which is a contradiction. Thus, $|y|_p = 1$, so $a(\frac{x}{y})^2 + b + c(\frac{z}{y})^2 = 0$, with $\frac{x}{y}, \frac{z}{y} \in \mathbb{Z}_p$. Hence,

$$b + c\left(\frac{z}{y}\right)^2 \equiv 0 \pmod{p}.$$

□

Let us assume now that p is odd and $p \mid a$. Then let $r_p \in \mathbb{Z}/p\mathbb{Z}$ be a solution to $b + cr_p^2 \equiv 0 \pmod{p}$, and impose the condition

$$r_p s + t \equiv 0 \pmod{p}.$$

Then

$$ar^2 + bs^2 + ct^2 \equiv bs^2 + ct^2 \equiv bs^2 + c(r_p s)^2 \equiv s^2(b + cr_p^2) \equiv 0 \pmod{p}.$$

If p is odd and $p \mid b$ or $p \mid c$, impose analogous conditions. By imposing this condition, we get an index p subgroup $(r, s, t) \in \mathbb{Z}^3$ such that $Q(\Lambda) \equiv 0 \pmod{p}$.

First assume that p is odd and WLOG, $p \mid a$. Then let $u_p \in \mathbb{Z}/p\mathbb{Z}$ be a solution to

$$b + cu_p^2 \equiv 0 \mod p,$$

and impose the condition

$$u_p s + t \equiv 0 \mod p.$$

Then

$$ar^2 + bs^2 + ct^2 \equiv bs^2 + ct^2 \equiv bs^2 + c(u_p s)^2 = s^2(b + cu_p^2) \equiv 0 \mod p.$$

Now assume that $p = 2$. Let $2 \mid a$, or by symmetry $2 \mid b$ or $2 \mid c$. We will write a condition to ensure that $ar^2 + bs^2 + ct^2 \equiv 0 \pmod{8}$. We saw in the proof of the claim that there exist $x, y, z \in \mathbb{Z}_2$ with $|x|_2 \leq 1$ and $|y|_2 = |z|_2 = 1$, and $ax^2 + by^2 + cz^2 = 0$. Then $ax^2 + by^2 + cz^2 \equiv ax^2 + b + c \equiv 0 \pmod{8}$.

- If $|x|_2 = 1$, then $a + b + c \equiv 0 \pmod{8}$. Impose

  $$s \equiv t \pmod{4}, \qquad r \equiv s \pmod{2}.$$

  Then either $r, s, t \equiv 1 \pmod{2}$, and $ar^2 + bs^2 + ct^2 \equiv a + b + c \equiv 0 \pmod{8}$, or $r, s, t \equiv 0 \pmod{2}$, and $ar^2 + bs^2 + ct^2 \equiv bs^2 + ct^2 \equiv 4(b + c) \equiv -4a \equiv 0 \pmod{8}$.

- If $|x|_2 < 1$, then $ax^2 + b + c \equiv b + c \pmod{8}$, that is $b + c \equiv 0 \pmod{8}$. Impose

  $$s \equiv t \pmod{4}, \qquad r \equiv 0 \pmod{2}.$$

  Then $ar^2 + bs^2 + ct^2 \equiv bs^2 + ct^2 \equiv bs^2 + cs^2 \equiv (b + c)s^2 \equiv 0 \pmod{8}$.

Now let $2 \nmid abc$. Let $(x, y, z) \in \mathbb{P}^2(\mathbb{Q}_2)$ with $ax^2 + by^2 + cz^2 = 0$. Assume that $\max(|x|_2, |y|_2, |z|_2) = 1$. Then $ax^2 + by^2 + cz^2 \equiv 0 \pmod 2$, that is $x^2 + y^2 + z^2 \equiv 0 \pmod 2$. Exactly one of x, y, z is even; without loss of generality, assume $|x|_2 < 1$ and $|y|_2 = |z|_2 = 1$. Then $0 = ax^2 + by^2 + cz^2 \equiv by^2 + cz^2 \equiv b + c \pmod 4$. Impose the conditions

$$r \equiv 0 \pmod 2, \qquad s \equiv t \pmod 2.$$

Then

$$ar^2 + bs^2 + ct^2 \equiv bs^2 + ct^2 \equiv (b + c)s^2 \equiv 0 \pmod 4.$$

We're done with the case working now!

To finish off, let $\Lambda \subseteq \mathbb{Z}^3$ be the subgroup defined by all of these congruence conditions for $p \in \Sigma$. Then

$$[\mathbb{Z}^3 : \Lambda] = 4|abc| = 4 \prod_{p|abc} p,$$

using the Chinese remainder theorem. If $(r, s, t) \in \Lambda$, then $ar^2+bs^2+ct^2 \equiv 0 \pmod{4|abc|}$. Thus, the proof is completed. □

Remark 3.2.3 — Note that nowhere in the proof did we use that there is a solution over $\mathbb{R}$. In fact, this is a consequence of solubility over all $\mathbb{Q}_p$ ($p \neq \infty$). This phenomenon is related to quadratic reciprocity, however we won't pursue it here. In fact, it is true that for any conic over $\mathbb{Q}$, the number of p (including ∞) for which there is no point over $\mathbb{Q}_p$ is always even (see Exercises in [3]).

Exercise 3.2.4. Determine whether

$$F(X, Y, Z) = 7X^2 + 3Y^2 - 2Z^2 + 4YZ + 6ZX + 2XY$$

has solutions over $\mathbb{Q}_p$ for all p, and so whether it has a rational solution. Further, find rational integers x, y, z not all divisible by 17 such that $F(x, y, z) = 0 \pmod{7^3}$.

Solution: If we apply the change of coordinates $Y \mapsto Y - \frac{4}{5}X$, $Z \mapsto Z + Y + \frac{7}{10}X$, we get the form

$$\left(\frac{83}{10}\right) X^2 + 5Y^2 - 2Z^2.$$

Multiplying by 10, we reduce to finding a solution to

$$83X^2 + 50Y^2 - 20Z^2 \equiv 0.$$

By absorbing $5Y \mapsto Y$, and $2Z \mapsto Z$, we finally reduce to

$$83X^2 + 2Y^2 - 5Z^2 = 0.$$

Now, we can try to solve this over $\mathbb{Q}_2$, $\mathbb{Q}_5$, and $\mathbb{Q}_{83}$, which are easy to analyze. We note that

$$\begin{aligned} 83 \cdot 1^2 + 2 \cdot 1^2 - 5 \cdot 1^2 &\equiv 0 \pmod 8, \\ 83 \cdot 1^2 + 2 \cdot 1^2 &\equiv 0 \pmod 5, \\ 2 \cdot 8^2 - 5 \cdot 3^2 &\equiv 0 \pmod{83}. \end{aligned}$$

Hence, by Hensel's lemma, we have solutions over $\mathbb{Q}_2$, $\mathbb{Q}_5$, and $\mathbb{Q}_{83}$. By Hensel's lemma, we can find a solution over $\mathbb{R}$. Alternatively, we can observe that $(1, 9, 7)$ is a solution over the rationals.

To find a solution to $F(x, y, z) = 0 \pmod{17^3}$, the strategy is to first find a solution modulo 17 in $\mathbb{F}_{17}^3$. $(2, 3, 0) \in \mathbb{F}_{17}^3$ is a solution. We then lift this to a solution in $\mathbb{Z}_{17}^3$. After iterating, we can find that $(2, 3 + 17 \cdot 2, 0)$ is a solution to the full congruence. Mapping this back to our original coordinates, we get the point $(20, 354, 384)$. This satisfies $F(20, 354, 384) = 0 \pmod{17^3}$. We can factor out the 2 to get the primitive solution $(10, 177, 192)$ as well. □

Exercise 3.2.5. For each of the following sets of a, b, c find the set of primes p (including ∞) for which the only solution of

$$ax^2 + by^2 + cz^2 = 0$$

in $\mathbb{Q}_p$ is the trivial one:

1. $(a, b, c) = (1, 1, -2)$
2. $(a, b, c) = (1, 1, -3)$
3. $(a, b, c) = (1, 1, 1)$
4. $(a, b, c) = (14, -15, 33)$

Do you observe anything about the parity of the number N of primes (including ∞) for which there is insolubility?

1. Prove your observation for the above in the special case $a = 1$, $b = -r$, $c = -s$, where r, s are distinct primes > 2. (Use quadratic reciprocity!)
2. Prove your observation for all $a, b, c \in \mathbb{Z}$ (this is hard!).

Solution: 1. By Hasse-Minkowski, we see that there is a solution $(1, 1, 1)$ in $\mathbb{Z}$. Therefore, there are nontrivial solutions for the completion of $\mathbb{Q}$ at all places.

For 2, the only solution is trivial by Exercise 3.0.10. At ∞, we can also find the nontrivial solution $(\sqrt{3}, 0, 1)$. At 3, there is only the trivial solution. At all other primes, there exists a nontrivial solution by plugging in some value (say 1) into z and using Exercise 3.0.8.

3. Only the nontrivial solution over $\mathbb{P}^2(\mathbb{R})$, since this is nonnegative. For $p = 2$, there is only the nontrivial solution by Exercise 3.0.10. By Exercise 3.0.8, we can find nontrivial solutions for other primes.

4. We can definitely find a nontrivial solution in $\mathbb{R}$. At 2, we can find some nontrivial solution by Exercise 3.0.10. By Exercise 3.0.8, for $p > 11$, we can always find nontrivial solutions. We need to see what happens at the odd primes $3, 5, 7, 11$ then. We won't show this here, but we can find nontrivial solutions for the primes 3 and 11. Thus, we only get trivial solutions at $p = 5, 7$.

It seems that this number is always even. We shall prove this when $a = 1, b = -r, c = -s$. For this, the equation we're looking at is $x^2 = ry^2 + cz^2$. We note that ∞ doesn't contribute to N. Also, for primes $p \nmid 2rs$, we can find nontrivial solutions by Exercise (3.0.8). Now, note that if r or s is congruent to 1 modulo 4, then the Legendre symbols match. This implies that we can find a nontrivial solution at the primes r and s. Also, at 2, then $-r$ or $-s$ is congruent to 3 modulo 4, so at 2, we also find nontrivial solutions. Thus, $N = 0$ when r or s or congruent to 1 modulo 4. If both r, s are congruent to 3 modulo 4, then we only find nontrivial solutions at 2. Also, the Legendre symbols are opposite, so this means that we can only find a nontrivial solution for one of these primes. Thus, $N = 2$ in this case.

Now, for general $a, b, c \in \mathbb{Z}$. Check Cassels' book 'Rational quadratic forms' Lemma 3.4. $\square$

4 Cubic Curves

§4.1 Plane Cubics

Definition 4.1.1. A **plane cubic** is a cubic equation

$$F(X, Y, Z) = 0$$

defined by a homogeneous cubic polynomial (cubic form) in three variables over a field k.

§4.1.i Singular Cubics

It turns out the k-points of singular cubic curves can be parametrized by k, much like how k-points of conics were.

Example 4.1.2 (Cuspidal Cubic)
For the cusp curve $Y^2Z = X^3$, we can use the parametrization

$$t \mapsto (t^2, t^3, 1).$$

Example 4.1.3 (Nodal Cubic)
For the node $Y^2Z = X^3 + X^2Z$, we can use the parametrization

$$t \mapsto (t^3 - t, t^2 - 1, 1).$$

Proposition 4.1.4
Suppose that C is an irreducible singular plane cubic over a field k of characteristic zero or suppose k is a finite field (or any perfect field). Then there is a unique singular point P with coordinates in k. If k is infinite, then there are infinitely many points on C over k, and we can parametrize them by looking at lines through P (with rational slope).

Proof. Suppose C has two distinct singular points P, Q, then the line joining P and Q intersects C at both P and Q with multiplicity ≥ 2. By Bézout's theorem, this can only happen if C contains the whole line, but this cannot happen for an irreducible curve. Thus, if C is singular, then it has a unique singular point over some algebraic extension of k.

A singular point is defined by the system of equations

$$F = 0, \quad \frac{\partial F}{\partial X} = 0, \quad \frac{\partial F}{\partial Y} = 0, \quad \frac{\partial F}{\partial Z} = 0.$$

Since C is defined over k, if a singular point P has coordinates in an extension of k, then any Galois conjugate of P must also be a singular point. Since there is only one singular point, its coordinates must be invariant under the Galois group, and thus must lie in k; here, we are really requiring k to be perfect.

Finally, the parametrization of C by k is achieved by consider lines ℓ through the singular point P. A generic line ℓ through P intersects C at P with multiplicity 2. By Bézout's Theorem, there must be a third point of intersection, which is also defined over k. Varying the slope of ℓ over k gives a parametrization of the k-points on C. $\square$

Remark 4.1.5 — If C is not irreducible, we may look at the irreducible components.

Remark 4.1.6 — Now let ℓ_P be the line through the unique singular point P in the setting above. Then, $(\ell_P \cdot C)_P = 2$, so Bézout tells us that $\ell_P \cap C = \{P, \Pi(\ell)\}$, where $\Pi(\ell)$ is the other intersection point. In fact, this other intersection point $\Pi(\ell)$ must also have coordinates in k, since P has multiplicity two inside a cubic.

§4.1.ii Smooth Cubics

We now turn our attention to smooth cubic curves.

Example 4.1.7 (Non-Example)

We note that not all smooth cubic curves over $\mathbb{Q}$ have a rational point. For example, for p prime,

$$C\colon X^3 + pY^3 + p^2Z^3 = 0$$

is a smooth cubic curve over $\mathbb{Q}$ but it doesn't have a rational point. Checking smoothness is straightforward, by taking derivatives. Suppose there exists a rational point on C; by multiplying with common denominators, we would obtain integers $a, b, c \in \mathbb{Z}$ such that $a^3 + pb^3 + p^2c^3 = 0$ and $\gcd(a, b, c) = 1$. Now, this implies $p \mid a$, but then $p^3 \mid pb^3 + p^2c^3$, hence $p^2 \mid pb^3$, so $p \mid b$. Then $p^3 \mid pb^3$, hence $p \mid c$. This is a contradiction to $\gcd(a, b, c) = 1$. These are not Elliptic curves!

Definition 4.1.8. An **elliptic curve** E is a smooth plane cubic over k equipped with a specified point $\mathcal{O} \in E(k)$ that is defined over k, i.e. $\mathcal{O}$ is a k-point.

Exercise 4.1.9.

1. Show that the cubic curve

$$Y^2Z = X^3 + AXZ^2 + BZ^3$$

is non-singular provided that

$$4A^3 + 27B^2 \neq 0.$$

2. If

$$4A^3 + 27B^2 = 0,$$

find a singularity and decide whether it is a cusp or a double point with distinct tangents.

Solution: 1. Let $F(X, Y, Z) = X^3 + AXZ^2 + BZ^3 - Y^2Z$. Then,

$$F_X = 3X^2 + AZ^2, \quad F_Y = -2YZ, \quad F_Z = 2AXZ + 3BZ^2 - Y^2.$$

Setting these equal to 0, we see that Y or Z is zero, so let $Z = 0$ to begin with. Then, $X = Y = 0$, which is not in $\mathbb{P}^2$. If $Y = 0$, then $Z = 1$ (up to scaling). Then,

$$3X^2 = -A, \quad 2AX = -3B.$$

We want $X \neq 0$, so we can deduce that $\frac{-A}{3} = \frac{9B^2}{4A^2}$, implying that $4A^3 + 27B^2 = 0$, so not possible.

2. If $X = 0$ in the above, then $A = B = 0$, so we have a singular point, and our equation is $Y^2Z = X^3$. This is cuspidal. If $X \neq 0$, then $X = \frac{-3B}{2A}$ is a singular point. If we apply the projective transformation $X \mapsto X - \frac{3B}{2A}Z$, then we find that

$$\left(X - \frac{3B}{2A}Z\right)^3 + A\left(X - \frac{3B}{2A}Z\right)Z^2 + BZ^3 = X^2(X - \frac{9B}{2A}Z).$$

This is a nodal singularity. □

Exercise 4.1.10.

1. Let
$$F(x) = a_1X_1^3 + a_2X_2^3 + a_3X_3^3 + dX_1X_2X_3,$$
where
$$a_1a_2a_3 \neq 0.$$
Show that $F(x) = 0$ is non-singular provided that
$$27a_1a_2a_3 + d^3 \neq 0.$$
2. If $a_1 = a_2 = a_3 = 1$, $d = -3$, show that any point (x_1, x_2, x_3) with
$$x_1^3 = x_2^3 = x_3^3 = x_1x_2x_3 = 1$$
is a singularity.
3. How does the result of 2. square with the result proved in the text that a cubic curve has at most one singularity?

Solution: 1. We calculate the first partials:

$$F_i = 3a_iX_i^2 + dX_jX_k$$

Here $j \neq k$ are indices which differ from i. If any of the X_i are 0, then all the others are 0, so assume these are all nonzero. Then, shift the dX_jX_k to the right and multiply them all together to get

$$27a_1a_2a_3X_1^2X_2^2X_3^2 = -d^3X_1^2X_2^2X_3^2,$$

so $27a_1a_2a_3 = -d^3$, but this is not possible.

2. We first see that $3X_i^2 = 3X_jX_k$ is satisfied. Multiplying by X_i, we find $X_i^3 = X_iX_jX_k$, and this holds for all permutations of i, j, k. This also shows that it satisfies F. Setting this equal to 1, it is a singularity. For a neater solution, we can see that

$$F(x) = (X_1 + X_2 + X_3)(X_1 + \xi X_2 + \xi X_3)(X_1 + \xi^2X_2 + \xi X_3),$$

where $\xi^3 = 1$.

3. This curve is not irreducible of $\overline{\mathbb{Q}}$ or any field containing ξ, and the result only holds for irreducible cubics. □

§4.2 Chord and Tangent

Notation 4.2.1. When we say $p, q \in E(\mathbb{Q})$ for a given elliptic curve E, we mean that the points p, q lie on E and are defined over $\mathbb{Q}$.

Let E be a smooth cubic curve defined over $\mathbb{Q}$. Let $p, q \in E(\mathbb{Q})$. Consider the line $\ell_{p,q}$ joining them. By Bézout's theorem, such a line will intersect E at a third point, in general distinct. Note that a priori Bézout only gives that such a point lies on an algebraic extension of $\mathbb{Q}$; however, intersecting the line $\ell_{p,q}$ with E, we get a cubic equation in one variable, two of whose roots are rational, hence the third is also rational. This is called a **chord process**.

There is also a variant of this which we saw in the first lecture, namely starting from a rational point $p \in E(\mathbb{Q})$, we can consider the tangent line ℓ_p to E at p; again this will generally intersect E at another point in $E(\mathbb{Q})$. This is called a **tangent process**.

We now consider the following question: Starting from a point $p \in E$, repeating this process of taking tangent lines and chords we can obtain several new points of E, but do we get all rational points of E this way? Do we even get infinitely many points on E this way?

Definition 4.2.2. Starting at p, if we can only get finitely many points of E through chord and tangent processes, we call the original point p to be **exceptional**.

Example 4.2.3

Inflection points are exceptional. The reason for this is because we can only take a tangent, but this tangent intersects the cubic with multiplicity three for flex points, so the process terminates.

Proposition 4.2.4

Let $a \geq 1$ be a cube-free integer and let

$$E\colon X^3 + Y^3 = aZ^3.$$

The point $[1 : -1 : 0]$ is exceptional. For $a = 1$, the points $[0 : 1 : 1]$, $[1 : 0 : 1]$ are also exceptional. For $a = 2$, the point $[1 : 1 : 1]$ is exceptional. No other rational point is exceptional.

Proof. The Hessian is given by the cubic $H(X, Y, Z) = XYZ$. Hence $[1 : -1 : 0]$ is an inflection point. We may also observe that the tangent line at $[1 : -1 : 0]$ is $X + Y = 0$, and this line intersects E only at $[1 : -1 : 0]$. We can similarly see that when $a = 1$, the points $[0 : 1 : 1]$ and $[1 : 0 : 1]$ are also inflection points. Finally, when $a = 2$, check that the tangent line to $[1 : 1 : 1]$ intersects the cubic at another point which is an inflection point, namely $[1 : -1 : 0]$. Hence, $[1 : 1 : 1]$ is an exceptional point. We now want to show that there are no other exceptional points.

Now, suppose $\mathbf{x}_0 = [x_0 : y_0 : z_0]$ is any other rational point. Note that this is not a flex point, since we've already ran through them. We compute the tangent line to be

$$\ell_{\mathbf{x}_0}\colon x_0^2 X + y_0^2 Y - a z_0^2 Z = 0.$$

Claim 4.2.5. $\ell_{\mathbf{x}_0}$ intersects E at another point

$$\mathbf{x}_1 = [x_1 : y_1 : z_1] = [x_0(x_0^3 + 2y_0^3) : -y_0(2x_0^3 + y_0^3) : z_0(x_0^3 - y_0^3)].$$

Proof. We can verify this claim by showing that $\mathbf{x}_1$ satisfies the equations of E and $\ell_{\mathbf{x}_0}$. This is straightforward.

$$\begin{aligned}\ell_{\mathbf{x}_0}(\mathbf{x}_1) &= x_0^2(x_0(x_0^3 + 2y_0^3)) - y_0^2(y_0(2x_0^3 + y_0^3)) - az_0^2(z_0(x_0^3 - y_0^3)) \\ &= x_0^6 - y_0^6 - az_0^6 x_0^3 + ay_0^3 z_0^3 \\ &= x_0^6 - y_0^6 - x_0^3(x_0^3 + y_0^3) + y_0^3(x_0^3 + y_0^3) = 0.\end{aligned}$$

We also check

$$\begin{aligned}F(\mathbf{x}_1) &= x_0^3(x_0^3 + 2y_0^3)^3 - y_0^3(2x_0^3 + y_0^3)^3 - az_0^3(x_0^3 - y_0^3)^3 \\ &= x_0^3(x_0^3 + 2y_0^3)^3 - y_0^3(2x_0^3 + y_0^3)^3 - (x_0^3 + y_0^3)(x_0^3 - y_0^3)^3 \\ &= G(x_0^3, y_0^3),\end{aligned}$$

for $G(x,y) = x(x+2y)^3 - y(2x+y)^3 - (x+y)(x-y)^3$. It is easy to see by expanding that $G(x,y) \equiv 0$ identically. Hence $\mathbf{x}_1$ lies on E. $\square$

Now, note that for any point $[x : y : z]$ on E, we may arrange that $x, y, z \in \mathbb{Z}$ are coprime. Then, we define the **height** $h(\mathbf{x})$ of a point $\mathbf{x} = [x : y : z]$ on E to be $|z|$, assuming we arranged $x, y, z \in \mathbb{Z}$ and $\gcd(x, y, z) = 1$.

Claim 4.2.6. $h(\mathbf{x}_1) > h(\mathbf{x}_0)$.

Proof. Let $[x_0 : y_0 : z_0]$ be such that $x_0, y_0, z_0 \in \mathbb{Z}$ and $\gcd(x_0, y_0, z_0) = 1$. Since $x_0^3 + y_0^3 = az_0^3$ and a is cube-free, it follows that x_0, y_0, z_0 are pairwise coprime. Then

$$h(\mathbf{x}_0) = |z_0|.$$

Put

$$d = \gcd(x_0(x_0^3 + 2y_0^3), -y_0(2x_0^3 + y_0^3), z_0(x_0^3 - y_0^3)).$$

If p is a prime dividing both x_0 and d, then since $p \mid y_0(2x_0^3 + y_0^3)$, it follows that $p \mid y_0$. Then, from the original equation, $p \mid z_0$, which contradicts $\gcd(x_0, y_0, z_0) = 1$. Hence $\gcd(d, x_0) = 1$. Similarly, $\gcd(d, y_0) = 1$. It follows that if d divides $x_0^3 + 2y_0^3$ and $2x_0^3 + y_0^3$; hence d divides $3x_0^3$ and $3y_0^3$. So $d = 1$ or 3. Hence

$$h(\mathbf{x}_1) = \begin{cases} |z_0||x_0^3 - y_0^3|, & d = 1, \\ \frac{1}{3}|z_0||x_0^3 - y_0^3|, & d = 3. \end{cases}$$

Thus the claim follows from $|x_0^3 - y_0^3| > 3$. It is easy to verify that the latter can only fail if $x_0, y_0 = 0, \pm 1$, but those points were already treated. $\square$

By repeating the tangent process, we get a sequence

$$|z_0| < |z_1| < |z_2| < \cdots.$$

Hence the points $\mathbf{x}_i$ are distinct, showing that $\mathbf{x}_0$ is not exceptional. $\square$

Exercise 4.2.7. Let $F(x)$ be the cubic

$$F(x) = a_1X_1^3 + a_2X_2^3 + a_3X_3^3 + dX_1X_2X_3$$

from Exercise 4.1.10 and suppose that $F(x) = 0$ is non-singular.

1. Let $F(x) = 0$. Show that the third intersection t of the tangent at x is given by

$$t_j = x_j \left(a_{j+1}x_{j+1}^3 - a_{j+2}x_{j+2}^3\right), \quad j = 1, 2, 3,$$

where the suffixes are taken modulo 3.

2. Let x, y be distinct points on $F(X) = 0$. Show that the third intersection z of the line joining them is given by

$$z_j = x_j^2 y_{j+1}y_{j+2} - y_j^2 x_{j+1}x_{j+2}.$$

This is called Desboves formulae.

Solution: 1. We first calculate the tangent. The partials are of the form

$$3a_iX_i^2 + dX_{i+1}X_{i+2}.$$

This is nonzero for some i, as we assume non-singularity. Then, the tangent line is given by

$$\sum_{i=1}^{3}(3a_ix_i^2 + dx_{i+1}x_{i+2})X_i = 0.$$

It then suffices to show that the given $t = [t_1 : t_2 : t_3]$ satisfies both equations. First is the tangent line check; by looking at symmetric parts, we find that

$$\sum_{i=1}^{3}(3a_ix_i^2 + dx_{i+1}x_{i+2}) \left(x_i(a_{i+1}x_{i+1}^3 - a_{i+2}x_{i+2}^3)\right) = 0.$$

Now for $F(t)$. We have that

$$\begin{aligned} F(t) = a_1(x_1(a_2x_2^3 - a_3x_3^3))^3 + a_2(x_2(a_3x_3^3 - a_1x_1^3))^3 + a_3(x_3(a_1x_1^3 - a_2x_2^3))^3 \\ + dx_1x_2x_3(a_2x_3^3 - a_3x_3^3)(a_3x_1^3 - a_1x_1^3)(a_1x_2^3 - a_2x_2^3) \end{aligned}$$

To see the vanishing more clearly, we need to replace $dx_1x_2x_3$ with $-a_1x_1^3 - a_2x_2^3 - a_3x_3^3$, and note that there is a general algebraic identity given as follows:

$$A(B - C)^3 + B(C - A)^3 + C(A - B)^3 = (A + B + C)(B - C)(C - A)(A - B).$$

We apply this with $A = a_1x_1^3$, $B = a_2x_2^3$, $C = a_3x_3^3$.

2. Recall/ note that the (projective) line going through $x = [x_1 : x_2 : x_3]$ and $y = [y_1 : y_2 : y_3]$ is given by

$$(x_2y_3 - x_3y_2)X + (x_3y_1 - x_1y_3)Y + (x_1y_2 - x_2y_1)Z = 0.$$

Then, we want to show that this $z = [z_1 : z_2 : z_3]$ satisfies the equation. First is the tangent line check:

$$\sum_{i=1}^{3}(x_{i+1}y_{i+2} - x_{i+2}y_{i+2})(x_i^2y_{i+1}y_{i+2} - y_i^2x_{i+1}x_{i+2})$$

By considering symmetries, this cancels. Next, for $F(z)$:

$$F(z) = a_1(x_1^2y_2y_3 - y_1^2x_2x_3)^3 + a_2(x_2^2y_3y_1 - y_2^2x_3x_1)^3 + a_3(x_3^2y_1y_2 - y_3^2x_1x_2)^3 \\ + d(x_1^2y_2y_3 - y_1^2x_2x_3)(x_2^2y_3y_1 - y_2^2x_3x_1)(x_3^2y_1y_2 - y_3^2x_1x_2)$$

This is zero, after some work. □

Exercise 4.2.8. Starting with the solution $(2, -1, -1)$ of

$$X^3 + Y^3 + 7Z^3 = 0,$$

find 10 distinct solutions.

Solution: First, by homogeneity of the equation, we should ensure that $\gcd(x, y, z) = 1$. Next, notice that if (x, y, z) is a solution, then so is (y, x, z). Now, all we do is just apply the results from the previous exercise (4.2.7). □

§4.3 Group Law

Theorem 4.3.1

Let E_f be an irreducible cubic over a field k defined by f. Assume that there exists at least one point over k where $f = 0$. Fix such a point $\mathcal{O} \in E$. Let $E_{ns}(k)$ be the set of non-singular (smooth) points of $f = 0$ with coordinates in k. We can give $E_{ns}(k)$ the structure of an abelian group with identity $\mathcal{O}$. Note that if E is smooth, then $E(k) = E_{ns}(k)$.

Notation 4.3.2. If E is smooth, then we write $E(k)$ instead of $E_{ns}(k)$.

Lemma 4.3.3 (Cayley-Bacharach)

Let k be an infinite field. If two cubics intersect in exactly nine points, then every cubic passing through eight of these also contains the ninth one.

Proof. Let f_1 and f_2 be the two cubics having distinct intersections at $p_1, \dots, p_9$. Any cubic is given by an equation

$$a_1X^3 + a_2X^2Z + a_3XZ^2 + a_4Z^3 + a_5Y^3 + a_6Y^2Z + a_7YZ^2 + a_8X^2Y + a_9XY^2 + a_{10}XYZ = 0.$$

The condition that $p_1, \dots, p_8$ are on the cubic gives eight linear equations in $a_1, \dots, a_{10}$. The main claim is that these eight linear equations are independent.

Consider now another cubic f containing $p_1, \dots, p_8$. We want to prove that $f = \lambda f_1 + \mu f_2$ for some $\lambda, \mu \in k$. If f, f_1 and f_2 are independent, then we can choose $\lambda, \mu, \nu \in k$ such that $\lambda f + \mu f_1 + \nu f_2$ passes through any two given points. The points p_i are not completely arbitrary: No four of them can lie on a line, for such a line would be a common component of f_1 and f_2, since there are a maximum of 3 points. Similarly, no seven of them can lie on a conic. Moreover, any five points lie on a unique conic: Indeed any conic is given by an equation

$$b_1X^2 + b_2Y^2 + b_3Z^2 + b_4XY + b_5XZ + b_6YZ$$

so we can find a conic through any given set of five points. However, if two conics have five common points, then they would have a common line component by Bézout, but the remaining components can only have one intersection not on this line, which implies that the other four points must be collinear. The rest of the argument is covered in the following cases.

1. Suppose that p_1, p_2, p_3 lie on a line L. Then $p_4, \ldots, p_8$ are on a unique conic Q. Let p_9 be a point on L distinct from p_1, p_2, p_3 and p_{10} be a point in $\mathbb{P}^2_k$ not on L or Q. Now, we can find λ, μ, ν so that $\lambda f + \mu f_1 + \nu f_2$ contains $p_1, \ldots, p_8, p_9, p_{10}$. Thus, it will necessarily have L as a component. The other component must therefore be Q. But p_{10} is not on L or Q, a contradiction.

2. Suppose that $p_1, \ldots, p_6$ lie on a conic Q. Let L be the line through p_7, p_8. Choose another point p_9 on Q and p_{10} not on Q or L. Then, we arrive at a contradiction as before since a combination $\lambda f + \mu f_1 + \nu f_2$ must be of the form $Q \cdot L$. Thus, we miss p_{10}.

3. Finally, assume no three of $p_1, \ldots, p_8$ is on a line and no six of them are on a conic. Let ℓ be the line passing through p_1, p_2, and Q be the conic passing through $p_3, \ldots, p_7$. Let p_9 and p_{10} be points on the line L. We again arrive at a contradiction as before since p_8 is neither on L nor Q.

Therefore, we get that f is linearly dependent on f_1, f_2, so we have that if f goes through $p_1, \ldots, p_8$, it goes through p_9 as well. □

> **Exercise 4.3.4.** We have seen that intersections of two cubics having no common component is not a collection of arbitrary nine points in $\mathbb{P}^2_k$. Conversely, show that if $p_1, p_2, \ldots, p_8 \in \mathbb{P}^2_k$ are distinct points such that no four of them lie on a line and no seven of them lie on a conic, then there are two cubics passing through $p_1, p_2, \ldots, p_8$ having no common component.

Proof of Theorem 4.3.1. Let $P + Q$ be the third point of intersection of the line $\overrightarrow{OR}$, where R is the third point of intersection of the line through $\overrightarrow{PQ}$. If $P = Q$, take the tangent at P.

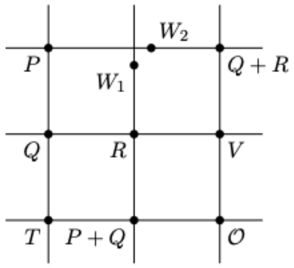
- To see that $P + Q \in E_{ns}(k)$, note that given any two points in $E_{ns}(k)$, the third point of intersection has coordinates in k, such as by explicitly solving equations, and must be non-singular, else there would be at least $2 + 1 + 1 = 4$ points of intersection with multiplicity.

- $P + Q = Q + P$ by definition.

- $P + \mathcal{O} = P$ by definition.

- To find an inverse $-P$ such that $P + (-P) = \mathcal{O}$, let Q be the third point of intersection of the tangent line at $\mathcal{O}$ with the elliptic curve. Then $-P$ is defined as the third point of intersection of $\overrightarrow{PQ}$ with $f = 0$.

- It remains to check that $(P + Q) + R = P + (Q + R)$. We follow Cassels' book, which almost proves it. Let P, Q, and R be given. Consider the following diagram:



Except for W_1 and W_2, all the points are intersections of the two lines. The whole figure is determined by P, Q, R, and O. Associativity is the statement that W_1 and W_2 are not as shown but coincide with the unlabeled intersection of the top line and the middle vertical line. Consider the eight points

$$\mathcal{O}, P, Q, R, T, V, P+Q, Q+R.$$

Also, consider the three cubics:

- C_1: $f = 0$
- C_2: the union of the three vertical lines
- C_3: the union of the three horizontal lines.

By Lemma 4.3.3, the eight points lie on three cubics, so the three cubics go through a ninth intersection point W. By Bézout's theorem, the ninth point of intersection of the first two cubics is W_1, so $W = W_1$. Similarly, $W = W_2$. Hence $W_1 = W_2$, and $(P+Q)+R = P+(Q+R)$.

Note that to apply Cayley-Bacharach, we require the eight points above to be distinct. This requires that $\mathcal{O}, P, Q, R, T, V, P+Q, Q+R$ are in general position. This is an open condition. The equation $P+(Q+R) = (P+Q)+R$ is a closed condition. To complete the proof, work in the Zariski topology. However, we can patch this up later via explicit formulae. □

Lemma 4.3.5

If $\mathcal{O}$ is a point of inflection, then $P+Q+R = \mathcal{O}$ if and only if P, Q, R are collinear.

Proof. Note that if $\mathcal{O}$ is an inflection point, then the tangent line of $\mathcal{O}$ intersects E at $\mathcal{O}$ only. $P+Q+R = \mathcal{O}$ if and only if $R = -(P+Q)$, if and only if R is the third point of intersection of the line through P and Q. □

Exercise 4.3.6. Let $\mathcal{O}, P$ be rational points on the nonsingular cubic C. Construct the point $-P$ with respect to the group law for which $\mathcal{O}$ is the neutral element.

Solution: This is in the proof above. Let Q be the third intersection of the tangent line of $\mathcal{O}$ to C. Then, $-P$ is the third intersection of $\overrightarrow{PQ}$ with C. □

Exercise 4.3.7. Let $\mathcal{O}, \mathcal{O}_1$ be rational points on the nonsingular cubic C. Show how the group law for which $\mathcal{O}_1$ is the neutral element can be expressed in terms of that for which $\mathcal{O}$ is the neutral element.

Solution: Let $+_1$ denote the group law associated to $\mathcal{O}_1$. The claim is that

$$x +_1 y = x + y - \mathcal{O}_1.$$

Let z be the third intersection of the line through x and y. Then, $x + y$ is the third intersection of z and $\mathcal{O}$. Similarly, $x +_1 y$ is the third intersection of z and $\mathcal{O}_1$. We now calculate $(x +_1 y) + \mathcal{O}_1$. For this, we take the third intersection of $(x +_1 y)$ and $\mathcal{O}_1$, which is just z, then take the third intersection of that (z) with $\mathcal{O}$. This is precisely $x + y$. Therefore,

$$(x +_1 y) + \mathcal{O}_1 = x + y,$$

so we are done. □

Exercise 4.3.8. Let $\mathcal{O}, P$ be rational points on the nonsingular cubic C and suppose that

$$3P = \mathcal{O}$$

with respect to the group law based on $\mathcal{O}$. Let $Q = 2P$. Show that each side of the triangle $\mathcal{O}, P, Q$ meets the tangent to C of the opposite vertex at a point of C. Take $\mathcal{O}, P, Q$ as the triangle of reference and express this condition in terms of the coefficients of the cubic form determining C.

Solution: The fact that the tangent to P meets $\overrightarrow{\mathcal{O}Q}$ at a point of C comes from $Q = 2P$. Particularly, $2P$ is calculated by first taking the tangent and its intersection with C, say R. Then, we consider the line through $\mathcal{O}$ and R, which gives $2P = Q$. Thus, the tangent to P meets the line through $\mathcal{O}$ and Q at some point in C. Analogously, the same fact for Q and $\overrightarrow{\mathcal{O}P}$ is clear from $2Q = P$. As for the condition for $\mathcal{O}$ and $\overrightarrow{PQ}$, this follows from $P + Q = \mathcal{O}$. The first part is done.

Next for the second part. We can assume that $P = [1 : 0 : 0]$, $\mathcal{O} = [0 : 1 : 0]$, and $2P = Q = [0 : 0 : 1]$ by applying a projective change of coordinates. Let

$$F(X, Y, Z) = cXYZ + a_1XY^2 + a_2YZ^2 + a_3ZX^2 + b_1Z^2X + b_2X^2Y + b_3Y^2Z$$

Note that there are no terms corresponding to X^3, Y^3, Z^3 by the condition that F passes through $P, \mathcal{O}, Q$. Now, we have the lines

$$\ell_{\mathcal{O},Q} = \{X = 0\}, \quad \ell_{P,Q} = \{Y = 0\}, \quad \ell_{P,\mathcal{O}} = \{Z = 0\}$$

and the tangent lines

$$t_P = \{b_2Y + a_3Z = 0\}, \quad t_{\mathcal{O}} = \{a_1X + b_3Z = 0\}, \quad t_Q = \{b_1X + a_2Y = 0\}.$$

The intersections of these lines are

$$\begin{aligned} \ell_{\mathcal{O},Q} \cap t_P &= [0 : a_3 : -b_2], \\ \ell_{P,Q} \cap t_{\mathcal{O}} &= [b_3 : 0 : -a_1], \\ \ell_{P,\mathcal{O}} \cap t_Q &= [a_2 : -b_1 : 0]. \end{aligned}$$

The condition that these lie on C is equivalent to

$$a_1b_1 = a_2b_2 = a_3b_3,$$

which is solved for by plugging in the above values. For instance,

$$0 = F(0, a_3, -b_2) = a_2a_3b_2^2 - b_3a_3^2b_2 \implies a_2b_2 = a_3b_3.$$

Doing this for the other ones, we get the above. □

Exercise 4.3.9. Let E be an elliptic curve over some field k and $P, Q, R \in E(k)$ be three points that lie on a line. Let $P', Q', R' \in E(k)$ be the three points that are the third intersections with E of the tangent lines at P, Q, R respectively. Show that P', Q', R' lie on a line.

Solution: Later, we shall see that the elliptic curve can be brought into Weierstrass form, which has a unique smooth point, which is flex, at infinity. Thus, the condition says that $P + Q + R = \mathcal{O}$. Let P', Q', R' be the third intersections. Then, it must be that $P' = -P$, $Q' = -Q$, and $R' = -R$.

$$-P - Q - R = \mathcal{O},$$

so we are done. □

Exercise 4.3.10. Let C be the curve

$$X^3 + Y^3 - XZ^2 - YZ^2 + 7XYZ = 0$$

and let $\mathbf{x} = (x, y, z)$ be a point on C defined over some $\mathbb{Q}_p$. Show that $\frac{y}{x} \to -1$ as $\mathbf{x} \to (0, 0, 1)$ with respect to the p-adic topology.

Solution: Since this equation is homogeneous, we can begin by assuming that the maximum p-adic norm of x, y, z is 1. We wish to show that for all $M > 0$, there exists some $N > M$ such that if $|x|_p, |y|_p, |z - 1|_p < p^{-N}$, then $|\frac{y}{x} + 1|_p < p^{-M}$. We will show that $N = M$ works.

Now, let $x = p^{-N}x'$, $y = p^{-N}y'$, $z = p^{-N}z' + 1$. Plugging this into the cubic, we find that

$$p^{3N}x'^3 + p^{3N}y'^3 + 7p^{2N}x'y'(1 + p^N z') = (p^N x' + p^N y')(1 + p^N z')^2.$$

It follows that depending on whether $x \mid y$ or $y \mid x$, ither $p^N x \mid x + y$ or $p^N y \mid x + y$. Equivalently, either

$$|x + y|_p < p^{-N}|x|_p \text{ or } |x + y|_p < p^{-N}|y|_p.$$

In either case, we see that

$$|x + y|_p < |x|_p \text{ or } |x + y|_p < |y|_p,$$

which by the non-archimedean property implies that $|x|_p = |y|_p$. Hence the two possibilities are the same. Thus, we conclude that

$$\left|\frac{y}{x} + 1\right|_p < p^{-N}$$

as required. □

Exercise 4.3.11. In this question everything is defined over $\mathbb{Q}_p$ for some p. Let P be a nonsingular point on the cubic curve

$$F(X, Y, Z) = 0$$

and let $t(X) = 0$ be the tangent. Let $l(X) = 0$, $m(X) = 0$ be lines through P distinct from the tangent. Show that there are d, e, f such that

$$d\ell(X) + em(X) + ft(X) = 0$$

identically, with $d \neq 0$, $e \neq 0$. Show that

$$\frac{m(x)}{l(x)} \to -\frac{d}{e}$$

as $x \to P$.

§4.4 Weierstrass Normal Form

Lemma 4.3.5 was very nice, giving an algebraic appearance for the the group law. It would be nice if we could use this identity more generally. We switch our focus to the Weierstrass normal form, which is where we attain a flex point as we will shortly see. Then, we will see in Nagell's algorithm that so long as our cubic curve is irreducible and our field has characteristic not 2 with some point over the field, then it can be brought into Weierstrass normal form. In particular, if we have an elliptic curve (a smooth planar cubic, so is necessarily irreducible), then it can be brought into Weierstrass normal form.

Recall that we have defined a Weierstrass form a cubic over a general field k as given by an equation of the form:

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

Over a field k of characteristic not equal 2, we can complete the square and simplify the equation to

$$E\colon y^2 = x^3 + a_2x^2 + a_4x + a_6$$

Over a field k of characteristic not equal to 2 nor 3, we can also complete the cube and simplify the equation to

$$E\colon y^2 = x^3 + a_4x + a_6$$

Lemma 4.4.1

Let E be given in the Weierstrass form.

(i) This has a unique point at infinity, which is non-singular, namely $\mathcal{O} = [0 : 1 : 0]$.

(ii) The point at infinity is a point of inflection.

(iii) This curve is always irreducible, and if $\operatorname{char}(k) \neq 2$ is given in the form $y^2 = f(x)$ then it is non-singular if and only if $f(x)$ has distinct roots.

Proof. (i): $Z = 0$ implies that $X^3 = 0$, so $X = 0$, so $[0 : 1 : 0]$ is the unique point at infinity, and $\frac{\partial}{\partial Z} = Y^2$, so evaluated at $[0 : 1 : 0]$, we have $\frac{\partial}{\partial Z} |_{[0:1:0]} = 1^2 = 1 \neq 0$, so this is a smooth point.

(ii): $\frac{\partial}{\partial X} = \frac{\partial}{\partial Y} = 0$ at $[0 : 1 : 0]$, so the tangent line is just $\{Z = 0\}$. Thus, the only point of the curve on this is $[0 : 1 : 0]$, by (i). Thus, P is an inflection point.

(iii): Reducible cubic curves consist of three lines or a conic and a line. In any case, reducible would mean that the equation should be a product of a conic and a linear polynomial. It is easy by expanding out such a product to see that no such factorization is possible.

Any singular point is a point of $y^2 = f(x)$, by (i). Then $\frac{\partial}{\partial y} = 2y = 0$ and $\frac{\partial}{\partial x} = f'(x) = 0$, and $\mathrm{char}(k) \neq 2$ implies that $y = 0$ and $f'(x) = 0$, so $f(x) = 0$. Thus $f(x) = f'(x) = 0$ has a solution if and only if $f(x)$ has a repeated root, since $f(x) = (x-\alpha)(x-\beta)(x-\gamma)$. In particular, we see that there is at most one singular point. This also implies irreducibility, since reducible cubics have at least two singular points (corresponding to intersections of the components). □

§4.4.i The Discriminant

Definition 4.4.2. The **discriminant** of $f(x) = x^3+ax^2+bx+c = (x-e_1)(x-e_2)(x-e_3)$ is

$$\mathrm{Disc}(f) = (e_1-e_2)^2(e_1-e_3)^2(e_2-e_3)^2 = \prod_{i\neq j}(e_i-e_j)^2.$$

This is a way of getting a quantity that vanishes if and only if x^3+ax^2+bx+c has a double root, and the main point of taking the squares is that this quantity is expressed in terms of coefficients of the original polynomial (that is, we don't need to know the roots). The discriminant is computed (up to a sign) as the resultant of $f(x)$ and $f'(x)$. That is,

$$\mathrm{Disc}(f) = -\begin{vmatrix} 1 & a & b & c & 0 \\ 0 & 1 & a & b & c \\ 3 & 2a & b & 0 & 0 \\ 0 & 3 & 2a & b & 0 \\ 0 & 0 & 3 & 2a & b \end{vmatrix} = -4a^3c + a^2b^2 + 18abc - 4b^3 - 27c^2$$

Exercise 4.4.3. For a general monic polynomial $p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0 = 0$. Let $\{f_k\}_{k=1}^{n-1}$ be the roots of $p'(x)$. Show that

$$\mathrm{Disc}(p) = (-1)^{\frac{n(n-1)}{2}} n^n \prod_{k=1}^{n-1} p(f_k).$$

Solution: Let $p(x) = (x-e_1)(x-e_2)\ldots(x-e_n)$ and $p'(x) = n(x-f_1)(x-f_2)\ldots(x-f_{n-1})$. Then

$$\mathrm{Disc}(p) = \pm\prod_{i\neq j}\prod(e_i-e_j) = \pm\prod_i p'(e_i) = \pm n^n \prod_i\prod_k(e_i-f_k) = \pm n^n\prod_k p(f_k).$$

□

Definition 4.4.4. For an elliptic curve $E : y^2 = f(x) = (x-e_1)(x-e_2)(x-e_3)$, we define its **discriminant** Δ by

$$\Delta := 16\,\mathrm{Disc}(f) = 16(e_1-e_2)^2(e_1-e_3)^2(e_2-e_3)^2.$$

Remark 4.4.5 — The factor of 16 simplifies some calculations later on, but is not all that important.

Remark 4.4.6 — In particular, $y^2 = x^3 + ax^2 + bx + c$ is an elliptic curve if and only if $\Delta \neq 0$.

Remark 4.4.7 — Note that over the reals $f(x) = x^3 + ax^2 + bx + c$ always has at least one root. Whether or not all of e_1, e_2, e_3 are real is determined by the sign of Δ. If $\Delta > 0$ then there are 3 distinct real roots, and if $\Delta < 0$, there is only one. This is the reason that, by convention, there is a sign difference between the discriminant and resultant of f and f'.

Definition 4.4.8. Given a Weierstrass equation $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, an **admissible change of variables** is one of the form

$$x = u^2x' + r, \quad y = u^3y' + su^2x' + t$$

where $u, r, s, t \in k$ and $u \neq 0$.

Exercise 4.4.9. Show that such a change of variables preserves the point $[0 : 1 : 0]$ and the tangent line $Z = 0$ at this point. Conversely, show that any change of variables which preserves $[0 : 1 : 0]$ and its tangent line is an admissible change of variable.

Solution: In projective coordinates, an admissible change of variables is the following mapping:

$$[X' : Y' : Z'] \mapsto [u^2X' + rZ' : u^3Y' + su^2X' + tZ' : Z'].$$

We can check that $[0 : 1 : 0] \mapsto [0 : u^3 : 0] = [0 : 1 : 0]$, since $u \neq 0$. Further, if we apply this to a Weierstrass equation

$$Y^2 + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3,$$

(where really everything should be in X', Y', Z'), we get another Weierstrass form

$$Y^2 + a_1'XYZ + a_3'YZ^2 = X^3 + a_2'X^2Z + a_4'XZ^2 + a_6'Z^3,$$

where these coefficients are now altered. However, the tangent line is still the same, since this is essentially still the same partials are satisfied.

Now for the converse. Let a general projective linear change of coordinates be

$$\begin{aligned} X &= \alpha X' + \beta Y' + \gamma Z', \\ Y &= \delta X' + \varepsilon Y' + \zeta Z', \\ Z &= \eta X' + \theta Y' + \kappa Z'. \end{aligned}$$

Suppose this change preserves the point $\mathcal{O} = [0 : 1 : 0]$ and $Z = 0$. Preserving $\mathcal{O}$ means that the image of $[0 : 1 : 0]$ is again $[0 : 1 : 0]$, so $\beta = \theta = 0$ and $\varepsilon \neq 0$. Preserving the line $Z = 0$ means that this line is mapped to itself. Thus the new Z-coordinate must be a nonzero scalar multiple of Z'. Therefore

$$Z = \kappa Z', \quad \kappa \neq 0.$$

So the projective change has the form

$$\begin{aligned} X &= \alpha X' + \gamma Z', \\ Y &= \delta X' + \varepsilon Y' + \zeta Z', \\ Z &= \kappa Z'. \end{aligned}$$

This is of the form we want, so we just need to show that it has the appropriate scalar exponent. Passing to affine coordinates

$$x = \frac{X}{Z}, \quad y = \frac{Y}{Z},$$

we obtain

$$x = \frac{\alpha X' + \gamma Z'}{\kappa Z'} = \frac{\alpha}{\kappa} x' + \frac{\gamma}{\kappa},$$

and

$$y = \frac{\delta X' + \varepsilon Y' + \zeta Z'}{\kappa Z'} = \frac{\varepsilon}{\kappa} y' + \frac{\delta}{\kappa} x' + \frac{\zeta}{\kappa}.$$

Thus the affine change has the form

$$x = ax' + r, \quad y = by' + cx' + t,$$

where

$$a = \frac{\alpha}{\kappa}, \quad r = \frac{\gamma}{\kappa},$$

$$b = \frac{\varepsilon}{\kappa}, \quad c = \frac{\delta}{\kappa}, \quad t = \frac{\zeta}{\kappa}.$$

Since the change of coordinates is invertible, we have

$$a \neq 0, \quad b \neq 0.$$

Now use the fact that the equation remains in Weierstrass form. The original equation is

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Substitute

$$x = ax' + r, \quad y = by' + cx' + t.$$

The coefficient of y'^2 is b^2, while the coefficient of x'^3 is a^3. For the transformed equation to again be a normalized Weierstrass equation, these must agree up to the same scalar multiple. Hence $b^2 = a^3$.Defining $u = \frac{b}{a}$, we have that

$$u^2 = \frac{b^2}{a^2} = \frac{a^3}{a^2} = a,$$

and

$$u^3 = u \cdot u^2 = \frac{b}{a} \cdot a = b.$$

Therefore $a = u^2$, and $b = u^3$. Finally, define $s = \frac{c}{u^2}$. Then $c = su^2$. Hence the change becomes

$$x = u^2x' + r, \quad y = u^3y' + su^2x' + t.$$

Therefore the change of variables is admissible. □

From the exercise above, we obtain another Weierstrass equation

$$y'^2 + a_1'x'y' + a_3'y' = x'^3 + a_2'x'^2 + a_4'x' + a_6'$$

which is related to the original equation by the rules

$$\begin{aligned}
ua_1' &= a_1 + 2s, \\
u^2a_2' &= a_2 - sa_1 + 3r - s^2, \\
u^3a_3' &= a_3 + ra_1 + 2t, \\
u^4a_4' &= a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st, \\
u^6a_6' &= a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1.
\end{aligned}$$

In the special case of $y^2 = x^3 + a_2x^2 + a_4x + a_6$ and $r = s = t = 0$. We have

$$y = u^3y', \quad x = u^2x', \quad u \neq 0,$$

then $u^6y'^2 = u^6x'^3 + u^4a_2x'^2 + u^2a_4x' + a_6$, so $y'^2 = x'^3 + a_2'x'^2 + a_4'x' + a_6'$. This takes

$$(a_2, a_4, a_6) \leftrightarrow (u^2a_2', u^4a_4', u^6a_6'), \quad \Delta \leftrightarrow u^{12}\Delta'.$$

These transformation rules clarify the reasoning behind the standard notation for a cubic in the Weierstrass form which emphasizes how the coefficients get transformed.

Definition 4.4.10. The **j-invariant** of $y^2 = x^3 + bx + c$ is

$$j = 1728\frac{4b^3}{4b^3 + 27c^2}.$$

Note that j is invariant under $(x, y) \leftrightarrow (u^2x, u^3y)$. The normalization of j may vary between different resources. The constant 1728 is suggested by the complex variable theory.

Proposition 4.4.11

If k is algebraically closed, then two elliptic curves of the form $y^2 = x^3 + bx + c$ are isomorphic if and only if they have the same j-invariant.

Proof. Assume $\text{char}(k) \neq 2, 3$. Suppose that we have two elliptic curves $y^2 = x^3 + bx + c$ and $y^2 = x^3 + b'x + c'$. These have the same j-invariants if and only if

$$\frac{b^3}{4b^3 + 27c^2} = \frac{b'^3}{4b'^3 + 27c'^2},$$

if and only if

$$\frac{c^2}{b^3} = \frac{c'^2}{b'^3}.$$

Now, if two curves in Weierstrass form are isomorphic then they are related by an admissible change of variables. Hence, the j-invariants are equal, as $(x, y) \leftrightarrow (u^2x, u^3y)$ does not change the j-invariant.

Conversely, assume $\frac{c^2}{b^3} = \frac{c'^2}{b'^3}$, that is $\left(\frac{b'}{b}\right)^3 = \left(\frac{c'}{c}\right)^2$. Since k is algebraically closed, there exists u such that $u^4 = \frac{b'}{b}$ and $u^6 = \frac{c'}{c}$. Then the transformation $(x, y) \leftrightarrow (u^2x, u^3y)$ takes $y^2 = x^3 + bx + c \leftrightarrow y^2 = x^3 + b'x + c'$. Implicitly assuming that $b, c, b', c' \neq 0$. More correctly, the equation is $c^2b'^3 = c'^2b^3$. If $b = 0$ then $c \neq 0$, as the curve is non-singular, so $b' = 0$, so $c' \neq 0$, just choose $u^6 = \frac{c'}{c}$. Similarly, if $c = 0$ then $b \neq 0$, so $c' = 0$ and $b' \neq 0$, and choose $u^4 = \frac{b'}{b}$. □

Remark 4.4.12 — This 'isomorphic' means isomorphic as algebraic varieties, or as function fields. A more elementary relation is to say that they're related under an admissible change of coordinates.

Example 4.4.13 (*j*-invariant is Surjective)

For $j_0 \neq 0, 1728$, consider the curve

$$E\colon y^2 = x^3 + \frac{3j_0}{1728 - j_0}x + \frac{2j_0}{1728 - j_0}.$$

An easy calculation shows that $j = j_0$. We also have $E\colon y^2 = x^3 + a_4 x$ with $j = 1728$, and $E\colon y^2 = x^3 + a_6$ with $j = 0$. Hence, we see that any $j_0 \in \bar{k}$ is the j-invariant of some elliptic curve.

Remark 4.4.14 — Note that j-invariant tells us when two elliptic curves are isomorphic over $\bar{k}$. Beware that if we work over $\mathbb{Q}$ (say) there are elliptic curves that have the same j-invariant but not isomorphic over $\mathbb{Q}$, such as the following.

Example 4.4.15

For example, $y^2 = x^3 - 25x$ and $y^2 = x^3 - 4x$ both have $j = 1728$ but are not isomorphic over $\mathbb{Q}$. The reason is that the former has the rational point $(-4, 6)$ which gives a non-torsion point in $E(\mathbb{Q})$ whereas the latter has only torsion points. (We may see a proof of these later in the course.)

Exercise 4.4.16. Show that the set of isomorphism classes of elliptic curves E over $\mathbb{Q}$ with fixed j-invariant $j = j_0 \in \mathbb{Q}$ and $j_0 \neq 0, 1728$ is in bijection with the group $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$. For $j_0 = 0$ and $j_0 = 1728$, the bijection is with $\mathbb{Q}^\times/(\mathbb{Q}^\times)^6$ and $\mathbb{Q}^\times/(\mathbb{Q}^\times)^4$, respectively.

Solution: We note that two elliptic curves have the same j-invariant if the two elliptic curves are of the form

$$y^2 = x^3 + ax + b, \quad y^2 = x^3 + cx + d,$$

where $d \in \mathbb{Q}$. If these two curves have the same j-invariant, this implies that

$$\frac{c^3}{d^2} = \frac{a^3}{b^2},$$

which provides

$$\left(\frac{c}{a}\right)^3 = \left(\frac{d}{b}\right)^2.$$

Thus, we can parametrize these values, by let's say z, so that $\frac{c}{a} = z^2$ and $\frac{d}{b} = z^3$. Therefore, $c = z^2 a$ and $d = z^3 b$. The question is now interested in when E_z and $E_{z'}$ are isomorphic, where z, z' are two isomorphism classes. Now, two elliptic curves (such as the above) are isomorphic if and only if $a = cu^4$ and $b = du^6$, for $u \in \mathbb{Q}$. Now consider

$$y^2 = x^3 + ax + b,$$

so that the following are two different elliptic curves, but with the same j-invariant.

$$y^2 = x^3 + az^2x + z^3b, \quad y^2 = x^3 + a(z')^2x + (z')^3b.$$

If these are isomorphic, then we must have that

$$az^2 = a(z')^2u^4, \quad z^3b = (z')^3bu^6.$$

This implies that $\frac{z}{z'} = u^2$, so $\frac{z}{z'} \in (\mathbb{Q}^\times)^2$.

Now we consider when the j-invariant is 0. In this case, we already have an elliptic curve $E_a \colon y^2 = x^3 + a$. Mimicking the above, we have that two elliptic curves E_a, E_b of this form are isomorphic if and only if $a = bu^6$, for some $u \in \mathbb{Q}^\times$. Therefore, it follows that $\frac{a}{b} = u^6$, and so the isomorphism classes are in bijection with $\mathbb{Q}^\times/(\mathbb{Q}^\times)^6$.

Next is when the j-invariant is 1728. In this case, we already have an elliptic curve $E_a \colon y^2 = x^3 + ax$. Then, E_a, E_b are isomorphic if and only if $a = bu^4$, for some $u \in \mathbb{Q}^\times$, so the isomorphism classes are in bijection with $\mathbb{Q}^\times/(\mathbb{Q}^\times)^4$. □

Exercise 4.4.17. Let F be a cubic polynomial defining an elliptic curve E with j-invariant $j \neq 0$, compute the j-invariant of the cubic given by the Hessian H_F.

We next show that every plane cubic with a rational point over k can be brought into a Weierstrass form over k. (Note that previously we proved this over $\bar{k}$.)

Theorem 4.4.18 (Nagell's algorithm)

Any irreducible cubic curve over an arbitrary field k of char$k \neq 2$ with a given smooth point $\mathcal{O}$ is birational to a cubic in the Weierstrass form

$$y^2 = f(x)$$

such that $\mathcal{O}$ corresponds to the unique point $[0 : 1 : 0]$ at infinity.

Remark 4.4.19 — Here **birational** means, more precisely, the following. Given an irreducible cubic defined by F. There is a rational map $\mathbb{P}^2_k \to \mathbb{P}^2_k$, that is, a map defined on a Zariski open set of $\mathbb{P}^2_k$ by $\phi : [X : Y : Z] \to [P(X, Y, Z) : Q(X, Y, Z) : R(X, Y, Z)]$ where P, Q, R are homogeneous polynomials of the same degree. Such a map ϕ is not defined on the **base locus** $B(\phi)$ (where all of P, Q, R vanish simultaneously). However, we can consider the image of $C_F \setminus (C_F \cap B(\phi))$ under ϕ where ϕ is defined, and take the Zariski closure of the image. For the appropriate choice of ϕ, this leads to an isomorphic elliptic curve that is embedded differently to $\mathbb{P}^2_k$ in a way that is now cut out by a Weierstrass equation.

Proof of Theorem 4.4.18. We give a sketch. First draw the tangent line to the cubic at $\mathcal{O}$. The tangent line will meet the cubic at three points, at least two of which will be $\mathcal{O}$.

Case 1: The third point of intersection is also $\mathcal{O}$, that is, $\mathcal{O}$ is an inflection point. In this case, apply a linear change of co-ordinates by a linear transformation that sends $\mathcal{O}$ to $[0 : 1 : 0]$ and the tangent line to the line $Z = 0$ in $\mathbb{P}^2_k$, and we see as in the proof of Corollary 2.2.35 that the resulting equation has to be in the Weierstrass form.

Case 2: Say the third point of intersection is $P \neq \mathcal{O}$. Make a linear change of variables so that $P = (0, 0)$ and the tangent line to $\mathcal{O}$ is the y-axis in the affine plane $\mathbb{A}^2$ with (x, y)-coordinates. Suppose the affine equation of the cubic is given by $f(x, y) = 0$ in these co-ordinates. The fact that $(0, 0)$ lies on the cubic implies that

$$f(x, y) = f_1(x, y) + f_2(x, y) + f_3(x, y),$$

where $f_i(x, y)$ is homogeneous of degree i. We know that the intersection of $f = 0$ with the line $x = 0$ (that is, y-axis) is $(0, 0)$ with multiplicity 1 and some other point (namely, $\mathcal{O}$) with multiplicity 2. Hence, setting $x = 0$, we get that equation

$$f(0, y) = yf_1(0, 1) + y^2 f_2(0, 1) + y^3 f_3(0, 1),$$

which must have a root at the origin and a double root at another point $\mathcal{O}$. Thus, the quadratic

$$f_1(0, 1) + yf_2(0, 1) + y^2 f_3(0, 1)$$

has a double root, which is to say that the discriminant vanishes. Hence, $f_2(0, 1)^2 = 4f_1(0, 1)f_3(0, 1)$.

Now, draw lines $y = tx$ through the origin (think of t as a constant). This line meets the curve at 3 points whose x-coordinates are the roots of

$$xf_1(1, t) + x^2 f_2(1, t) + x^3 f_3(1, t) = 0.$$

Now, this line hits a k-point other than $(0, 0)$ if and only if the quadratic $x^2 f_3(1, t) + xf_2(1, t) + f_1(1, t)$ has roots in k, which happens if and only if $f_2(1, t)^2 - 4f_1(1, t)f_3(1, t)$ is a square in k. Hence, we write

$$s^2 = f_2(1, t)^2 - 4f_1(1, t)f_3(1, t).$$

We're almost at the form we want. The formula for the roots of the quadratic equation gives us, for every solution to this equation, a root of $x^2 f_3(1, t) + xf_2(1, t) + f_1(1, t)$, namely

$$x = (-f_2(1, t) + s)/2f_3(1, t).$$

Now, we claim though that the coefficient of t^4 is 0. Defining $G(t) := f_2(1, t)^2 - 4f_1(1, t)f_3(1, t)$, this looks like an equation of degree 4 in t, but is actually of degree 3 because the coefficient of t^4 is $f_2(0, 1)^2 - 4f_1(0, 1)f_3(0, 1)$ which we know is zero. Hence the cubic has now become an equation of the form

$$s^2 = G(t)$$

with G a cubic, which is what we were after. □

Remark 4.4.20 — The t-value such that $f_1(1, t) = 0$ corresponds to the slope of the tangent line at $(0, 0)$. Check that under the constructed isomorphism, P goes (s, t) such that $f_1(1, t) = 0$, and $s = f_2(1, t)$, and $\mathcal{O}$ goes to $[0 : 1 : 0]$. On the other hand, if there are t-values in k such that $f_3(1, t) = 0$, these correspond to points at infinity of the cubic $f(x, y) = 0$, and they go to (s, t) such that $f_3(1, t) = 0$ and $s = f_2(1, t)$.

Exercise 4.4.21. Consider the elliptic curves $X^3 + Y^3 + dZ^3 = 0$ over a field k with $\operatorname{char} k \neq 2, 3$ with a given point $P = [1 : -1 : 0]$ and $d \in k$. Transform these to the Weierstrass canonical form such that $\mathcal{O}$ becomes the point at infinity.

Solution: We first check if P is an inflection point. What we need to do is look at the tangent line and consider if this intersects the elliptic curve again. The tangent line at $[1 : -1 : 0]$ is $X + Y = 0$. We observe that this intersects the curve at only $P = [1 : -1 : 0]$.

Thus, P is an inflection point!

We now want to find a transformation that sends P to $O = [0 : 1 : 0]$ and the line $X + Y = 0$ to $Z = 0$. We use the transformation $[X : Y : Z] \mapsto [Y + Z : Z - Y : X]$ which on points maps $[x : y : z] \mapsto [2z : x - y : x + y]$, hence it sends P to O. Note that if we apply a change on coordinate unctions, then the coordinate change on points is the matrix inverse. The equation then becomes

$$(Y + Z)^3 + (Z - Y)^3 + dX^3 = 2Z^3 + 6ZY^2 + dX^3 = 0$$

The affine equation then becomes

$$6y^2 = -dx^3 - 2$$

We can adjust the constants by multiplying with 6^3d^2, which gives the equation

$$6^4d^2y^2 = -6^3d^3x^3 - 26^3d^2.$$

We then let $(-6dx, 6^2dy) \mapsto (x, y)$, then the equation becomes

$$y^2 = x^3 - 2^4 3^3 d^2$$

which is the desired Weierstrass form. The full projective transformation is

$$[x : y : z] \mapsto [-12dz : 36d(z - y) : x + y].$$

$\square$

> **Exercise 4.4.22.** Let $F(X, Y, Z) = X^2Y + XZ^2 + 2Y^3 + Z^3$. Find a birational transformation defined over $\mathbb{Q}$ taking the curve $F = 0$ into canonical form with the point $[1 : 0 : 0]$ going to the point at infinity.

Solution: We take $\mathcal{O} = [1 : 0 : 0]$. Check that this is not an inflection point. We compute the derivatives

$$\frac{\partial F}{\partial X} = 2XY + Z^2, \quad \frac{\partial F}{\partial Y} = X^2 + 6Y^2, \quad \frac{\partial F}{\partial Z} = 2XZ + 3Z^2$$

We see that the tangent line $t(\mathcal{O}) = Y = 0$, and we see that intersects the curve defined by F at another point $P = [1 : 0 : -1]$, if we consider the equation

$$F(X, 0, Z) = Z^2(X + Z).$$

We want to work in an affine chart where both these points are visible and P is at the origin and $\mathcal{O}$ is in the y-axis. We achieve this using the transformation given by the point-wise map

$$[X : Y : Z] \mapsto [Y : X + Z : X].$$

This sends $\mathcal{O}$ to $[0 : 1 : 1]$ and P to $[0 : 0 : 1]$. Coordinate-wise, we apply the transformation $[X : Y : Z] \to [Z : X : Y - Z]$ to F. Then, we see that the tangent line $Y = 0$ is now $X = 0$. Note that the tangent line is therefore the y-axis in the affine coordinates. Applying this change of coordinates, we find the new equation

$$\begin{aligned} 0 = F(X, Y, Z) &= Z^2X + Z(Y - Z)^2 + 2X^3 + (Y - Z)^3 \\ &= 2X^3 + Y^3 - 2Y^2Z + YZ^2 + XZ^2 \end{aligned}$$

We can now apply the method given above. Setting $Z = 1$ and reorganizing according to degree gives

$$F(X,Y) = F_3(X,Y) + F_2(X,Y) + F_1(X,Y)$$

with $F_3(X,Y) = 2X^3 + Y^3$, $F_2(X,Y) = -2Y^2$ and $F_1(X,Y) = X + Y$. Next, we set $s^2 = F_2(1,t)^2 - 4F_1(1,t)F_3(1,t)$. This gives the equation

$$s^2 = 4t^4 - 4(1+t)(2+t^3).$$

The degree 4 terms vanish (as expected), giving

$$s^2 = -4t^3 - 8t - 8.$$

We just need to rescale the coefficient on t^3 now. Letting $x = -t$ and $y = \frac{s}{2}$, we arrive at

$$y^2 = x^3 + 2x - 2$$

Chasing through the transformations to verify that if $[x : y : z]$ satisfies the equation

$$y^2z = x^3 + 2xz^2 - 2z^3,$$

then

$$[X : Y : Z] = [2xz - y^2 : yz + x^2 : -2z^2 - xy]$$

satisfies the original equation $X^2Y + XZ^2 + 2Y^3 + Z^3 = 0$.

We can always check our calculation in GP. We get the Weierstrass equation from a cubic equation $F(x,y)$ as follows:

```
ellfromeqn(x^2*y + y*x + 2*y^3 + 3)
```

This returns

$$[0, 0, 0, 2, -2].$$

□

Exercise 4.4.23. Consider the cubic curve $C\colon X^3 + 2Y^3 + 4Z^3 - 7XYZ = 0$. Show that none of the inflection points of C are defined over $\mathbb{Q}$. Transform C into the Weierstrass form. Can you find the base locus $B(\phi)$ of your transformation? Which points on the Weierstrass form do these correspond to?

Solution: First, we note that there are no singular points. We can calculate the partials of the cubic as follows:

$$F_X = 3X^2 - 7YZ, \quad F_Y = 6Y^2 - 7XZ, \quad F_Z = 12Z^2 - 7XY.$$

If a point was singular, these would all vanish, so $3X^3 = 6Y^3 = 12Z^3 = 7XYZ$, and then plugging into the original cubic, we see there are no solutions. To calculate the absence of flex points, we use the Hessian.

$$H = \det \begin{pmatrix} 6X & -7Z & -7Y \\ -7Z & 12Y & -7X \\ -7Y & -7X & 24Z \end{pmatrix},$$

so

$$H = -588X^3 - 1176Y^3 - 2352Z^3 - 686XYZ.$$

We want to show that there are no points in $\mathbb{Q}$ satisfying both the Hessian and F. To simplify H, we observe that 98 divides all coefficients. This gives

$$0 = 6X^3 + 12Y^3 + 24Z^3 + 7XYZ.$$

Then, it is clearly not possible for a point to lie on both curves. We have that $X^3 + 2Y^3 + 4Z^3 = 7XYZ$, so we have that a point lies on both curves if the following is satisfied:

$$42XYZ + 7XYZ = 49XYZ = 0.$$

Therefore, $X = 0$, $Y = 0$, or $Z = 0$. If any one of these are zero, then we are left to solve a two variables cubic. Since we're finding solutions over $\mathbb{Q}$, we can just assume we are solving for coprime integer solutions. However, the power of 2 is not a cubic, so we can never find a solution pair in the remaining two variables. Thus, there are no inflection points.

Now, let's put this into Weierstrass form. Note that $P = [1 : 1 : 1]$ lies on C. Then, if we calculate the tangent line, we find that this is $-4X - Y + 5Z = 0$. Intersecting this with the cubic C again, we find the remaining intersection point as $[2 : -3 : 1] = O$. We want to map O to the origin $[0 : 0 : 1]$ and P to $[0 : 1 : 1]$, and the tangent line to $X = 0$. If we use these conditions, then we find that the coordinate change $[X : Y : Z] \to [-Y + 2Z : -X + 4Y - 3Z : Z]$ works. We can check that all the conditions indeed satisfy this. Applying this transformation to the original cubic, this spits out

$$-2X^3 + 24X^2Y - 18X^2 - 96XY^2 + 137XY - 40X + 127Y^3 - 254Y^2 + 127Y = 0.$$

Therefore,

$$\begin{aligned}
F_3 &= -2X^3 + 24X^2Y - 96XY^2 + 127Y^3, \\
F_2 &= -18X^2 + 137XY - 254Y^2, \\
F_1 &= -40X + 127Y.
\end{aligned}$$

Thus, we may set

$$s = 2F_3(1, t)x + F_2(1, t), \quad G(t) = F_2(1, t)^2 - 4F_1(1, t)F_3(1, t).$$

This gives $s^2 = G(t)$, which is the following:

$$\begin{aligned}
F_3(1, t) &= -2 + 24t - 96t^2 + 127t^3, \\
F_2(1, t) &= -18 + 137t - 254t^2, \\
F_1(1, t) &= -40 + 127t.
\end{aligned}$$

Therefore,

$$s^2 = -508t^3 + 361t^2 - 76t + 4.$$

Now, we find the base locus of our transformation. Recall that we first applied $[X : Y : Z] \to [-Y + 2Z : -X + 4Y - 3Z : Z]$, then went into the affine coordinates where $Y = tX$. Note that here, we set $Z = 1$, and the Nagell map is

$$[X : Y] \to \left[t = \frac{Y}{X} : s = 2F_3\left(1, \frac{Y}{X}\right) + F_2(1, \frac{Y}{X})\right].$$

Now, this last map vanishes if $X = Y = 0$. To see what point maps to $(0, 0)$, we find the point-wise inversion of the first map. The pointwise map of the inverse is $[X : Y : Z] \mapsto [-Y + 2Z : -X + 4Y - 3Z : Z]$, so $[0 : 0 : 1] \mapsto [2 : -3 : 1]$. So this is the base locus. $\square$

Remark 4.4.24 — Moving forward, we will usually assume that the characteristic of the base field k will not be 2. We may also not assume that the characteristic 2 or 3.

Exercise 4.4.25. Transform the following to canonical form.

$$X^3 + Y^3 + Z^3 - 3mXYZ = 0, \quad m \in k.$$

Solution: We note that $[1 : -1 : 0]$ lies on the curve. We check if this is flex or not. Let F be the cubic on the LHS. The derivatives are

$$\frac{\partial F}{\partial X} = 3X^2 - 3mYZ, \quad \frac{\partial F}{\partial Y} = 3Y^2 - 3mXZ, \quad \frac{\partial F}{\partial Z} = 3Z^2 - 3mXY.$$

Then, it is clear that if $m^3 = 1$, we get singular points, so we avoid these. Further, the tangent line to P is $X + Y + mZ = 0$. We apply this relation to F to get that

$$(m^3 - 1)(X + Y)^3 = 0.$$

Thus, this is a flex point. What we want to do is then map P to $[0 : 1 : 0]$ and the tangent line to $Z = 0$. One such transformation is the mapping

$$[X : Y : Z] \to [Y : -mX - Y + Z : X].$$

It is clear that the tangent line maps to $Z = 0$. As for P, this maps $[1 : -1 : 0] \mapsto [0 : 1 : 0]$, since the point wise map is $x \mapsto z$, $y \mapsto x$, $z \mapsto y + mz + x$. Applying this coordinate change to F, we find the following:

$$Y^3 + (-mX - Y + Z)^3 + X^3 = 3mXY(-mX - Y + Z)^3$$

This simplifies to the following:

$$(m^3 - 1)X^3 - 3m^2X^2Z + 3mXZ^2 - Z^3 = 3Y^2Z + 3mXYZ - 3YZ^2.$$

We first multiply by -1 and multiply by $3^3(m^3 - 1)$. Then, we coordinate change via $[(m^3 - 1)X : 3^2(m^3 - 1)Y : Z]$.

$$Y^2Z + 9mXYZ - 9(m^3 - 1)YZ^2 = 27X^3 - 81m^2X^2Z + 81m(m^3 - 1)XZ^2 - 27(m^3 - 1)^2Z^3.$$

If $\operatorname{char} k \neq 3$, we can simplify this further via $[X : Y : Z] \to \left[\frac{X}{3} : Y - \frac{9m}{2}X + \frac{9(m^3-1)}{2}Z : Z\right]$. This gives the following:

$$Y^2Z = X^3 - \frac{27m^2}{4}X^2Z + \frac{27m(m^3 - 1)}{2}XZ^2 - \frac{27(m^3 - 1)^2}{4}Z^3.$$

If I also go through $[X : Y : Z] \to [X + \frac{9m^2}{4}Z : Y : Z]$, then we can cancel the quadratic X term to get the following:

$$Y^2Z = X^3 - \frac{27m(m^3 + 8)}{16}XZ^2 + \frac{27(m^6 - 20m^3 - 8)}{32}Z^3.$$

Multiplying by 2^6 and rescaling X, Y accordingly, we find the following:

$$Y^2Z = X^3 - 27m(m^3 + 8)XZ^2 + 54(m^6 - 20m^3 - 8)Z^3.$$

We can plug in $Z = 1$ to get the affine version too. □

Exercise 4.4.26. Transform the following to canonical form.

$$Y^2 - kT^2 = X^2, \quad Y^2 + kT^2 = Z^2$$

Solution: This is an intersection of quadratics, so we bring it into the pure quadratic plus mixed term form. Consider the transformation $[X : Y : Z : T] \mapsto [X + T : T : Z + T : Y]$. This gives

$$X^2 + kY^2 + 2XT = 0, \quad kY^2 - Z^2 - 2ZT = 0.$$

Now, we can eliminate T and bring it into the following form:

$$kY^2Z + kXY^2 = -X^2Z + XZ^2.$$

We have the rational point $[0 : 1 : 0]$ lands on the curve. We check if this is flex or not. The tangent line is $X + Z = 0$, and this doesn't provide any new solutions, so this point is flex. This point is already in the desirable position, so all we wish to do is send its tangent line to $Z = 0$. For this, we map $Z \mapsto Z - X$. This provides the following cubic:

$$kY^2Z = -X^2(Z - X) + X(Z - X)^2 \implies kY^2Z = 2X^3 - 3X^2Z + XZ^2.$$

Multiplying the entire equation by $4k^3$ and rescaling X and Y, we arrive at the Weierstrass form as below:

$$Y^2Z = X^3 - 3kX^2Z + 2k^2XZ^2.$$

If we wish to further simplify, we map $X \mapsto X + 3kZ$ to get the following:

$$Y^2Z = X^3 - k^2XZ.$$

□

Exercise 4.4.27. Transform the following to canonical form.

$$X_1^2X_2 - X_1X_2^2 - X_1X_3^2 + X_2^2X_3 = 0$$

Solution: Map $[X_1 : X_2 : X_3] \mapsto [Y : X : Z]$. We consider

$$Y^2X - YX^2 - YZ^2 + X^2Z = 0.$$

We note that $P = [0 : 1 : 0]$ is on this cubic. Calculating the partials, we find that the tangent line to this point is $X = 0$. This tangent intersects the cubic again at $[0 : 0 : 1]$. Now, we want to bring these points so that the tangent remains at the x axis but the point P is now on the x-axis. For this, we apply the coordinate change $[X : Y : Z] \mapsto [X : Y : Z - Y]$. This maps P to $[0 : 1 : 1]$. Furthermore, we have as follows:

$$Y^2X - YX^2 - Y(Z - Y)^2 + X^2(Z - Y) = 0$$

The next step in Nagell's algorithm is to set $Z = 1$. We then reorganize this using by degree;

$$\begin{aligned}
F_3(X, Y) &= -Y^3 + Y^2X - 2YX2 \\
F_2(X, Y) &= X^2 + 2Y^2 \\
F_1(X, Y) &= -Y
\end{aligned}$$

As in the proof for Nagell's algorithm, we set

$$s = 2F_3(1, t)x + F_2(1, t)$$

and

$$G(t) = F_2(1,t)^2 - 4F_1(1,t)F_3(1,t).$$

Then, we find that our curve is equivalent to $s^2 = G(t)$. Then, we arrive at

$$s^2 = 4t^3 - 4t^2 + 1.$$

Rescaling and relabeling then gives

$$y^2 = x^3 - 4x^2 + 16.$$

Finally, we can eliminate the x^2 term via $x \mapsto \frac{4}{3}x$ to attain

$$y^2 = x^3 - \frac{16}{3}x + \frac{16 \cdot 19}{27}.$$

Then, we rescale finally to attain:

$$y^2 = x^3 - 2^4 \cdot 3^3 x + 2^4 \cdot 3 \cdot 19.$$

□

§4.4.ii Explicit Formulae

When an elliptic curve is given in the Weierstrass form, we can compute formulae that express the group law quite explicitly. These formulae can then be used to give another, more explicit, proof of the associativity law for the group operation. We consider an elliptic curve given by

$$E\colon y^2 = x^3 + ax^2 + bx + c$$

First, let us note that if $P = (x_0, y_0)$, then the line through P and $\mathcal{O}$ is $x = x_0$, so $-P = (x_0, -y_0)$.

Now, given two points $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$ on E, we're interested in finding the coordinates of $P_3 = P_1 + P_2 = (x_3, y_3)$. That is, we want to find the unique point $-P_3 = (x_3, -y_3)$ on E such that P_1, P_2, and $-P_3$ are collinear. If $P_1 \neq P_2$, then we write the equation of the line passing through P_1 and P_2 by

$$y = \lambda x + \nu$$

where

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{and} \quad \nu = y_1 - \lambda x_1 = y_2 - \lambda x_2.$$

Note that $-P_3 = (x_3, -y_3)$. The group law says that $-P_3$, P_1, and P_2 all lie on the same line. Now, we plug this line equation into the equation of the cubic to get

$$(\lambda x + \nu)^2 = x^3 + ax^2 + bx + c$$

This is a cubic equation in x with roots x_1, x_2, x_3. Hence, we have

$$x^3 + (a - \lambda^2)x^2 + (b - 2\lambda\nu)x + c - \nu^2 = (x - x_1)(x - x_2)(x - x_3)$$

From which we conclude that

$$x_1 + x_2 + x_3 = \lambda^2 - a$$

or

$$x_3 = -x_1 - x_2 - a + \lambda^2, \quad y_3 = -\lambda x_3 - \nu$$

The case of $P_1 = P_2$ can be worked out in a similar fashion, by using the tangent line at $P_1 = P_2$ instead:

$$x_3 = -2x_1 - a + \left(\frac{3x_1^2 + 2ax_1 + b}{2y_1}\right)^2, \quad y_3 = -y_1 - \frac{3x_1^2 + 2ax_1 + b}{2y_1}(x_3 - x_1)$$

To summarize, we have the following formulae:

- If $x_1 \neq x_2$:

$$x_3 = -x_1 - x_2 - a + \left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2, \quad y_3 = -y_1 - \frac{y_2 - y_1}{x_2 - x_1}(x_3 - x_1)$$

- If $(x_1, y_1) = (x_2, y_2)$:

$$x_3 = -2x_1 - a + \left(\frac{3x_1^2 + 2ax_1 + b}{2y_1}\right)^2, \quad y_3 = -y_1 - \frac{3x_1^2 + 2ax_1 + b}{2y_1}(x_3 - x_1)$$

- If $(x_1, y_1) = (x_2, -y_2)$:

$$P_3 = \mathcal{O}.$$

In particular, we get that if $P = (x, y)$, then the x-coordinate of $2P$ is given by

$$x' = \frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4(x^3 + ax^2 + bx + c)} = \frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4y^2}.$$

This is known as the **duplication formula**. Note that we may also rewrite this as

$$x' = -2x - a + \left(\frac{f'(x)}{2y}\right)^2$$

We can write the y-coordinate of $2P$ by

$$y' = \frac{f'(x)}{2y}(x - x') - y$$

§4.4.iii Group Laws for Singular Curves

Example 4.4.28 (Cusp Curves)

Let E be the cubic curve $y^2 = x^3$ over a field k with char$k \neq 2, 3$. The curve has a cusp at $(0, 0)$, which is the only singular point. Let $\mathcal{O} = [0 : 1 : 0]$ be the point at infinity.

We find the group structure on E_{ns}. By Lemma 4.3.5 three points $P, Q, R \in E_{\text{ns}}$ sum to the identity $\mathcal{O}$ if and only if P, Q, R are collinear. Consider a line L given by $ax + by = 1$, which does not pass through the singular point $(0, 0)$. We want to find the intersection points $L \cap E$. To do this, we parameterize the non-singular points. Let $u = \frac{x}{y}$. From $y^2 = x^3$, we substitute $x = uy$, giving $y^2 = (uy)^3 = u^3y^3$. Since we are not at $(0, 0)$, we can assume $y \neq 0$. Dividing by y^2 gives $1 = u^3y$, so $y = \frac{1}{u^3}$. This also gives $x = uy = u(\frac{1}{u^3}) = \frac{1}{u^2}$. Now, substitute $x = 1/u^2$ and $y = 1/u^3$ into the line equation $ax + by = 1$:

$$a\left(\frac{1}{u^2}\right) + b\left(\frac{1}{u^3}\right) = 1$$

Multiplying by u^3 (since $u = 0$ would imply $x \to \infty$ or $y \to \infty$), we obtain

$$u^3 - au - b = 0$$

The three roots u_1, u_2, u_3 of this cubic polynomial correspond to the three intersection points P, Q, R. Since the u^2 coefficient is 0, we have:

$$u_1 + u_2 + u_3 = 0$$

This shows that the group operation on the cubic E_{ns} corresponds to addition in the field k under the parametrization $u = x/y$. We have a group isomorphism:

$$\varphi\colon E_{ns} \to (k, +), \quad (x, y) \mapsto \frac{x}{y}.$$

Example 4.4.29 (Nodal Curves)
Let E be the cubic $y^2 = x^3 + x^2$ over a field k with $\operatorname{char} k \neq 2$. This curve has a node (a self-intersection) at $(0,0)$. The non-singular points $E_{ns} = E \setminus \{(0,0)\}$ again form a group.

The two tangent directions at the node are given by $y^2 = x^2$, i.e., $y = x$ and $y = -x$. This suggests the substitution $t = \dfrac{y+x}{y-x}$. From $t(y-x) = y+x$, we get $y(t-1) = x(t+1)$. This gives $\dfrac{y}{x} = \dfrac{t+1}{t-1}$. From the curve equation $y^2 = x^2(x+1)$, we have $\left(\dfrac{y}{x}\right)^2 = x+1 \quad (\text{for } x \neq 0)$. Substituting our ratio: $\left(\dfrac{t+1}{t-1}\right)^2 = x+1$. This gives

$$x = \left(\frac{t+1}{t-1}\right)^2 - 1 = \frac{(t+1)^2 - (t-1)^2}{(t-1)^2} = \frac{4t}{(t-1)^2}.$$

We also have

$$y = x\left(\frac{t+1}{t-1}\right) = \frac{4t}{(t-1)^2} \cdot \frac{t+1}{t-1} = \frac{4t(t+1)}{(t-1)^3}.$$

Now, consider the intersection of E_{ns} with a line L given by $ax + by = 1$. Substituting the parametric equations:

$$a\left(\frac{4t}{(t-1)^2}\right) + b\left(\frac{4t(t+1)}{(t-1)^3}\right) = 1$$

Multiplying by $(t-1)^3$ to clear denominators and rearranging gives a cubic polynomial in t:

$$t^3 - (3 + 4a + 4b)t^2 + (3 + 4a - 4b)t - 1 = 0$$

Let the three roots corresponding to the intersection points P, Q, R be t_1, t_2, t_3. The product of the roots is:

$$t_1 t_2 t_3 = 1$$

This shows that the group operation on E_{ns} corresponds to multiplication in the field $k^\times$ under the parameter t. We have a group isomorphism:

$$\varphi \colon E_{ns} \to (k^\times, \times), \quad (x,y) \mapsto \frac{y+x}{y-x}.$$

Exercise 4.4.30. Generalize the previous example as follows. Let $y^2 = x^3 + dx^2$ over a field k with $\operatorname{char} k \neq 2$ such that d is not a square in k, then show that there is an isomorphism

$$\varphi \colon \{y^2 = x^3 + dx^2\} \setminus \{(0,0)\} \to \{u + v\sqrt{d}v \colon u, v \in k,\ u^2 - dv^2 = 1\}$$

given by

$$(x,y) \mapsto \frac{y + \sqrt{d}x}{y - \sqrt{d}x},$$

where the points of the conic $u^2 - dv^2 = 1$ is a group under multiplication thanks to Brahmagupta's identity

$$(u + v\sqrt{d})(w + t\sqrt{d}) = (uw + dvt) + (ut + vw)\sqrt{d}.$$

Solution: $\{u + v\sqrt{d} \mid u, v \in k,\ u^2 - dv^2 = 1\}$ is the set of elements in K with norm

1.Brahmagupta's identity confirms that it is a group under multiplication:

$$(u + v\sqrt{d})(w + t\sqrt{d}) = (uw + dvt) + (ut + vw)\sqrt{d}.$$

The norm of the product is

$$(uw + dvt)^2 - d(ut + vw)^2 = (u^2 - dv^2)(w^2 - dt^2) = 1.$$

We must verify that φ lands in the norm 1 subgroup. Let $N(u + v\sqrt{d}) = (u + v\sqrt{d})(u - v\sqrt{d}) = u^2 - dv^2$ be the norm map.

$$N\left(\frac{y + \sqrt{d}x}{y - \sqrt{d}x}\right) = \frac{N(y + \sqrt{d}x)}{N(y - \sqrt{d}x)} = \frac{y^2 - dx^2}{y^2 - dx^2} = 1.$$

As $y^2 - dx^2 = x^3$, and $(0, 0)$ is excluded, $x \neq 0$, so $y^2 - dx^2 \neq 0$.

φ is a homomorphism: This involves showing that if P, Q, R are collinear (so $P+Q+R = \mathcal{O}$), then $\varphi(P)\varphi(Q)\varphi(R) = 1$. This is analogous to the previous examples. Finally, to see that φ is a bijection, one would construct an inverse map, by parameterizing (x, y) in terms of $u + v\sqrt{d}$. $\square$

Remark 4.4.31 — For singular, irreducible cubics, the group of rational points $E_{ns}(\mathbb{Q})$ is not finitely generated. This is a key difference from elliptic curves, for which the Mordell–Weil theorem states the group of rational points is finitely generated. For example, the group of rational points on the unit circle $u^2 + v^2 = 1$, which is isomorphic to the group on $y^2 = x^3 - x^2$, is infinitely generated.

§4.4.iv Examples over Finite Fields

Example 4.4.32

Let $y^2 = x^3 + 1$, and let $k = \mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. The discriminant is $27 \neq 0$ in $\mathbb{F}_5$. The squares modulo five are $0, 1, 4$, and

x	0	1	2	3	4
y	± 1	$\emptyset$	± 2	$\emptyset$	0

There is also the point $\mathcal{O}$ at infinity. So six points, so $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/6\mathbb{Z}$.

- $2P = \mathcal{O}$ if and only if $y = 0$, since if $P = (x, y)$, then $-P = (x, -y)$. So $(4, 0)$ has order two.
- Claim that $(0, 1)$ is a point of order three. Then $3P = \mathcal{O}$, so $2P = -P$. so we just need a line which only meets the curve at this point. Check that $Y = Z$ in projective coordinates only goes through this point.
- The remaining points are $(0, -1)$ (given by negation, which has order 3. Finally, we get $(2, 3)$ and $(2, -3)$ which are of order 6.

Exercise 4.4.33. Compute $n(2, -2)$ for $n = 0, \ldots, 5$.

Exercise 4.4.34.

1. Show that the intersection of two quadric surfaces, that is, the simultaneous solutions to two homogeneous equations of degree 2 in 4 variables, can be put in the Weierstrass form, if it is irreducible and a smooth point is known. (You may assume that the intersection is smooth if you like.)

2. Find a birational transformation defined over $\mathbb{Q}$ taking the intersection of two quadric surfaces in $\mathbb{P}^3_{\mathbb{Q}}$ given by the equations

$$X_1^2 - 2X_2^2 + X_3^2 = 0, \quad X_2^2 - 2X_3^2 + X_4^2 = 0$$

into the canonical (Weierstrass) form of an elliptic curve in $\mathbb{P}^2_{\mathbb{Q}}$, with $(1, 1, 1, 1)$ going to the point at infinity.

Solution: 1. Let the four variables be X, Y, Z, W, and suppose the common smooth point is $[0 : 0 : 0 : 1]$. Then, we know that if F and G are our quadric surfaces, we have that

$$F = WL + R, \quad G = WM + S,$$

where L, M are linear in X, Y, Z, and R, S are quadratic terms. Now, first suppose that L and M are linearly dependent. Then, WLOG, we may assume that $M = 0$. We can then solve for W, giving

$$W = \frac{R}{L},$$

This provides a degenerate case, and so we get a conic. Otherwise, we can eliminate W to get that

$$LS - RM = 0,$$

which is a homogeneous cubic equation in variables X, Y, Z. This has a rational smooth point, as assumed. Therefore, by Nagell's algorithm, we can deduce that this cubic curve can be transformed into Weierstrass form.

2. We wish to mimic what we did above, so we first bring the curve into a form similar to the above. We apply the projective change of coordinates $[X_1 : X_2 : X_3 : X_4] \mapsto [X_1 + X_2 : X_2 : X_3 + X_2 : X_4 + X_2]$. Then, the two equations become

$$X_1^2 + X_3^2 + 2X_2(X_1 + X_3) = 0$$
$$X_4^2 - 2X_3^2 + 2X_2(X_4 - 2X_3) = 0.$$

In these coordinates, $[1 : 1 : 1 : 1]$ transformed to $[0 : 1 : 0 : 0]$. We can now eliminate X_2. We also just let $X_1 = X$, $X_3 = Y$, $X_4 = Z$ moving forward. The resulting cubic is

$$F(X, Y, Z) := (X^2 + Y^2)(Z - 2Y) - (X + Y)(Z^2 - 2Y^2) = 0.$$

We also wish to know where $[0 : 1 : 0 : 0]$ goes. Well, $[0 : 0 : 0]$ doesn't lie on this curve, so we take a step back and look at our initial two equations. We set $X_2 = 1$ there, and look at the tangent directions. This is given by the linear parts $X_1 + X_3 = 0$ and $X_4 - 2X_3 = 0$, so $[0 : 1 : 0 : 0]$ gets sent to $[1 : -1 : -2]$; we call this point $\mathcal{O}$. We're now in the position to run through Nagell's algorithm. Now, we're interested in the partials:

$$\frac{\partial F}{\partial X} = 2XZ - 4XY - Z^2 + 2Y^2,$$
$$\frac{\partial F}{\partial Y} = -2X^2 + 2YZ + 4XY - Z^2,$$
$$\frac{\partial F}{\partial Z} = X^2 + Y^2 - 2XZ - 2YZ.$$

Then, the tangent line at $\mathcal{O}$ is
$$X + 3Y - Z.$$
This intersects the curve defined by F at $P := [1 : 0 : 1]$. Now, we wish to move $\mathcal{O}$ to $[0 : 1 : 1]$, and P to $[0 : 0 : 1]$ and the tangent line at $\mathcal{O}$ to $X = 0$, so we apply the transformation $(X, Y, Z) \mapsto (X + Z, -Y, Z - 3Y)$. We also wish to move $\mathcal{O}$ to $[0 : 1 : 0]$ and the tangent line $t(\mathcal{O})$ to $Z = 0$. Thus, the total transformation is given by $(X, Y, Z) \mapsto (X + Y + Z, -Y, X - 2Y)$. Then the equation becomes

$$((X + Z)^2 + Y^2)(Z - Y) - ((Z - 3Y)^2 - 2Y^2)(X + Z - Y) = 0.$$

Setting $Z = 1$ and reorganizing according to the degree, we get

$$F_3(X, Y) + F_2(X, Y) + F_1(X, Y)$$

with

$$F_3(X, Y) = 6Y^3 - 7XY^2 - X^2Y, \quad F_2(X, Y) = X^2 + 4XY - 12Y^2, \quad F_1(X, Y) = X + 6Y.$$

Let
$$s = 2F_3(1, t)x + F_2(1, t)$$
and
$$G(t) = F_2(1, t)^2 - 4F_1(1, t)F_3(1, t).$$
Our curve is now equivalent to
$$s^2 = G(t).$$
Expanding out $G(t)$, we get the equation

$$s^2 = 48t^3 + 44t^2 + 12t + 1.$$

Multiplying by 36 and redefining $y = 6s$ and $x = 12t$, we get

$$y^2 = x^3 + 11x^2 + 36x + 36.$$

We notice that -2 is a root of the right hand side, so sending $x \mapsto x - 2$ simplifies the equation to
$$y^2 = x(x + 1)(x + 4).$$
By a further change of variables by sending $x \mapsto x - (5/3)$ and multiplying both sides with 3^6 and rescaling, one can get to the form

$$y^2 = x^3 - 351x + 1890.$$

□

Example 4.4.35 (Intersecting Quartics)
Let $Y^2 = X^4 + 2X^3 + 3X^2 + 4X + 2$. Find a rational point on this such and put this into Weierstrass normal form.

Solution: Note that this has a point at infinity. We put this into the form $Y^2 = G(X)^2 + H(X)$, where $G(X)$ is some quadratic and $H(X)$ is some linear polynomial. We can see that we can set $G(X) = X^2 + X + 1$, which takes care of all degree 2 and up terms of the RHS. Then, we just set $H(X) = 2X + 1$. Then, we find that

$$(Y - G(X))(Y + G(X)) = H(X).$$

Set $T = Y + G(X)$, so that we get $Y - G(X) = \frac{H(X)}{T}$, which gives

$$T - \frac{H(X)}{T} = 2G(X)$$

in turn. Multiply this by T^2 then to get

$$T^3 - H(X)T = 2G(X)T^2.$$

We now expand this where $H(X)$ and $G(X)$ are present and set $TX = S$.

$$T^3 - (2\frac{S}{T} + 1)T = 2(\frac{S^2}{T^2} + \frac{S}{T} + 1)T^2.$$

This leads to the following:

$$T^3 - 2T^2 - T - 2ST - 2S - 2S^2 = 0$$

In other words,

$$2S^2 + 2S + 2ST = T^3 - 2T^2 - T,$$

so this is in Weierstrass form. $\square$

Remark 4.4.36 — In general, if we have something like $Y^2 =$ quartic, then if there is a rational point at (a, b), we first apply $X \to \frac{1}{X-a}$, $Y \to \frac{1}{(X-a)^2}$, so that it gets sent to infinity. Then, we rescale our resulting quartic so that it becomes a monic quartic, and we proceed similarly to the above example.

5 Points of Finite Order

The main goal of the course is to take an elliptic curve E over $\mathbb{Q}$, and see what we can say about $E(\mathbb{Q})$. The main theorem is the Mordell-Weil theorem, that $E(\mathbb{Q})$ is a finitely generated abelian group. That is, there exists $\mathbb{Z}^N \to E(\mathbb{Q})$, that is there exist $g_1, \ldots, g_N \in E(\mathbb{Q})$ such that every element is of the form $\sum_{i=1}^N a_i g_i$ for $a_i \in \mathbb{Z}$. The group $\{\mathbb{Q}, +\}$ is not finitely generated. The structure theorem for finitely generated abelian groups is that any finitely generated abelian group is of the form $H \oplus T$, where $H \cong \mathbb{Z}^r$ for some $r \geq 0$, and T is finite. Both r and T are uniquely determined.

Definition 5.0.1. Call r the **rank** of the group, and T the **torsion subgroup**, that is the subgroup of elements of finite order.

§5.1 The group $E(\mathbb{Q})[m]$

Let $E\colon y^2 = x^3 + bx + c$ be an elliptic curve over $\mathbb{Q}$, so $4b^3 + 27c^2 \neq 0$. Making a change of variables $(x, y) \mapsto (u^2 x, u^3 y)$ for some $u \in \mathbb{Q}^\times$, we can send $(b, c) \mapsto (u^4 b, u^6 c)$, and we can assume $b, c \in \mathbb{Z}$ and $4b^3 + 27c^2 \neq 0$.

We proved in the previous section that $E(\mathbb{Q})$ is an abelian group. For a given positive integer m, consider the multiplication map

$$m\colon E(\mathbb{Q}) \to E(\mathbb{Q})$$

sending P to $mP = P + P + \cdots + P$. This is clearly a group homomorphism, and its kernel

$$E(\mathbb{Q})[m] = \{P \in E(\mathbb{Q})\colon mP = \mathcal{O}\}$$

is the subgroup of points of order dividing m.

§5.1.i Points of Order 2

We have already seen how to understand $E[2]$ before, namely $P \neq \mathcal{O}$ such that $2P = \mathcal{O}$ are given by $(x, 0) \in E$ such that $x^3 + bx + c = 0$. Hence, depending on whether the number of solutions of $x^3 + bx + c = 0$ over $\mathbb{Q}$, we get that $E[2]$ is either the trivial group, $\mathbb{Z}/2\mathbb{Z}$ or the Klein 4-group $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. When we have a trivial group, then we have no rational solutions. If all of them are rational solutions, then we have the Klein 4-group. If we only have $\mathbb{Z}/2\mathbb{Z}$, then there is only one rational solution.

§5.1.ii Points of Order 3

We also have a good understanding of $E[3]$. Indeed, $3P = \mathcal{O}$, implies $2P = -P$. Following the group law, this condition means that when we draw the tangent line at P, then take the third intersection point and connect it with $\mathcal{O}$, we get $-P$. But, this means that the third intersection of the tangent to P is also P, hence $3P = \mathcal{O}$, if and only if P is an inflection point. Note that $\mathcal{O}$, the point at infinity, is an inflection point, so $E[3]$ can be the trivial group. Now, by Bézout's theorem, we know that there are 9 inflection points over $\mathbb{C}$ (by intersecting with the Hessian). Thus, $E(\mathbb{C})[3] = \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, as this is the unique abelian group of order 9 where every non-zero element has order 3. Note that this is over $\mathbb{C}$, so we naturally ask if all these points can be defined over $\mathbb{Q}$. The answer is no. Let $P = (x, y)$ such that $2P = -P$, then, using the duplication formula for $y^2 = f(x) = x^3 + ax^2 + bx + c$, the x-coordinate of $2P$ and $-P = (x, -y)$, we get

$$-2x - a - \left(\frac{f'(x)}{2y}\right)^2 = \frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4(x^3 + ax^2 + bx + c)} = x$$

Hence, x has to be a root of

$$\psi_3(x) = 3x^4 + 4ax^3 + 6bx^2 + 12cx + 4ac - b^2 = 2f(x)f''(x) - (f'(x))^2$$

The claim is that over $\mathbb{R}$ or $\mathbb{Q}$, this polynomial doesn't have four roots. In particular, we claim that this has exactly two real roots. This follows from examining the discriminant. We have that

$$\psi_3'(x) = 2f'(x)f''(x) + 2f(x)f'''(x) - 2f'(x)f''(x) = 12f(x).$$

We now compute the discriminant. Writing $f(x) = (x - e_1)(x - e_2)(x - e_3)$, we have as follows:

$$4^4\psi_3(e_1)\psi_3(e_2)\psi_3(e_3) = -4^4(f'(e_1))^2(f'(e_2))^2(f'(e_3))^2 = -4^4\mathrm{Disc}(f)^2 = -27\Delta^2 < 0.$$

Now, if we have a degree four polynomial with negative discriminant, it has exactly two real roots (this is a general fact about degree four polynomials). From here, we need to perform calculations to see if y is still in $\mathbb{Q}$, but the key here is that we only have 2 possible x-values, so we don't have the entire group $E(\mathbb{C})[3]$. Therefore, $E(\mathbb{Q})[3]$ can only either be the trivial group or $\mathbb{Z}/3$.

Remark 5.1.1 — Note that we've used the following lemma. Let $p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0 = (x - r_1)\ldots(x - r_n)$, with derivative $p'(x) = n(x - f_1)\ldots(x - f_{n-1})$. Then,

$$\mathrm{Disc}(p) = \prod_{i \neq j}(r_i - r_j)^2 = (-1)^{\frac{n(n-1)}{2}} n^n \prod_{i=1}^{n-1} p(f_i).$$

The proof of this is very simple, and was covered previously.

Exercise 5.1.2. By a direct calculation, verify that $8\psi_3(x)$ is equal to the restriction of the Hessian determinant H to an affine point (x, y) of $y^2 = f(x)$. Using this, give another proof that roots of $\psi_3(x)$ are x-coordinates of inflection points.

§5.2 Nagell-Lutz

The main theorem that we are after in this section is the following.

Theorem 5.2.1 (Nagell-Lutz)
Let $E\colon y^2 = x^3 + bx + c$ with $4b^3 + 27c^2 \neq 0$ be an elliptic curve. The group of points of finite order in $E(\mathbb{Q})$ is finite. If $(x, y) \neq \mathcal{O}$ is of finite order, then

$$x, y \in \mathbb{Z}$$

and

$$y = 0 \quad \text{or} \quad y^2 \mid 4b^3 + 27c^2.$$

Note that when $y = 0$, we have the order 2 points.

Remark 5.2.2 — Note that Nagell-Lutz theorem does not assert that every point (x, y) with integer co-ordinates and $y^2 \mid 4b^3 + 27c^2$ must have finite order.

The core of the theorem is the statement that any odd torsion in $E(\mathbb{Q})$ has integer affine co-ordinates. We have the following claim which we will establish in the following subsection.

Claim 5.2.3. For any p, the torsion in $E(\mathbb{Q}_p)$ has coordinates $x, y \in \mathbb{Z}_p$.

Then, we use the fact that a rational number which is in $\mathbb{Z}_p$ for all p has to be an integer.

Conditional on the claim (5.2.3). If $(x, y) \in E(\mathbb{Q})$ is a torsion point, then by Claim (5.2.3) $x, y \in \mathbb{Z}_p$ for all p. So $x, y \in \mathbb{Z}$. If (x, y) is a torsion point, then $2(x, y)$ is also a torsion point, since $nP = \mathcal{O}$ implies that $n \cdot (2P) = 2(nP) = 2\mathcal{O} = \mathcal{O}$. Then if $2(x, y) = (x_2, y_2)$, then $x_2, y_2 \in \mathbb{Z}$. We will show that this implies that $y = 0$ or $y^2 \mid (4b^3 + 27c^2)$. Suppose that $y \neq 0$. By the duplication formula, we have

$$x_2 + 2x = \left(\frac{3x^2 + b}{2y}\right)^2 = \frac{(3x^2 + b)^2}{4(x^3 + bx + c)}$$

and so $y^2 = x^3 + bx + c$ divides $(3x^2 + b)^2$. Thus, $y \mid f'(x)$. We can them compute the discriminant of $f(x)$ to get

$$(3x^2 + b)(6bx^2 - 9cx + 4b^2) - (x^3 + bx + c)(18bx - 27c) = 4b^3 + 27c^2$$

to say that $y \mid (4b^3 + 27c^2)$. Note that $\mathrm{Disc}(f) = -\mathrm{Res}(f, f')$. However, we can get the stronger conditions

$$(3x^2 + b)^2(3x^2 + 4b) - (x^3 + bx + c)(27x^3 + 27bx - 27c) = 4b^3 + 27c^2$$

to get $y^2 \mid (4b^3 + 27c^2)$, as required. □

We will return back to proving Claim 5.2.3 but first we shall compute some torsion.

Example 5.2.4

Find the torsion group over $\mathbb{Q}$ of $y^2 = x^3 + 1$.

$4b^3 + 27c^2 = 27$. So, the only possible y-coordinates for torsion elements are $\{0, \pm 1, \pm 3\}$. We obtain the following rational points for these:

$$(-1, 0), (0, 1), (0, -1), (2, 3), (2, -3)$$

together with O (the point at infinity) we get 6 elements. $(-1, 0)$ has order 2. If you consider the line $y = 1$, then you get that it only intersect the curve at $(0, 1)$, hence $(0, 1)$ has order 3. Now, if you consider the line $y = x + 1$, it hits the curve at $(2, 3)$, $(-1, 0)$ and $(0, 1)$, hence $(2, 3)$ has order 6. Thus we can start with $P = (2, 3)$ and compute

$$2P = (0, 1), \quad 3P = (-1, 0), \quad 4P = (0, -1), \quad 5P = (2, -3), \quad 6P = O.$$

Hence, $(2, 3)$ generates the torsion subgroup and the final answer is

$$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}.$$

Example 5.2.5

If $y^2 = x^3 + 17$, then $4b^3 + 27c^2 = 27(17)^2 = 3^3 17^2$, so $y = 0, \pm 1, \pm 3, \pm 17, \pm 51$. If $17 \mid y$ then $17 \mid x^3 = y^2 - 17$ then $17^2 \mid y^2 - x^3 = 17$, a contradiction. We can easily rule out $y \in \{0, \pm 1\}$. Then $y = \pm 3$, so $x^3 = 9 - 17 = -8$. So if T denotes the torsion subgroup, then

$$T \subseteq \{(-2, \pm 3), \mathcal{O}\}.$$

We have not verified (in)equality yet, which is what we do. Using the doubling formula if $P = (-2, \pm 3)$, we see that the x-coordinate of $2P$ is 8. Thus, P is not torsion in the first place, since it's not contained in $\{(-2, \pm 3), \mathcal{O}\}$. We already saw that any torsion point has $x = -2$, so this cannot be a torsion point after all. So the torsion subgroup is just $\{\mathcal{O}\}$.

Remark 5.2.6 — This proves that there are infinitely many elements of $E(\mathbb{Q})$, that is infinitely many $x, y \in \mathbb{Q}$ such that $y^2 = x^3 + 17$. Indeed the points $n(2, 3)$ are all distinct for $n \in \mathbb{Z}$.

Remark 5.2.7 — $(-2, \pm 3)$, $(-1, \pm 4)$, $(2, \pm 5)$, $(4, \pm 9)$, $(8, \pm 23)$, $(43, \pm 282)$, $(52, \pm 375)$, $(5234, \pm 378661)$ all lie on $y^2 = x^3 + 17$ and none of them are torsion. Siegel's theorem says that on a smooth affine cubic curve there are only finitely many points with integer co-ordinates. Unfortunately, we won't have time to cover Siegel's theorem in this course.

Remark 5.2.8 — More generally, given an elliptic curve and a candidate of points (x, y), where $y = 0$ or $y \mid 4b^3 + 27c^2$, then we must verify whether each points is actually torsion, i.e. of finite order.

Exercise 5.2.9. Compute an elliptic curve with 5-torsion.

Solution: Suppose that we have an elliptic curve $E\colon y^2 = x^3 + bx + c$ which has torsion group $\mathbb{Z}/5$. We want $5P = \mathcal{O}$, or $4P = -P$. Let $P = (x, y)$. Denote by P_x to be the x-coordinate of P. Then, $-P_x = x$, and

$$x' := (2P)_x = \frac{x^4 - 2bx^2 - 8cx + b^2}{y^2} = 2x - \left(\frac{f'(x)}{y}\right)^2.$$

Setting

$$y' = (2P)_y = -y - \frac{3x^2 + b}{2y}(x' - x),$$

we have that

$$(4P)_x = \frac{(x')^4 - 2b(x')^2 - 8c(x') + b^2}{(y')^2} = 2x' - \left(\frac{f'(x')}{y'}\right)^2.$$

Therefore, we want

$$x = 2x' - \left(\frac{f'(x')}{y'}\right)^2,$$

but this is just finding roots of $\psi_3(x') = 2f(x')f''(x') - (f'(x'))^2$. Next, we have

$$x = 2f\left(2x - \left(\frac{f'(x)}{y}\right)^2\right) f''\left(2x - \left(\frac{f'(x)}{y}\right)^2\right) - \left(f'\left(2x - \left(\frac{f'(x)}{y}\right)^2\right)\right)^2.$$

Note that $f'(x) = 3x^2 + b$ and $f''(x) = 6x$.. □

Let's now return back to establishing the Claim 5.2.3.

§5.3 Reduction modulo p

The next aim is to study elliptic curves over $\mathbb{Q}_p$ in order to show that torsion points have coordinates in $\mathbb{Z}_p$. Recall that we have $\mathbb{Z}_p \subset \mathbb{Q}_p$ and $\mathbb{Z}_p$ is a local ring with maximal ideal (p), so we have the quotient $\mathbb{Z}_p/(p) \cong \mathbb{F}_p$ denoted by $a \mapsto \bar{a}$. We can use this to construct a reduction map $\mathbb{P}^n(\mathbb{Q}_p) \to \mathbb{P}^n(\mathbb{F}_p)$ as follows.

Given a point $[a_0 : \cdots : a_n] \in \mathbb{P}^n(\mathbb{Q}_p)$, by multiplying $a_0, \ldots, a_n$ by the same element of $\mathbb{Q}_p$, we can ensure that

$$\max(|a_0|_p, \ldots, |a_n|_p) = 1.$$

Then, $a_0, \ldots, a_n \in \mathbb{Z}_p$ and at least one of a_i is invertible, then we can use this to define a map

$$\mathbb{P}^n(\mathbb{Q}_p) \to \mathbb{P}^n(\mathbb{F}_p), \quad [a_0 : \cdots : a_n] \mapsto [\bar{a}_0 : \cdots : \bar{a}_n], \quad \bar{a}_i = a_i \bmod p.$$

One can also reduce lines in $\mathbb{P}^2(\mathbb{Q}_p)$ modulo p. Write any line as $aX + bY + cZ = 0$. Rescale so that $a, b, c \in \mathbb{Z}_p$, and at least one is in $\mathbb{Z}_p^\times$. Then $\bar{a}X + \bar{b}Y + \bar{c}Z = 0$ is a well-defined line in $\mathbb{P}^2(\mathbb{F}_p)$. More generally, if

$$C\colon F(X_1, X_2, X_3) = 0$$

is a cubic curve given by $F(X_1, X_2, X_3) = \sum_{i \le j \le k} f_{ijk} X_i X_j X_k$ with $f_{ijk} \in \mathbb{Q}_p$ and $\max_{i,j,k} |f_{ijk}|_p = 1$, then the **reduction modulo** p is given by

$$\overline{C}\colon \overline{F}(X_1, X_2, X_3) = 0$$

where

$$\overline{F}(X_1, X_2, X_3) = \sum_{i \leq j \leq k} \overline{f}_{ijk} X_i X_j X_k \in \mathbb{F}_p[X_1, X_2, X_3].$$

If a point $\mathbf{x} = (x_1, x_2, x_3)$ lies on C, we have $\overline{\mathbf{x}} = (\overline{x}_1, \overline{x}_2, \overline{x}_3)$ lies on $\overline{C}$.

Notation 5.3.1. In this section, we will usually refer to C as $E(\mathbb{Q}_p)$ and $\overline{C}$ as $E(\mathbb{F}_p)$, or $\overline{E}(\mathbb{F}_p)$.

On the other hand, note that even if F defines a smooth curve, it is possible that $\overline{F}$ becomes singular (and even reducible when F is not in the Weierstrass form).

Example 5.3.2

Let $p = 3$ and E be $y^2 = x^3 - 18$. Modulo 3, this is $y^2 = x^3$, which has a singular point $(0, 0)$. All the coefficients are in $\mathbb{Z}_3$ and at least one is a unit in $\mathbb{Z}_3$, which is why we can do this. For example, $(3, 3) \in E(\mathbb{Q}_3)$ is $[3 : 3 : 1] \mapsto [0 : 0 : 1] \pmod 3$; again, we can do this because the maximum of the 3-adic norms of $3, 3, 1$ is 1. We shall note that a smooth point in $E(\mathbb{Q}_3)$ which mapped to a singular point in $E(\mathbb{F}_3)$.

We now claim that there is a point (x, y) in $E(\mathbb{Q}_3)$ with $x = \frac{1}{9}$, since $(\frac{1}{9})^3 - 18 = \frac{1}{27^2}(1 - 18 \cdot 27^2)$ is a square, by Hensel's lemma for $y^2 - \frac{1}{27^2}(1 - 18 \cdot 27^2)$. Then $y = \frac{1}{27}u$ for some $u \in \mathbb{Z}_3^\times$. Thus,

$$[\frac{1}{9} : \frac{1}{27}u : 1] = [3 : u : 27] = [0 : 1 : 0]$$

in $E(\mathbb{F}_3)$. Thus there is a point which is not at infinity in $E(\mathbb{Q}_3)$ which is a point at infinity in $E(\mathbb{F}_3)$.

We can classify the possible reductions as follows: Given an elliptic curve $y^2 = f(x) = x^3 + ax^2 + bx + c$ over $\mathbb{Q}_p$, then we denote by $\overline{f}(x) = x^3 + \overline{a}x^2 + \overline{b}x + \overline{c}$.

- **Good reduction** if $\overline{E}(\mathbb{F}_p)$ is smooth if and only if $\overline{f}(x)$ has 3 distinct roots.
- **Multiplicative reduction** if $\overline{E}(\mathbb{F}_p)$ has a node $\iff \overline{f}(x)$ has a double root.
- **Additive reduction** if $\overline{E}_{ns}(\mathbb{F}_p)$ has a cusp if and only if $\overline{f}(x)$ has a triple root.

Remark 5.3.3 — Note that the naming of this comes from the group law associated with each case coming from Examples 4.4.28 and 4.4.29.

Example 5.3.4

Let $p \geq 5$. Then the elliptic curve $y^2 = x^3 + px^2 + 1$ has a good reduction mod p, while $y^2 = x^3 + x^2 + p$ has multiplicative reduction and $y^2 = x^3 + p$ has additive reduction.

Lemma 5.3.5

Let $\overline{\mathbf{b}}$ be a smooth point of $\overline{C} = E(\mathbb{F}_p)$. Then there is an $\mathbf{a}$ on $C = E(\mathbb{Q}_p)$ such that $\overline{\mathbf{a}} = \overline{\mathbf{b}}$.

Proof. Let F be the defining equation for C and $\overline{F}$ its reduction. Since $\overline{\mathbf{b}}$ is smooth on $\overline{C}$, we may suppose that $\frac{\partial \overline{F}}{\partial X}(\overline{\mathbf{b}}) \neq 0$. Let $b_1, b_2, b_3 \in \mathbb{Z}_p$ such that $\overline{\mathbf{b}} = [\overline{b}_1 : \overline{b}_2 : \overline{b}_3]$. Consider the polynomial $G \in \mathbb{Z}_p[X]$ defined by $G(X) = F(X, b_2, b_3)$. Now, apply Hensel with $x_0 = b_1$. Then, we get a solution $\mathbf{a} = (x, b_2, b_3)$ with $x \in \mathbb{Z}_p$ such that $G(x) = F(x, b_2, b_3) = 0$ and $\overline{\mathbf{a}} = \overline{\mathbf{b}}$ as required. □

Example 5.3.6

As an example, consider $y^2 = x^3 + p$. This is an additive reduction. In fact, the lemma above does not apply to the point $(0,0) \in E(\mathbb{F}_p)$. We cannot lift to any $a, b \in \mathbb{Z}_p$ such that $a^2 = b^3 + p$. If such a, b exists, then $p \mid a, b$ (since (a, b) reduces to $(0,0)$), so we reach a contradiction, since $p^2 \nmid p$.

For the rest of this section, let us fix the notation

$$E\colon Y^2Z = X^3 + aX^2Z + bXZ^2 + cZ^3, \quad a, b, c \in \mathbb{Z}_p$$

an elliptic curve defined over $\mathbb{Q}_p$, and

$$\overline{E}\colon Y^2Z = X^3 + \overline{a}X^2Z + \overline{b}XZ^2 + \overline{c}Z^3 \quad \overline{a}, \overline{b}, \overline{c} \in \mathbb{F}_p$$

be its reduction to $\mathbb{F}_p$. We write $f(x) = x^3 + ax^2 + bx + c \in \mathbb{Z}_p[x]$ and $\overline{f}(x) = x^3 + \overline{a}x^2 + \overline{b}x + \overline{c} \in \mathbb{F}_p[x]$ for its reduction.

Lemma 5.3.7

If $(x_0, y_0) \in E(\mathbb{Q}_p)$ then either $x_0, y_0 \in \mathbb{Z}_p$ or there exists $n \geq 1$ such that $|x_0|_p = p^{2n}$ and $|y_0|_p = p^{3n}$. Moreover, if $f(x) \equiv x^3 \pmod p$, and $(x_0, y_0) \in E(\mathbb{Q}_p)$ with $x_0, y_0 \in \mathbb{Z}_p$, then either $|x_0|_p = |y_0|_p = 1$ or $|x_0|_p, |y_0|_p < 1$.

Proof. If $|x_0|_p > 1$, then $|x_0^3|_p > |ax_0^2|_p, |bx_0|_p, |c|_p$, since $a, b, c \in \mathbb{Z}_p$. So $|x_0^3 + ax_0^2 + bx_0 + c|_p = |x_0|_p^3$. Then $|y_0|_p^2 = |x_0|_p^3$, and so $|x_0|_p = p^{2n}$ and $|y_0| = p^{3n}$.

Next, if $|x_0|_p \leq 1$, then $|y_0|_p^2 = |x_0^3 + ax_0^2 + bx_0 + c|_p \leq 1$.

For the final claim, the assumption tells us that $p \mid a, b, c$. So

$$p \mid x_0 \quad \iff \quad p \mid (x_0^3 + ax_0^2 + bx_0 + c) \quad \iff \quad p \mid y_0^2 \quad \Leftrightarrow \quad p \mid y_0.$$

That is, $|x_0|_p < 1$ if and only if $|y_0|_p < 1$. □

§5.3.i The Groups $E(\mathbb{Q}_p)^{(n)}$

We define

$$E(\mathbb{Q}_p)^{(0)} := \{P \in E(\mathbb{Q}_p) : \overline{P} \in \overline{E}_{ns}(\mathbb{F}_p)\},$$

and

$$E(\mathbb{Q}_p)^{(1)} := \{P \in E(\mathbb{Q}_p) : \overline{P} = \mathcal{O} \in \overline{E}(\mathbb{F}_p)\} = \{(x, y) \in E(\mathbb{Q}_p) : |x|_p, |y|_p > 1\} \cup \{\mathcal{O}\}.$$

Remark 5.3.8 — Note that the second equality really comes from considering a point $[x : y : 1]$ in $\mathbb{P}^2_{\mathbb{Q}_p}$. Then, if $x, y \in \mathbb{Z}_p$, then we can't reduce the third coordinate to 0, so we really require $|x|_p, |y|_p > 1$.

Lemma 5.3.9

1. $E(\mathbb{Q}_p)^{(0)}$ is a subgroup of $E(\mathbb{Q}_p)$.
2. Reduction modulo p is a surjective group homomorphism from $E(\mathbb{Q}_p)^{(0)}$ onto the group of non-singular points in $E_{ns}(\mathbb{F}_p)$. The kernel of this homomorphism is $E(\mathbb{Q}_p)^{(1)}$.

Proof. Need to show that if $P, Q \in E(\mathbb{Q}_p)^{(0)}$ then $P+Q$ is also in $E(\mathbb{Q}_p)^{(0)}$. Consider the line ℓ giving $P + Q + R = \mathcal{O}$. If we consider the reduced line $\overline{\ell}$ it intersects the reduced curve $\overline{E} = E(\mathbb{F}_p)$ at the smooth points $\overline{P}$ and $\overline{Q}$ (by definition of P and Q being in $E(\mathbb{Q}_p)^{(0)}$). Therefore, by Bézout, $\overline{R}$ must also be smooth, and so $\overline{R} \in \overline{E}_{ns}(\mathbb{F}_p)$. Similarly, considering the line through R and $\mathcal{O}$, we get that $P+Q = -R$, so $-\overline{R}$ is also in $E(\mathbb{Q}_p)^{(0)}$.

This also shows that the reduction mod p is a group homomorphism and the kernel is $E(\mathbb{Q}_p)^{(1)}$ by definition. Finally, the surjectivity follows from Lemma 5.3.5. $\square$

Remark 5.3.10 — Note that the proof used the fact that reduction of the line ℓ is not a component of $E_{ns}(\mathbb{F}_p)$. This holds because $\overline{E}$ is given in the Weierstrass form. It can be singular but is always irreducible.

We next define

$$E(\mathbb{Q}_p)^{(n)} := \{(x, y) \in E(\mathbb{Q}_p) \colon |x|_p \geq p^{2n}, |y|_p \geq p^{3n}\} \cup \{\mathcal{O}\}, \qquad n \geq 1.$$

This gives the following sequence:

$$E(\mathbb{Q}_p) \supset E(\mathbb{Q}_p)^{(0)} \supset E(\mathbb{Q}_p)^{(1)} \supset E(\mathbb{Q}_p)^{(2)} \supset \cdots \supset E(\mathbb{Q}_p)^{(n)} \supset \cdots \supset \{\mathcal{O}\}$$

The claim is that all of the groups $E(\mathbb{Q}_p)^{(n)}$ are subgroups of $E(\mathbb{Q}_p)$, for all n.

Definition 5.3.11. We say that (x, y) has **level** n if $|x|_p = p^{2n}$, so $|y|_p = p^{3n}$, i.e. $(x, y) \in E(\mathbb{Q}_p)^{(n)} \setminus E(\mathbb{Q}_p)^{(n+1)}$.

Remark 5.3.12 — The idea is if P has level n, then $P = [x : y : 1] = [p^{3n}x : p^{3n}y : p^{3n}]$ is close to $[0 : 1 : 0]$. Indeed, normalizing the y-coordinate, we get the point $[\frac{x}{y} : 1 : \frac{1}{y}]$, and noting that $|x| = p^{2n}$ and $|y| = p^{3n}$, the two coordinates (not y) are getting small. We are looking at points which are close to $\mathcal{O}$, and the group law is simpler here. Near the identity in a Lie group, the group law becomes simpler. For example, in $\mathrm{GL}_n(\mathbb{C})$,

$$(1 + \epsilon X)(1 + \epsilon Y) = 1 + \epsilon(X + Y) + O(\epsilon^2).$$

Lemma 5.3.13

$E(\mathbb{Q}_p)^{(n)}$ is a subgroup of $E(\mathbb{Q}_p)$ for all n.

Proof. Let E_n be the elliptic curve obtained by the (pointwise) change of variables $(x, y) \mapsto (p^{2n}x, p^{3n}y)$, so it is given by the equation:

$$E_n\colon Y^2Z = X^3 + p^{2n}aX^2Z + p^{4n}bXZ^2 + p^{6n}cZ^3$$

This is isomorphic to the curve $E\colon Y^2Z = X^3 + aX^2Z + bXZ^2 + cZ^3$ in $\mathbb{Q}_p$, since they're equivalent up to an admissible change of coordinates

$$\begin{array}{ccc} E & \longrightarrow & E_n \\ (x, y) & \longmapsto & (p^{2n}x, p^{3n}y). \end{array}$$

This then provides a bijection $E(\mathbb{Q}_p)^{(n+1)} = E_n(\mathbb{Q}_p)^{(1)}$, which is a group by Lemma 5.3.9. □

Lemma 5.3.14

For each $n \geq 1$ there is an isomorphism of abelian groups

$$E(\mathbb{Q}_p)^{(n)}/E(\mathbb{Q}_p)^{(n+1)} \cong \mathbb{Z}/p\mathbb{Z}.$$

Proof. As before use the identification $E \to E_n$ that takes $E(\mathbb{Q}_p)^{(n)}$ to a subgroup of $E_n(\mathbb{Q}_p)^{(0)}$, and $E(\mathbb{Q}_p)^{(n+1)}$ to $E_n(\mathbb{Q}_p)^{(1)}$. The curve E_n reduces to $y^2 = x^3 \pmod p$. So it suffices to show that if E' is an elliptic curve with reduction $y^2 = x^3$, then we have an isomorphism $E'(\mathbb{Q}_p)^{(0)}/E'(\mathbb{Q}_p)^{(1)} \cong E'_{ns}(\mathbb{F}_p)$. But reduction modulo p is a homomorphism $E'(\mathbb{Q}_p)^{(0)}/E'(\mathbb{Q}_p)^{(1)}$ to the non-singular points in $y^2 = x^3 \pmod p$. By Example 4.4.28, the right-hand side is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ via $(x, y) \mapsto x/y$. □

To summarize, we have a chain of subgroups

$$E(\mathbb{Q}_p)^{(0)} \supset E(\mathbb{Q}_p)^{(1)} \supset E(\mathbb{Q}_p)^{(2)} \supset \dots$$

with $\bigcap E(\mathbb{Q}_p)^{(n)} = \{\mathcal{O}\}$ and short exact sequences

$$1 \to E(\mathbb{Q}_p)^{(1)} \to E(\mathbb{Q}_p)^{(0)} \to E_{ns}(\mathbb{F}_p) \to 1$$

for $n = 0$, and

$$1 \to E(\mathbb{Q}_p)^{(n+1)} \to E(\mathbb{Q}_p)^{(n)} \to \mathbb{Z}/p\mathbb{Z} \to 1$$

for $n \geq 1$.

Definition 5.3.15. This sequence of groups is called **p-adic filtration of** $E(\mathbb{Q}_p)$.

This immediately implies the following:

Corollary 5.3.16

Let $(x, y) \in E(\mathbb{Q}_p)$ be a point of finite order m such that $(m, p) = 1$. Then $x, y \in \mathbb{Z}_p$.

Proof. By Lemma 5.3.7, if this doesn't hold, then (x, y) is of some level $n \geq 1$. Then $(x, y) \in E(\mathbb{Q}_p)^{(n)} \setminus E(\mathbb{Q}_p)^{(n+1)}$, and so maps to $\mathbb{Z}/p\mathbb{Z}$. But this is of order p. □

This is almost all we need for completing the proof of Nagell-Lutz. It only remains to get rid of the assumption $(m,p)=1$.

Let us now make some simple observations: $E(\mathbb{Q}_p)\setminus E(\mathbb{Q}_p)^{(1)}$ consists of (x,y) such that $|x|_p,|y|_p\leq 1$ hence to show all the torsion has co-ordinates in $\mathbb{Z}_p$, it suffices to show $E(\mathbb{Q}_p)^{(1)}$ has no torsion. Since for $n\geq 1$, $E(\mathbb{Q}_p)^{(n)}/E(\mathbb{Q}_p)^{(n+1)}\simeq \mathbb{Z}/p\mathbb{Z}$, the filtered chain of subgroups

$$E(\mathbb{Q}_p)^{(1)}\supset E(\mathbb{Q}_p)^{(2)}\supset\dots$$

looks a bit like the filtered chain of subgroups

$$p\mathbb{Z}_p\supset p^2\mathbb{Z}_p\supset\dots$$

Thus, we see if the first chain can be identified with the second. Here, we have in mind the additive group structure on $p\mathbb{Z}_p$. In fact, there is a map

$$\begin{aligned}u\colon E(\mathbb{Q}_p)^{(1)}&\to p\mathbb{Z}_p\\ \mathcal{O}&\mapsto 0\\ (x,y)&\mapsto -\frac{x}{y}.\end{aligned}$$

This respects the filtrations, that is,

$$|u(x,y)|_p\leq p^{-n}\quad\text{if}\quad |x|_p\geq p^{2n},|y|_p\geq p^{3n}$$

However, the issue is that u does not respect group law. We will fix this, by equipping the set $p\mathbb{Z}_p$ with a group structure that is a bit more intricate which forces u to be a group homomorphism, and that's for the next section.

Exercise 5.3.17.

1. Let C be the curve
$$Y^2=X^3+p$$
over $\mathbb{Q}_p$. Show that the point $(0,0)$ on the mod p curve does not lift to a point of C.
2. Find an example of an elliptic curve C over $\mathbb{Q}_p$ such that the mod p curve has a cusp which is the reduction of a point on C.

Solution: 1. Firstly, it is clear that $(0,0)$ lands on the elliptic curve mod p. However, this does not lift. If such a lift (a,b) did exist, then $a,b\in\mathbb{Z}_p$. Then, $p\mid a,b$. Then, going back to the original equation, $p^2\nmid a^3+p$, so we are done.

2. Consider $y^2=x^3+px$. This has a cusp at $(0,0)$. Another example is $yr=x^3+p^2$. This has a cusp at $(0,p)$. $\square$

Exercise 5.3.18. Find examples of curves C over $\mathbb{Q}_p$ such that the mod p curve has a double point with distinct tangents which

1. lifts,
2. does not lift,

to C.

Solution: 1. $y^2=x^3+x^2+p^2$ is a curve whose reduction mod p has a double point at $(0,0)$ which lifts to $(0,p)$. $y^2=x^3+x^2+px$ is also another example, with lift $(0,0)$.

2. $y^2=x^3+x^2+p$ is curve whose reduction mod p has a double point at $(0,0)$ which doesn't lift. If it did lift to (a,b), they're in $\mathbb{Z}_p$ and $p\mid a,b$. However, $p^2\nmid a^3+a^2+p$. $\square$

§5.4 Formal Group of an Elliptic Curve

Given an elliptic curve in the Weierstrass form

$$E\colon y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

with $a_1, a_2, a_3, a_4, a_6 \in \mathbb{Z}_p$.

Proposition 5.4.1

The map $u\colon E(\mathbb{Q}_p)^{(1)} \to p\mathbb{Z}_p$ given by $u(\mathcal{O}) = 0$ and $u(x, y) = -\frac{x}{y}$ is a bijection with inverse $t \mapsto (x(t), y(t))$ given by

$$\begin{aligned} x(t) &= \frac{t}{w(t)} = \frac{1}{t^2} - \frac{a_1}{t} - a_2 - a_3t - (a_4 + a_1a_3)t^2 + \dots \\ y(t) &= \frac{-1}{w(t)} = -\frac{1}{t^3} + \frac{a_1}{t^2} + \frac{a_2}{t} + a_3 + (a_4 + a_1a_3)t + \dots \end{aligned}$$

Here, $w(t)$ will be defined in the proof.

Proof. Let $t = -\frac{x}{y}$, $w = -\frac{1}{y}$ so that $\mathcal{O} = (0, 0)$ in the (t, w) co-ordinates (since in projective coordinates, $w = \frac{Z}{Y}$). In these co-ordinates $E(\mathbb{Q}_p)^{(1)}$ is precisely the set of points with $t, w \in p\mathbb{Z}_p$. The equation of E becomes

$$w = t^3 + a_1tw + a_2t^2w + a_3w^2 + a_4tw^2 + a_6w^3 = f(t, w)$$

For each fixed $t \in p\mathbb{Z}_p$, consider the polynomial $F(w) = w - f(t, w) \in \mathbb{Z}_p[w]$. We have $\overline{F}(0) = 0 - \overline{F}(t, 0) = 0 \in \mathbb{F}_p$. But $\frac{\partial \overline{F}}{\partial w}(0) = 1$. Hence, by Hensel's lemma, there exists a unique solution $w(t) \in p\mathbb{Z}_p$ such that

$$w(t) = f(t, w(t)).$$

Here is a procedure to obtain $w(t)$ explicitly. We can repeatedly substitute $f(t, w)$ in place of w and get

$$\begin{aligned} w &= t^3 + (a_1t + a_2t^2)[t^3 + (a_1t + a_2t^2)w + (a_3 + a_4t)w^2 + a_6w^3] \\ &\quad + (a_3 + a_4t)[t^3 + (a_1t + a_2t^2)w + (a_3 + a_4t)w^2 + a_6w^3]^2 \\ &\quad + a_6[t^3 + (a_1t + a_2t^2)w + (a_3 + a_4t)w^2 + a_6w^3]^3 \\ &\quad \vdots \\ &= t^3 + a_1t^4 + (a_1^2 + a_2)t^5 + (a_1^3 + 2a_1a_2 + a_3)t^6 \\ &\quad + (a_1^4 + 3a_1^2a_2 + 3a_1a_3 + a_2^2 + a_4)t^7 + \dots \\ &= t^3(1 + A_1t + A_2t^2 + \dots), \end{aligned}$$

where each $A_i \in \mathbb{Z}[a_1, a_2, a_3, a_4, a_6]$ is a polynomial in the coefficients of E. Note that this converges for $t \in p\mathbb{Z}_p$ as all the A_i are integral. To clarify the process, let $f_0(t, w) = f(t, w)$ and $f_{m+1}(t, w) = f_m(t, f(t, w))$. Let $w_m(t) = f_m(t, 0)$. Then, one can check that

$$w_m = f(t, w_m) + O(t^{m+4}).$$

Letting $m \to \infty$, we arrive at our analytic $w(t)$ with $w(t) = f(t, w(t))$ for $t \in p\mathbb{Z}_p$. We have the result. $\square$

Remark 5.4.2 — Over $\mathbb{Q}_p$, these series converge.

We next study the group law under this bijection.

Proposition 5.4.3

There is a unique power series

$$\mathcal{F}_E(t_1, t_2) = t_1 + t_2 - a_1 t_1 t_2 - a_2(t_1^2 t_2 + t_1 t_2^2) + \cdots \in \mathbb{Z}_p[[t_1, t_2]]$$

such that for $t_1, t_2 \in p\mathbb{Z}_p$, $t_3 = \mathcal{F}_E(t_1, t_2) \in \mathbb{Z}_p$ and

$$(x(t_1), y(t_1)) + (x(t_2), y(t_2)) = (x(t_3), y(t_3)) \in E(\mathbb{Q}_p)^{(1)},$$

and a unique power series $\iota_E(t) \in \mathbb{Z}_p[[t]]$ such that for $t \in p\mathbb{Z}_p$,

$$-(x(t), y(t)) = (x(\iota_E(t)), y(\iota_E(t))).$$

Proof. Consider the line $w = \lambda t + \nu$ through $(t_1, w(t_1))$ and $(t_2, w(t_2))$. We have

$$\lambda = \frac{w(t_2) - w(t_1)}{t_2 - t_1} = \frac{t_2^3 - t_1^3}{t_2 - t_1} + A_1 \frac{t_2^4 - t_1^4}{t_2 - t_1} + A_2 \frac{t_2^5 - t_1^5}{t_2 - t_1} + \cdots \in \mathbb{Z}_p[[t_1, t_2]]$$

and

$$\nu = w(t_1) - \lambda t_1 \in \mathbb{Z}_p[[t_1, t_2]]$$

Note that $\lambda \in (t_1, t_2)^2$ ((t_1, t_2) is an ideal) and $\nu \in (t_1, t_2)^3$. Substituting this into $w(t) = f(t, w(t))$ we find

$$\lambda t + \nu = t^3 + a_1 t(\lambda t + \nu) + a_2 t^2(\lambda t + \nu) + a_3(\lambda t + \nu)^2 + a_4 t(\lambda t + \nu)^2 + a_6(\lambda t + \nu)^3$$

Two roots of this equation are t_1, t_2 and call the third root t_3. The coefficient of t^3 is $1 + a_2\lambda + a_4\lambda^2 + a_6\lambda^3$ (note that this is 1 mod (t_1, t_2), hence is invertible) and the coefficient of t^2 is $a_1\lambda + a_2\nu + a_3\lambda^2 + 2a_4\lambda\nu + 3a_6\lambda^2\nu$. Hence, we get that

$$\tau = -t_1 - t_2 - \frac{a_1\lambda + a_3\lambda^2 + a_2\nu + 2a_4\lambda\nu + 3a_6\lambda^2\nu}{1 + a_2\lambda + a_4\lambda^2 + a_6\lambda^3} \in \mathbb{Z}_p[[t_1, t_2]]$$

Now, consider the special case $t_1 = t$, $t_2 = 0$. Making use of $\lambda = \frac{w(t)}{t}$, and $\nu = 0$, we find that the w-coordinate of the inverse point of $(t, w(t))$ is

$$\begin{aligned}\iota_E(t) &= -t - \frac{a_1(\frac{w(t)}{t}) + a_3(\frac{w(t)}{t})^2}{1 + a_2(\frac{w(t)}{t}) + a_4(\frac{w(t)}{t})^2 + a_6(\frac{w(t)}{t})^3} \\ &= -t - a_1 t^2 - a_1^2 t^3 - (a_1^3 + a_3)t^4 - a_1(a_1^3 + 3a_3)t^5 + \cdots \in \mathbb{Z}_p[[t]].\end{aligned}$$

Next, the t-coordinate of the sum $(t_1, w(t_1)) + (t_2, w(t_2))$ is given by

$$t_3 = \iota_E(\tau) = t_1 + t_2 - a_1 t_1 t_2 - a_2(t_1^2 t_2 + t_1 t_2^2) - (2a_3 t_1^2 t_2 - (a_1 a_2 - 3a_3)t_1^2 t_2^2 + 2a_3 t_1 t_2^2) + \cdots,$$

which is $\mathbb{Z}_p[[t_1, t_2]]$, and is what we wanted. □

Definition 5.4.4. A **(one-parameter, commutative) formal group** $\mathcal{F}$ over a ring R is a power series $\mathcal{F} \in R[[X, Y]]$, such that

1. $\mathcal{F}(X,Y) \in X+Y+(X,Y)^2$,
2. $\mathcal{F}(X,\mathcal{F}(Y,Z)) = \mathcal{F}(\mathcal{F}(X,Y),Z)$ (associativity)
3. $\mathcal{F}(X,Y) = \mathcal{F}(Y,X)$ (commutativity)
4. $\mathcal{F}(X,0) = X$, $\mathcal{F}(0,Y) = Y$ (identity)
5. There exists a unique $i(T) \in TR[[T]]$ such that $\mathcal{F}(T,i(T)) = 0 = \mathcal{F}(i(T),T)$ (inverses)

Remark 5.4.5 — In fact, the identity and inverse properties are consequences of the previous ones.

Remark 5.4.6 — A formal group is a *group law without any elements*, but if R is a complete discrete valuation ring and $\mathfrak{m}$ is its maximal ideal, then $\mathcal{F}$ induces an abelian group structure on $\mathfrak{m}$.

Example 5.4.7

For $\mathcal{F}(X,Y) = X+Y$, we get the additive group $(\mathfrak{m},+)$, and $\mathcal{F}(X,Y) = X+Y+XY$, gives the multiplicative group $(1+\mathfrak{m},\times)$.

In our case, we have $R = \mathbb{Z}_p$ and the formal group

$$\mathcal{F}_E(X,Y) = X+Y-a_1XY-a_2(XY^2+X^2Y)+\dots$$

For general $x,y \in \mathbb{Z}_p$, $\mathcal{F}_E(x,y)$ does not converge, but if we restrict to the maximal ideal $\mathfrak{m} = p\mathbb{Z}_p$ we get convergence, hence we get an abelian group structure on $p\mathbb{Z}_p$ associated to E, that gives the following identifications.

$$\begin{tikzcd}
E(\mathbb{Q}_p)^{(1)} \times E(\mathbb{Q}_p)^{(1)} \arrow[r, "(P,Q)\mapsto P+Q"] \arrow[d, "(x,y)\mapsto -\frac{x}{y}"', "\simeq"] & E(\mathbb{Q}_p)^{(1)} \arrow[d, "\simeq"] \\
p\mathbb{Z}_p \times p\mathbb{Z}_p \arrow[r, "(t_1,t_2)\mapsto \mathcal{F}_E(t_1,t_2)"'] & p\mathbb{Z}_p
\end{tikzcd}$$

and similarly for the inverses.

Finally, we are ready to complete the proof that $E(\mathbb{Q}_p)^{(1)}$ has no torsion.

Theorem 5.4.8

Let E be an elliptic curve over $\mathbb{Q}_p$ with Weierstrass form $y^2 = x^3+a_2x^2+a_4x+a_6$. Then $E(\mathbb{Q}_p)^{(1)}$ has no p-torsion.

Proof. $a_1 = 0$, so we have $\mathcal{F}_E(t_1,t_2) = t_1+t_2+(t_1,t_2)^3$, hence

$$[n]\cdot t = nt + (t^3)$$

So, if $\tau \in p\mathbb{Z}_p \setminus \{0\}$ with $|\tau|_p = p^{-k}$, then

$$[p]\cdot\tau = p\tau + (p^{3k})$$

but $|p\tau|_p = p^{-(k+1)}$, hence if $k+1 < 3k$, $[p]\cdot\tau \neq 0$. $\square$

This concludes the proof of Nagell-Lutz.

Exercise 5.4.9. For more general equation $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, same argument applies for odd p (as we can complete the square by $y \to y - \frac{a_1}{2}x$). For $p = 2$, again same argument applies if $a_1 = 0 \pmod 2$. If a_1 is odd, then show that $E(\mathbb{Q}_2)^0$ cannot have elements of order 4 (but can have order 2 elements). Show that $y^4 + xy^2 + 4x + 1$ is an elliptic curve over $\mathbb{Q}_2$ and $P = (\frac{-1}{4}, \pm 1)$ is a point of order 2.

Corollary 5.4.10

Let $E\colon y^2 = x^3 + ax^2 + bx + c$ be an elliptic curve over $\mathbb{Q}_p$. If $(x, y) \in E(\mathbb{Q}_p)$ is a torsion point, then $x, y \in \mathbb{Z}_p$.

Corollary 5.4.11

If $E(\mathbb{Q})$ is an elliptic curve of the form $y^2 = x^3 + ax^2 + bx + c$ for $a, b, c \in \mathbb{Z}$, and $p \nmid 2(4a^3c - a^2b^2 - 18abc + 4b^3 + 27c^2) = 2\mathrm{Disc}(f)$, then the subgroup of $E(\mathbb{Q})$ of torsion elements injects into $E(\mathbb{F}_p)$.

Proof. $p \nmid 2(a^4c^2 - a^2b^2 - 18abc + 4b^3 + 27c^2)$ implies that $E(\mathbb{Q}_p)^{(0)} = E(\mathbb{Q}_p)$ since $\overline{E}(\mathbb{F}_p)$ is then smooth. By Theorem 5.4.8 and Corollary 5.3.16, the intersection of $E(\mathbb{Q})_{tors}$ with $E(\mathbb{Q}_p)^{(1)} = \ker(E(\mathbb{Q}_p) \to E(\mathbb{F}_p))$ is trivial, since we showed that $E(\mathbb{Q}_p)^{(1)}$ has no torsion. ☐

Example 5.4.12

Consider $y^2 = x^3 + 105^{100}x + 3$. The discriminant of this is

$$\Delta = 4(105)^{300} + 27 \cdot 3^2.$$

First, we see that $5 \nmid 2\Delta$. Then, the previous corollary tells us that $E(\mathbb{Q})_{tors} \hookrightarrow \overline{E}(\mathbb{F}_5)$. Thus, we can consider the reduced curve $y^2 = x^3 + 3$. We have the following table:

x	0	1	2	3	4
$x^2 + 3$	3	4	1	0	2
$\#y$	0	2	2	1	0

Thus, $\#\overline{E}(\mathbb{F}_5) = 6$, Since we always have the point at infinity. Thus, $\overline{E}(\mathbb{F}_5) = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$. This restricts our choice for $E(\mathbb{Q})_{tors}$. Next, we consider 7, since $7 \nmid 2\Delta$. Again, $E(\mathbb{Q})_{tors} \hookrightarrow \overline{E}(\mathbb{F}_7)$. Modulo 7, we get the reduced curve $y^2 = x^3 + 3$.

x	0	1	2	3	4	5	6
$x^2 + 3$	3	4	4	2	4	2	2
$\#y$	0	2	2	2	2	2	2

Therefore, $\#\overline{E}(\mathbb{F}_7) = 13$, so $\overline{E}(\mathbb{F}_7) = \mathbb{Z}/13\mathbb{Z}$. It follows that $E(\mathbb{Q})_{tors} = \{1\}$.

Example 5.4.13

Let $E\colon y^2 = x^3 + 10^{2010}x + 3$. Neither five nor seven divides $4(10^{1510})^3 + 27(3)^2$.

- Modulo 5, $y^2 = x^3 + 3$. The squares modulo five are $0, 1, 4$, so

x	0	1	2	3	4
$x^2 + 3$	3	4	1	0	2
$\#y$	0	2	2	1	0

  Then $\#E(\mathbb{F}_5) = 5 + 1 = 6$, so $E(\mathbb{F}_5) \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$.

- Modulo 7, $y^2 = x^3 + 3$. The squares modulo seven are $0, 1, 2, 4$, so

x	0	1	2	3	4	5	6
$x^2 + 3$	3	4	4	2	4	2	2
$\#y$	0	2	2	2	2	2	2

  Then $\#E(\mathbb{F}_7) = 12 + 1 = 13$, so $E(\mathbb{F}_7) \cong \mathbb{Z}/13\mathbb{Z}$.

Thus, the torsion subgroup is just $\{0\}$.

Example 5.4.14

Let $y^2 = x^3 + 8$. Then, $\Delta = 4b^4 + 27c^2$, so the discriminant is $\Delta = 1728 = 2^6 3^3$. If $y = 0$, then $x = -2$, so we get $(-2, 0)$ as a 2-torsion point. If $y \neq 0$, then $y^2 \mid 1728$, so $y \mid 24$. Going through various possibilities, we find that the only solutions with $y \mid 24$ are given by $(1, \pm 3)$ and $(2, \pm 4)$. We are not done yet! We have only found candidates at this point; we need to verify that these are actually torsion points. However, we can use the doubling formula $x \mapsto \frac{x^4 - 64x}{4y^2}$ to compute

$$2(1, \pm 3) = \left(-\frac{7}{4}, -\frac{13}{8}\right), \quad 2(2, \pm 4) = \left(-\frac{7}{4}, \frac{13}{8}\right).$$

Sincere these points don't have integer coordinates, $(1, \pm 3)$ and $(2, \pm 4)$ cannot be torsion. It follows that the torsion subgroup of $E(\mathbb{Q})$ is $\{\mathcal{O}, (-2, 0)\}$.

Exercise 5.4.15. Let $A \in \mathbb{Z}$ be 4-th power free. Then, all the torsion points on $Y^2 = X(X^2 + A)$ are given by

- $(0, 0)$ of order 2.
- If $A = 4$, the points $(2, \pm 4, 1)$ of order 4.
- If $A = -C^2$, $C \in \mathbb{Z}$, the points $(\pm C, 0)$ of order 2.

Solution: It is clear that $(0, 0)$ is 2-torsion. Now, if $(x, y) = 2(a, b)$, then

$$x = -2a + \frac{(3a^2 + A)^2}{4b^2} = \frac{(a^2 - A)^2}{4b^2}.$$

Now, if $A = -C^2$, then $(\pm C, 0)$ are 2-torsion points as well. If $A \neq -C^2$, for some C, then the quadratic is not separable, and so these are the only 2-torsion points. Finally, suppose we have some 4-torsion. If $A = -C^2$, then $x = \frac{(a^2 + C^2)^2}{4b^2}$. If $2(a, b) = (\pm C, 0)$, this would imply that we must get $\pm C = \frac{(a^2+C^2)^2}{4b^2}$, so C itself a square, but then A contains a 4-th power; a contradiction. Thus, if we have 4-torsion, we must $2(a,b) = (0,0)$. In this case, we must have that $A = a^2$. Then, if we use the doubling formula on y, we find that $b^2 = 2a^3$. This implies that $a = 2$, since if not, then there is a mismatch of factors. Therefore, $b = \pm 4$ and $A = 4$. The reverse is clear. Thus, we have verified that these are torsion points.

Before moving on to odd torsion points, we must verify that there are no 8-torsion points. In this case, our elliptic curve is concrete, as $A = 4$, so $Y^2 = X(X^2+4)$. We also need a solution to $2 = \frac{(a^2-4)^2}{4b^2}$, but it is clear that this is not possible, since 2 is not a square.

Now, assume that we have a $2m+1$-torsion point (a,b). Then, note that $2m(a,b) = -(a,b) = (a,-b)$, so we find that

$$a = \frac{(a_m^2 - A)^2}{4b_m^2},$$

where $m(a,b) = (a_m, b_m)$. Thus, a is a square. Let $d = \gcd(a,b)$, and divide out by this factor of d to get

$$b_0^2 = a_0(da_0^2 + A_0),$$

where these new variables are just the original divided by d. Note that d divides b because there is a factor of d^2 on the RHS. Because $\gcd(a_0, (da_0^2 + A_0)) = 1$, we find that each of these are squares, say f^2 and g^2, respectively. Since a is a square, $d = h^2$ as well. Then, we can find that

$$a^2 - A = 2h^4f^4 - h^2g^2, \quad b = h^2fg.$$

Further, since $2(a,b)$ has integer coordinates, $2b \mid a^2 - A$ by the doubling formula. Thus, we must have $2h^2fg \mid 2h^4f^4 - h^2g^2$. In particular, $h^2f \mid a^2 - A$, so $f \mid g^2$. However, $\gcd(f, g^2) = 1$, so $f = 1$. Then, the divisibility condition says that $2h^2 \mid 2h^4 - h^2g^2$, so $2 \mid g^2$. Then, $2g \mid 2h^4 - h^2g^2$, so $g \mid h^4$; thus, $2 \mid h$. We bring this all together now. Now, we have

$$A = a^2 - 2h^4f^4 - h^2g^2,$$

so $2^4 \mid A$, which is a contradiction. Thus, there are no odd torsion points. Therefore, we have found all the torsion points. □

Exercise 5.4.16. Let $B \in \mathbb{Z}$ be 6-th power free. Then, all the torsion points on $Y^2 = X^3 + B$ are given by

- If $B = C^2$, the points $(0, \pm C)$ of order 3.
- If $B = D^3$, the points $(-D, 0)$ of order 2.
- If $B = 1$, the points $(2, \pm 3)$ of order 6.
- If $B = -432 = -2^4 3^3$, the points $(12, \pm 36)$ of order 3.

Solution: First note that in each of these cases, the points do lie on the elliptic curve. Now, let $2(a,b) = (x,y)$. Then,

$$x = -2a - \left(\frac{3a^2}{2b}\right)^2 = a(\frac{9a^3}{4b^2} - 2).$$

Order 2 elements solve $X^3 + B = 0$. This has a rational solution if and only if $B = D^3$, for some D. Then, we can write

$$X^3 + D^3 = (X + D)(X^2 - DX + D^2).$$

However, the quadratic on the right has discriminant $D^2 - 4D^2 = -3D^2$, which can never be a square. Thus, $(-D, 0)$ is the only 2-torsion point. If we want a 4-torsion point, then we must have $2(a, b) = (-D, 0)$, so

$$-D = a(\frac{9a^3}{4b^2} - 2).$$

This must be an integer, so $3 \mid b$ and $2 \mid a$. We must also have

$$0 = \left(\frac{3a^2}{2b}\right)(-D - a) + b.$$

If $a = 0$, then $b = 0$, which is not possible. Thus, $a \neq 0$, so $D = \frac{2b^2}{3a^2} - a$. We can equate D to get

$$\frac{2b^2}{3a^2} - a = -a(\frac{9a^3}{4b^2} - 2) \implies 8b^4 - 36a^3b^2 + 27a^6 = 0.$$

This can be viewed as a quadratic in b^2, which has discriminant $(36a^3)^2 - 4 \cdot 8 \cdot 27a^6 = 432a^6$. This can never be a square, since 432 is not a square. Thus, there are no more 2-power torsion points.

Next, we look for 3-torsion. For 3-torsion, we want $2(a, b) = (a, -b)$. Thus,

$$a = a(\frac{9a^3}{4b^2} - 2) \implies a\left(\frac{9a^3}{4b^2} - 3\right) = 0.$$

If $a = 0$, we get $B = C^2$, and the points $(0, \pm C)$ are 3-torsion points.

We now let (a, b) be some odd-torsion point. The strategy is to show that $w = \frac{9a^3}{4b^2}$ is an integer. We do this in some steps. Let $(x, y) = 2(a, b)$.

1. Suppose $p \mid B$, $p \nmid a$. Then, if $4b^2x = a(9a^3 - 8b^2) = a^4 - 8aB$, so $p \nmid x$.

2. Also if $p \mid B$, $p \nmid x$, then from the same equation in the previous step, $p \nmid a$.

3. Now suppose $3^\ell \mid b$, $3^m \mid a$. If $\ell \geq 3$, then $4b^2x + 8ab^2 = 9a^4$ implies that $3^6 \mid 9a^4$, so $3 \mid a$. If $|a|_3 = 3^{-1}$, then $3 \nmid x$; however, this contradicts the previous step because $3 \mid B = a^3 - b^2$, $3 \nmid x$, and $3 \mid a$. Otherwise, $|a|_3 \leq 3^{-2}$. Then, $3^6 \mid B$, which is a contradiction to B being 6-th power free. Thus $\ell = 0, 1, 2$. $m \geq 1$ because $a \in \mathbb{Z}$. If $\ell = 0, 1$, $w \in \mathbb{Z}_3$ clearly. If $\ell = 2$, then $m \geq 1$, so $w \in \mathbb{Z}_3$.

4. Now, if $2^\ell \mid b$, then $2^{2\ell+2} \mid 9a^4$, so $2^{\lceil \frac{\ell+1}{2} \rceil} \mid a$. If $\ell \geq 3$, then $2^2 \mid a$, so $2^6 \mid B$, a contradiction. If $|b|_2 = 2^{-2}$, then $2^2 \mid a$. Thus, $w \in \mathbb{Z}_2$. Next, suppose $|b|_2 = 2^{-1}$. If $|a|_2 = 2^{-1}$, then $2 \mid B$, $2 \nmid x$, and $2 \mid a$, a contradiction to step 2. Thus, $|a|_2 2^{-2}$, so $w \in \mathbb{Z}_2$. If $2 \nmid b$, then $2 \mid a$, so $w \in \mathbb{Z}_2$.

5. Now let q be some prime dividing B which is not 2 nor 3. If $q \mid b$, then clearly $q \mid a$. If $q^2 \nmid b$, then clearly $w \in \mathbb{Z}_q$. if $q^2 \mid b$, then $q \nmid x$ and $q \mid B$, a contradiction to step 2. If $q^3 \mid b$, then $q^2 \mid a$. However, then $q^6 \mid B$, a contradiction to the assumption. Therefore, the only valid case is if $|b|_q = q^{-1}$, and in this case, $w \in \mathbb{Z}_q$.

6. Now, we assumed (a, b) was odd and torsion, so $b^2 \mid 27B^2$. Thus, if p is a prime such that $p \mid b$, then $p \mid 3B$. Since we showed that $w \in \mathbb{Z}_p$ for all primes $p \mid B$, we get that $w \in \mathbb{Z}$.

Now that we know $w \in \mathbb{Z}$, we eliminate the cases for $w \neq 1, 2, 3$. Then, we must have that

$$|w - 2|_\infty 1 \implies |x|_\infty > |a|_\infty.$$

We assumed that w is odd torsion, so it must be that $2^n(a, b) = (a, b)$ for some large enough n. However, the inequality above shows that the absolute value of the x-coordinate (as we double), gets larger and larger each time, so it is not possible to come back to a. In fact it gets multiplied every single time by some integer. Note that if it gets multiplied by 0, then we are actually in the setting where we get 3-torsion.

Thus, $w = 1, 2, 3$, so we just case work. Let $w = 1$ to begin with. Then, $9a^3 = 4b^2$. Then, we can parametrize a and b as $a = 4k^2$ and $b = 12k^3$, where $k \in \mathbb{Z}$. If $|k| > 1$, then $k^6 \mid B$, so $k = \pm 1$. Therefore, $a = 4, b = \pm 12$, so $B = 80 = 2^4 \cdot 5$. Then, $(x, y) = (-4, \pm 4)$. If we double once more, we find that $2(x, y) = (44, 292)$. However, this doesn't satisfy $w \in \{1, 2, 3\}$. Alternatively, we could double again to see that we don't get integer coordinates now.

Next, let $w = 2$. Then, $x = 0$, so we fall into the case when $B = C^2$ and we have the 3-torsion points $(0, \pm C)$, for some C. Now, $9a^3 = 8b^2$, so we can parametrize $a = 2k^2$ and $b = 3k^3$, for $k \in \mathbb{Z}$. Again, if $k > 1$, then $k^6 \mid B$, so $|k| = \pm 1$. Then, $a = 2$, $b = \pm 3$ and $B = 1$. Thus, we actually have 2-torsion as well: $(-1, 0)$. To show that this is actually 6-torsion, we can calculate that $2(2, \pm 3) = (0, 1)$, and this is a 3-torsion points as seen. Thus, this is of 6-torsion. Note that the discriminant of $x^3 + 1$ is 27, so consider the reduction of the elliptic curve at 5. Then, the torsion group injects into $E(\mathbb{F}_5)$, and it is an easy calculation that $\#E(\mathbb{F}_5) = 6$.

Finally, let $w = 3$. Then, $9a^3 = 12b^2$. Thus, we can parametrize as $a = 12k^2$ and $b = 36k^3$. Again, if $|k| > 1$, $k^6 \mid B$, which is a contradiction. Therefore, $k = \pm 1$ and $a = 12$, $b = \pm 36$, giving $B = -432 = -2^4 \cdot 3^3$. We calculate that $2(12, 36) = (12, -36) = -(12, 36)$. Thus, this is 3-torsion. Since the discriminant of the cubic $x^3 - 2^4 \cdot 3^3$ is $2^8 \cdot 3^3$, consider reduction at 5. Then, $\#E(\mathbb{F}_5) = 6$, so the torsion group injects into $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Since there is no 2-torsion, we the torsion subgroup is of order 3, and we have found all. ☐

Exercise 5.4.17. Show that $X^3 + Y^3 + dZ^3 = 0$ is birationally equivalent to

$$Y^2 = X^3 - 2^4 \cdot 3^3 \cdot d^2.$$

If $d > 0$, $d \in \mathbb{Z}$ is cube free, deduce from the preceding exercise that the only cases of torsion are

$$d = 1, \qquad (1, 0, -1) \text{ and } (0, 1, -1) \text{ of order } 3,$$

and

$$d = 2, \qquad (1, 1, -1) \text{ of order } 2.$$

Solution: We have seen the birational transformation in Exercise 4.4.21; it is the pointwise map

$$[x : y : z] \mapsto [-12dz : 36d(z - y) : x + y].$$

Thus, the inverse has the pointwise map

$$[x : y : z] \mapsto [36dz + y : 36dz - y : -6x].$$

Note that this is the point-wise map from the transformed curve to the original one! We then just see what d are valid in the above exercise.

If $d = 1$, we get $(12, \pm 36)$ which are of order 3. Then,

$$[12 : \pm 36 : 1] \mapsto [36 \pm 36 : 36 \mp 36 : -72] = [1 : 0 : -1], [0 : 1 : -1].$$

If $d = 2$, $B = (-12)^3$, so the torsion point is $(12, 0)$. Then,

$$[12 : 0 : 1] \mapsto [72 : 72 : -72] = [1 : 1 : -1].$$

□

§5.5 Parametrization of Elliptic Curves with a Prescribed Torsion Point

We'd like to mention the celebrated theorem of Mazur (1977) which shows that the following are the only possible groups that may arise as a torsion subgroup of a $E(\mathbb{Q})$.

$$\mathbb{Z}/n\mathbb{Z} \text{ for } n = 1, 2, \ldots, 10 \text{ and } 12, \quad \text{or} \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2n\mathbb{Z} \text{ for } n = 1, \ldots, 4.$$

Unfortunately, there isn't an elementary proof of Mazur's theorem that is known to humanity at this point in time.

Remark 5.5.1 — We can't use this theorem in the exam.

The following table lists some elliptic curves over $\mathbb{Q}$ along with the possible torsion subgroups $E(\mathbb{Q})_{\text{tors}}$.

Elliptic Curve	Torsion Subgroup
$y^2 = x^3 + 2$	$\{0\}$
$y^2 = x^3 + x$	$\mathbb{Z}/2\mathbb{Z}$
$y^2 = x^3 + 4$	$\mathbb{Z}/3\mathbb{Z}$
$y^2 = x^3 + 4x$	$\mathbb{Z}/4\mathbb{Z}$
$y^2 + y = x^3 - x^2$	$\mathbb{Z}/5\mathbb{Z}$
$y^2 = x^3 + 1$	$\mathbb{Z}/6\mathbb{Z}$
$y^2 = x^3 - 43x + 166$	$\mathbb{Z}/7\mathbb{Z}$
$y^2 + 7xy = x^3 + 16x$	$\mathbb{Z}/8\mathbb{Z}$
$y^2 = x^3 - 219x + 1654$	$\mathbb{Z}/9\mathbb{Z}$
$y^2 + xy = x^3 - 45x + 81$	$\mathbb{Z}/10\mathbb{Z}$
$y^2 + 43xy - 210y = x^3 - 210x^2$	$\mathbb{Z}/12\mathbb{Z}$
$y^2 = x^3 - x$	$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$
$y^2 = x(x+1)(x+4)$	$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$
$y^2 + xy + y = x^3 - 19x + 26$	$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z}$
$y^2 = x(x+81)(x+256)$	$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$

Exercise 5.5.2. Find all the torsion elements of the above curves and determine their order (you may want to check your calculations in GP.)

Over $\mathbb{Q}$, we have seen that every elliptic curve can be brought into a form

$$y^2 = x^3 + bx + c \quad \text{with} \quad 4b^3 + 27c^2 \neq 0.$$

What if we ask the general form of elliptic curves with a specified torsion structure? For example, elliptic curves over $\mathbb{Q}$ with a rational point of order 2 can be parametrized as

$$y^2 = x(x^2 + ax + b)$$

for a, b such that $a^2b^2 - 4b^3 \neq 0$ and the 2-torsion point is at $(0, 0)$. This comes from the observation that 2-torsion points have $y = 0$, so after a change of coordinates, we may have a root at $x = 0$. The condition $a^2b^2 - 4b^3 \neq 0$ is really for smoothness.

It turns out the space of pairs (E, P) where $P \in E(\mathbb{Q})$ is a fixed torsion element of some fixed order can always be parametrized by a "modular curve of genus 0". We won't explain what this means here, but will give two other examples.

Proposition 5.5.3 (Parametrizing 3-torsion Elliptic Curves)
Elliptic curves over $\mathbb{Q}$ with a rational point of order 3 can be parametrized as

$$Y^2 = X^3 + (lX + m)^2$$

for l, m such that $4l^3m^3 - 27m^4 \neq 0$ and the inflection points are $(0, \pm m)$.

Proof. We find a necessary and sufficient condition that a line $Y = aX + b$ should be an inflexional tangent to

$$Y^2 = X^3 + AX + B$$

If the line $Y = aX + b$ intersects the cubic $Y^2 = X^3 + AX + B$ at an inflection point (x, y) that means that (x, y) is the only intersection point between these curves. Let $F(X) = X^3 + AX + B - (aX + b)^2$. Then we must have that $F(X)$ has a unique triple root at x. We can express this as:

$$F(x) = x^3 - a^2x^2 + (A - 2ab)x + B - b^2 = 0$$
$$F'(x) = 3x^2 - 2a^2x + (A - 2ab) = 0$$
$$F''(x) = 6x - 2a^2 = 0$$

We're working over a field of characteristic $\neq 2, 3$, so the last equation gives $x = a^2/3$. The second equation gives $A = \frac{a^4}{3} + 2ab$ and the first one gives $B = b^2 - \frac{a^6}{7}$. Thus, the general such curve should have the form

$$Y^2 = X^3 + \left(\frac{a^4}{3} + 2ab\right) X + b^2 - \frac{a^6}{27}$$

Conversely, plugging in $x = \frac{a^2}{3}$ to the right hand side gives $\frac{a^6}{9} + \frac{2a^3b}{3} + b^2 = \left(\frac{a^3}{3} + b\right)^2$. Hence, $(x, y) = \left(\frac{a^2}{3}, \frac{a^3}{3} + b\right)$ is an order 3 point on such an elliptic curve.

We note that by doing a coordinate change $X \to X + \frac{a^2}{3}$, the equation becomes

$$Y^2 = (X + \frac{a^2}{3})^3 + \left(\frac{a^4}{3} + 2ab\right)(X + \frac{a^2}{3}) + b^2 - \frac{a^6}{27} = X^3 + (aX + \frac{a^3}{3} + b)^2$$

The tangent line then also becomes $Y = aX + \frac{a^3}{3} + b$. Hence, letting $(l, m) = \left(a, b + \frac{a^3}{3}\right)$, we arrive at the equation

$$Y^2 = X^3 + (lX + m)^2$$

with the inflection point at $(x, y) = (0, \pm m)$ and the tangent line $Y = lX + m$. $\square$

Exercise 5.5.4. Find a necessary and sufficient condition that a line $Y = lX + m$ should be an inflectional tangent to $Y^2 = X(X^2 + aX + b)$. Hence find a general formula for curves in canonical form having a point of order 6.

Solution: We argue as before. Let $F(X) = X(X^2 + aX + b) - (lX + m)^2$. Then if (x, y) is an inflection point we must have:

$$F(x) = x^3 + (a - l^2)x^2 + (b - 2lm)x - m^2 = 0$$
$$F'(x) = 3x^2 + 2(a - l^2)x + (b - 2lm) = 0$$
$$F''(x) = 6x + 2(a - l^2) = 0$$

Hence, over a field of characteristic not equal to 2 or 3, the third equation gives $x = (l^2 - a)/3$. The second equation gives $b = (l^2 - a)^2/3 + 2lm$. The first equation gives $(l^2 - a)^3/27 = m^2$. Thus, we find out that $\frac{l^2-a}{3}$ should be a square. Then, we can pick t such that $\frac{l^2-a}{3} = t^2$ and $m = t^3$. We find out that $a = l^2 - 3t^2$ and $b = t^3(2l + 3t)$. Thus, this leads to the canonical form

$$Y^2 = X\big(X^2 + (l^2 - 3t^2)X + t^3(2l + 3t)\big)$$

and the tangent line at the inflection point is given by $Y = lX + t^3$.

Plugging in $X = t^2$, we find out that the order 3 points are at $(t^2, \pm t^2(l + t))$. The given order 2 point is at $(0, 0)$ so the order 6 points are given by

$$(0, 0) \pm (t^2, \pm(l + t)t^2) = (t(2l + 3t), \pm(l + t)t(2l + 3t))\,.$$

□

Exercise 5.5.5. Let (a, b, c) be a Pythagorean triple, i.e. $a^2 + b^2 = c^2$. Show that the elliptic curve $y^2 = x(x + a^4)(x + b^4)$ has torsion subgroup $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$.

Solution: The 2-torsion points are given by $(0, 0)$, $(-a^4, 0)$, and $(-b^4, 0)$. Now, the discriminant of the cubic on the right is $\big((-a^4)(a^4 - b^4)(b^4)\big)^2 = a^8b^8(a^4 - b^4)^2$. Thus, if (x, y) is a torsion point, then we must have that $y \mid a^4b^4(a^4 - b^4)$. We want to show that there is a point of order 8 in $E(\mathbb{Q})_{\text{tors}}$. By the 2-descent homomorphism (which we see later), we will see that $(0, 0) \in \ker\delta = 2E(\mathbb{Q})$. Therefore, there are 4-torsion points. In particular, the following points are 4-torsion.

$$\Big(a^2b^2, \pm a^2b^2(a^2 + b^2)\Big)$$

Now, we note that these points map to the identity under δ as well, so in fact these points are halvable as well. Therefore, there are 8-torsion points.

We actually have more than just this, which are also 4-torsion.

$$\Big(-a^2b^2, \pm a^2b^2(a^2 - b^2)\Big)$$

In the image of the 2-descent map, the first one maps to

$$\Big(a^2b^2, \pm a^2b^2(a^2 + b^2)\Big) \mapsto (a^2b^2, a^2b^2 + a^4, a^2b^2 + b^4) = (1, 1, 1)$$

as stated and the second one maps to

$$\Big(-a^2b^2, \pm a^2b^2(a^2 - b^2)\Big) \mapsto (-a^2b^2, -a^2b^2 + a^4, -a^2b^2 + b^4) = (-1, a^2 - b^2, a^2 - b^2),$$

so these are indeed 4-torsion. Explicitly, one can find these to take the following form.

$$\begin{aligned}
&(ab(a+c)(b+c), \pm abc(a+b)(a+c)(b+c)),\\
&(ab(a-c)(b-c), \pm abc(a+b)(a-c)(b-c)),\\
&(ab(a+c)(b-c), \pm abc(a-b)(a+c)(b-c)),\\
&(ab(a-c)(b+c), \pm abc(a-b)(a-c)(b+c)).
\end{aligned}$$

To be precise, one must check that $2(x,y)$ is one of the 4-torsion points. □

Exercise 5.5.6. Let $s \in \mathbb{Q}$. Show that if there is one $k \in \mathbb{Q}$ such that

$$X^3 + sX + k = 0$$

has 3 rational roots, then there are infinitely many. *Hint.* Let u be a rational root. Find the condition, in terms of s, u, k, that the two remaining roots are rational.

Solution: Write u be a rational roots. Then, writing

$$X^3 + sX + k = (X-u)(X^2 + aX + b),$$

we can find that $a = u$, $b = s + u^2$, and $k = -u(s+u^2)$. The point now is to find infinitely many u and s, so we get infinitely many k. Our cubic is now

$$X^3 + sX + k = (X-u)(X^2 + uX + s + u^2).$$

Now, we want there to be three rational roots, so we want the discriminant of the quadratic on the right to be a square. That is, we want

$$u^2 - 4(s+u^2) = -3u^2 - 4s$$

to be a square in $\mathbb{Q}$. Suppose this is equal to v^2 now, where $v \in \mathbb{Q}$. Note that in this case, we can explicitly find the two rational roots. The product of the two roots is $\frac{u^2-v^2}{4}$, and they sum to $-u$. Therefore, it is clear that the two roots are $-\frac{u \pm v}{2}$, so we find that the cubic is

$$X^3 + sX + k = (X-u)(X + \frac{u+v}{2})(X + \frac{u-v}{2}).$$

Returning, we can solve for s to find

$$s = -\frac{3u^2 + v^2}{4}.$$

This implies that if we can find infinitely many rational solutions (u, v) to

$$3U^2 + V^2 + 4s = 0,$$

we then get infinitely many k, since k is parametrized by u and v. In fact, this is true, and this follows from the fact that we have one pair (u, v) which works. The point is that this conic is birational to $\mathbb{P}^1$, because this is just the affine cut of a smooth projective conic. The way we find more points on this conic, given a point is by parametrizing

$$V = t(U-u) + v, \quad t \in \mathbb{Q}.$$

We can plug this into the conic to get that

$$3U^2 + (t(U-u)+v)^2 + 4s = 0 \implies (t^2+3)U^2 + 2t(v-tu)U + ((t^2-3)u^2 - 2tuv) = 0.$$

This is a quadratic in U, and we know that u is one solution on it. Therefore, we can solve for the other by noting that the sum of roots must equal $\frac{2t(v-tu)}{t^2+3}$. Thus, the other root is

$$\frac{2t(v-tu)}{t^2+3} - u = \frac{u(t^2-3)+2tv}{t^2+3}.$$

Note that this is in $\mathbb{Q}$ because $u, t, v \in \mathbb{Q}$. Thus, the resulting V is in $\mathbb{Q}$ as well, so we are done. □

Exercise 5.5.7. Let $k \in \mathbb{Q}$, $k \neq 0$. Show that if there are two $s \in \mathbb{Q}$ such that

$$X^3 + sX + k = 0$$

has 3 rational roots, then there are infinitely many.

Solution: We proceed similarly to before, but instead of parametrizing k by some rational root u and s, we try to parametrize s by k and u. We can write the cubic as

$$X^3 + sX + k = (X-u)(X + uX + -\frac{k}{u}).$$

Also, we note that s satisfies the relation

$$s = -\frac{k}{u} - u^2.$$

Again, we wish for three rational roots, so we want the discriminant to be a square, say of $v \in \mathbb{Q}$.

$$v^2 := u^2 + \frac{4k}{u}.$$

Thus, if we can find infinitely many rational solution (u, v) to the cubic

$$V^2U - U^3 - 4k = 0,$$

then we are good. We have two valid s by assumption, so we are given two pairs $(u_0, \pm v_0)$ and $(u_1, \pm v_1)$. This looks awfully like an elliptic curve, so we put it in the right form. For this, we homogenize; if we homogenize, we find that $[u_i : \pm v_i : 1]$ lie on the curve. Setting $U = 1$, we get

$$V^2 = 4kW^3 + 1$$

and $(\pm\frac{v_i}{u_i}, \frac{1}{u_i})$ lie on the curve now. This is now an elliptic curve, so if we can show that this has nonzero rank, we are good. For this, we show that one of our initial points was not torsion in the first place. Before we do this, we put our elliptic curve into a form we're much more used to, by multiplying by $16k^2$ and rescaling V, W by $4k$. Then, the elliptic curve is

$$V^2 = W^3 + 16k^2$$

and now the point $(v, w) = (\pm 4k\frac{v_i}{u_i}, \frac{4k}{u_i})$ lie on the curve. Now, we must analyze the torsion points of such elliptic curves.

We note that this is in the general form $Y^2 = X^3 + C^2$, where C is some constant, and in such elliptic curves we have the torsion points $(0, \pm C)$ which are of order 3. Furthermore, if $k \neq \frac{m^3}{4}$ (so the elliptic curve does not look like $Y^2 = X^3 + C^6$), these are the only torsion points. If $k = \frac{m^3}{4}$, then we have a 2-torsion point $(-m^2, 0)$ (here, the elliptic curve is equivalent to $Y^2 = X^3 + 1$). Furthermore, we can calculate that the order 6 points are $(2m^2, \pm 3m^2)$. We must now transfer this all back to U and V. $U = \frac{4k}{u}$, $V = \frac{V}{W}$. Noting that $4k = m^3$, we can attain the torsion points

$$(m, 0), (\frac{1}{2}m, \pm\frac{3}{2}m)$$

of the original cubic

$$U^3 = V^2U + 1.$$

Then, we can calculate s and k explicitly using u, v and we can even find the roots explicitly. Therefore, these torsion points lead to the polynomial

$$X^3 - \frac{3}{4}m^2X + \frac{m^3}{4} = (X + m)\left(X - \frac{m}{2}\right)\left(X - \frac{m}{2}\right).$$

Thus, all three of these solutions give $s = -\frac{3s^2}{4}$. We conclude that $k = \frac{m^3}{4}$ does not satisfy the hypothesis of the problem. Hence, in this case, there is a unique value of s such that

$$X^3 + sX + \frac{m^3}{4}$$

has 3 rational roots, namely $s = -\frac{-3m^2}{4}$, and this is not possible by assumption. ☐

Exercise 5.5.8. Give the general form of an elliptic curve with a rational point of order 4.

Solution: We first require a 2-torsion point. WLOG, suppose it is at $(0, 0)$, for we can just shift the x-value if not. Thus, our elliptic curve takes the following form:

$$y^2 = x(x^2 + Ax + B) = x^3 + Ax^2 + Bx.$$

We want to determine A and B now. Suppose $P = (u, v)$ is our 4-torsion point; we can suppose that $2(u, v) = (0, 0)$, since if not, we can just transform the elliptic curve again so that this is the case. We use the doubling formula on the x-coordinate:

$$x_{2P} = -2u - A + \left(\frac{3 \cdot u^2 + 2A \cdot u + B}{2v}\right)^2.$$

We want this to be 0. We can simplify this, using that $v^2 = u^3 + Au^2 + Bu$. We have that this is equal to

$$\frac{(3u^2 + 2Au + B)^2 - 4A(u^3 + Au^2 + Bu) - 2u(u^3 + Au^2 + Bu)}{4v^2} = \frac{(u^2 - B)^2}{4v^2}.$$

Therefore, $B = u^2$; let us set $u := b$. Then, we have that

$$v^2 = b^3 + Ab^2 + b^3 = b^2(2b + A).$$

$2b + A$ must be a square, so call it a. Then, $2b + A = a^2$, so $A = a^2 - 2b$. Then, $v = \pm a$. Therefore, an elliptic curve with 4-torsion looks like

$$y^2 = x^3 + (a^2 - 2b)x^2 + b^2x,$$

with the 4-torsion points $(b, \pm ab)$. ☐

6 The Mordell-Weil Theorem

It is time to launch into the most important theorem of the course. This result was proved by Mordell (1922) over $\mathbb{Q}$ and it was greatly generalized by Weil (1928) who proved it for elliptic curves over number fields as well as for abelian varieties (higher dimensional analogues of elliptic curves).

Theorem 6.0.1 (Mordell-Weil)
The group $E(\mathbb{Q})$ of rational points on an elliptic curve over $\mathbb{Q}$ is finitely generated.

We first discuss a way of proving finite generation for an abelian group. Let A be an abelian group. Write mA for the image of the homomorphism $A \to A$ given by $P \to m \cdot P$.

Definition 6.0.2. A **norm function** on an abelian group A is a function $|\cdot|\colon A \to [0, \infty)$ that satisfies:

- $\{P \in A : |P| \leq c\}$ is finite for all $c \in \mathbb{R}$.
- $|mP| = m|P|$ for all $P \in A$ and $m \in \mathbb{Z}$.
- $|P + Q| \leq |P| + |Q|$ for all $P, Q \in A$.

Proposition 6.0.3
An abelian group A is finitely generated if and only if A/mA is finite for some $m > 1$ and the group A admits a norm function.

Proof. The restriction of the Euclidean norm on $\mathbb{R}^n$ to $\mathbb{Z}^n$ is a norm. In particular, any finitely generated abelian group A has a norm since A is isomorphic to $\mathbb{Z}^n \oplus A_{\text{tors}}$.

Conversely, suppose $Q_1, Q_2, \ldots, Q_r$ are coset representatives of A/mA. Let $c = (\max_i |Q_i|) + 1$. Let S be the finite set of all $P \in A$ with $|P| < c$ and let $B = \langle S \rangle \subset A$ be the subgroup of A generated by S. In particular, each $Q_i \in B$. We claim that $B = A$. If not, let $P \in A \setminus B$. In particular, $|P| > c$. We can write it as $P = Q_i + mP_1$ for some Q_i and $P_1 \in A$. We have

$$m|P_1| = |mP_1| \leq |P| + |Q_i| < |P| + c < m|P|,$$

where the last follows because $|P| > c$, and $m \neq 1$ in particular. Thus, $|P_1| < |P|$. We can iterate this to get a sequence of $P_i \in A \setminus B$ with

$$|P| > |P_1| > |P_2| > \ldots$$

However, the number of elements of norm less than $|P|$ is finite, so this descent must stop once there is some P_i with $|P_i| \leq c$. Thus, $P_i \in B$, but this is a contradiction then. $\square$

Thus, to prove Mordell–Weil, we need two things:

1. The Weak Mordell–Weil Theorem: $E(\mathbb{Q})/mE(\mathbb{Q})$ is finite for some $m > 1$
2. A norm function on $E(\mathbb{Q})$.

These two parts of the proof use different ideas and techniques. The first one is related to descent, and the second one to the theory of heights.

§6.1 The Weak Mordell–Weil theorem

The weak Mordell–Weil theorem is the statement that $E(\mathbb{Q})/2E(\mathbb{Q})$ is a finite group. Before proving this, we work out an example.

§6.1.i An Example

Let $y^2 = x^3 - x = x(x-1)(x+1)$. First, assume $y \neq 0$. Write

$$x - 1 = au^2, \quad x = bv^2, \quad x + 1 = cw^2, \quad u, v, w \in \mathbb{Q},$$

for $a, b, c \in \mathbb{Z}$ square-free.

Claim 6.1.1. $a, b, c \in \{\pm 1, \pm 2\}$.

Proof. We need to show that if p is an odd prime, then $p \nmid abc$. Since $y^2 = x(x-1)(x+1) = abc(uvw)^2$, so abc is a square. If $p \mid abc$, then p divides at least two of a, b, c (because abc is a square). Suppose $|x|_p > 1$, then $|x|_p = |x \pm 1|_p$. Then $|y|_p^2 = |x||x-1||x+1|_p = |x|_p^3$, so $|x|_p = |x \pm 1|_p = p^{2n}$ for some $n \geq 1$. Now, since we assumed $p \mid abc$, this implies that either a or b or c is not square free. So $|x|_p \leq 1$. Then $|x \pm 1|_p \leq 1$, and if $p \mid abc$, then p divides at least two of a, b, c, hence p divides at least two of x, $x-1$, $x+1$. But, the difference of these two is either one or two. So $p \mid 2$. $\square$

We now ask which triples (a, b, c) can occur. Since abc is a square, c is determined by a and b. For example, if $a = -1$ and $b = -2$, then $c = 2$. So there are at most sixteen possibilities for (a, b, c). Since the product of $x + 1 > x > x - 1$ is a square, either all three are positive, or $x + 1 > 0 > x > x - 1$. This leaves eight possibilities:

$$\{(1,1,1), (1,2,2), (2,1,2), (2,2,1), (-1,-1,1), (-1,-2,2), (-2,-1,2), (-2,-2,1)\}.$$

Now, we want to solve

$$x - 1 = au^2, \quad x = bv^2, \quad x + 1 = cw^2$$

Therefore,

$$bv^2 - au^2 = 1 = cw^2 - bv^2$$

This is the descent step. If x has roughly N digits in its numerator and denominator, then u, v, w will have roughly $N/2$ digits in their numerators and denominators. Hence, if we are searching for points (x, y) we can instead search for smaller numbers (u, v, w).

Claim 6.1.2. There are no solutions to $y^2 = x^3 - x$ which give $(a, b, c) = (1, 2, 2)$.

Proof. Let $x - 1 = u^2$, $x = 2v^2$, and $x + 1 = 2w^2$, so $u^2 - 2v^2 = -1$ and $2w^2 - 2v^2 = 1$. Want to show that these have no solutions with $u, v, w \in \mathbb{Q}$. Writing $u = U/Z$, $v = V/Z$, and $w = W/Z$, it is enough to show that there are no non-zero solutions in $\mathbb{Z}$ to $U^2 - 2V^2 = -Z^2$ and $2W^2 - 2V^2 = Z^2$. Then, $2 \mid Z$ and $2 \mid U$, so $2 \mid V$, so $2 \mid W$. However, they're all divisible by 2, so by descent, we get only the zero solution. Repeating, $U = V = W = Z = 0$. $\square$

Similar arguments work if $(a,b,c) \in \{(-1,-2,2),(2,2,1),(2,2,1),(-2,-2,1)\}$. We can also incorporate the solutions with $y=0$. We claim that the remaining cases actually provide solutions. Then $(0,0)$ and $(\pm 1, 0)$ are also solutions of $y^2 = x^3 - x$.

- If $x = 0$, then $-1 = au^2$ and $1 = cw^2$, so $a = -1$ and $c = 1$. abc must be a square, so $b = -1$.
- If $x = -1$, then $-2 = au^2$ and $-1 = bv^2$, so $a = -2$ and $b = -1$. Similarly, $c = 2$.
- If $x = 1$, then $1 = bv^2$ and $2 = cw^2$, so $b = 1$ and $c = 2$. Similarly, $a = 2$.

Finally, $\mathcal{O}$ corresponds to $a = 1$, $b = 1$, and $c = 1$, so

$$(a,b,c) \in \{(1,1,1),(2,1,1),(2,1,2),(-1,-1,1),(-2,-2,1)\}.$$

We will prove below that in fact we have defined a group homomorphism

$$\delta\colon E(\mathbb{Q}) \to \{\pm 1, \pm 2\}^3 \subset \mathbb{Q}^\times/(\mathbb{Q}^\times)^2, \quad (x,y) \mapsto (a,b,c), \quad \mathcal{O} \mapsto (1,1,1),$$

where $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is a group, and $\{\pm 1, \pm 2\} = \{\pm(\mathbb{Q}^\times)^2, \pm 2(\mathbb{Q}^\times)^2\}$ is a subgroup under multiplication. We're taking this quotient because we want a, b, c to be square free.

For example, $(2)(2) = 1$, $(2)(-2) = -1$, and $(-2)(-2) = 1$. In particular, the image of δ is a subgroup of $\{\pm 1, \pm 2\}^3$, so it has order a power of two. Then $\delta(2P) = \delta(P)^2 = 1$. In fact, we will show that the induced homomorphism $E(\mathbb{Q})/2E(\mathbb{Q}) \to \{\pm 1, \pm 2\}^3$ is injective (weak Mordell-Weil theorem). So the calculation we made shows that $E(\mathbb{Q})/2E(\mathbb{Q})$ has order four. We will see later that the fact that we eliminated all combinations except those coming from known points shows that we have found all the points, except possibly those of odd order, which can be computed via Nagell-Lutz theorem. In summary, we have $E(\mathbb{Q}) = \{\mathcal{O}, (0,0), (\pm 1, 0)\}$.

§6.1.ii The 2-descent Homomorphism

Theorem 6.1.3

Let K be a field of characteristic $\neq 2$, and let E be the elliptic curve

$$y^2 = (x - e_1)(x - e_2)(x - e_3),$$

for $e_1, e_2, e_3 \in K$ distinct. Define

$$\begin{aligned} \delta\colon E(K) &\to (K^\times/(K^\times)^2)^3 \\ \mathcal{O} &\longmapsto (1,1,1) \\ T_1 = (e_1, 0) &\longmapsto ((e_1 - e_2)(e_1 - e_3), e_1 - e_2, e_1 - e_3) \\ T_2 = (e_2, 0) &\longmapsto (e_2 - e_1, (e_2 - e_1)(e_2 - e_3), e_2 - e_3) \\ T_3 = (e_3, 0) &\longmapsto (e_3 - e_1, e_3 - e_2, (e_3 - e_1)(e_3 - e_2)) \\ (x,y) &\longmapsto (x - e_1, x - e_2, x - e_3), \quad y \neq 0. \end{aligned}$$

Then:

1. δ is a group homomorphism.
2. $\ker \delta = 2E(K)$.

Remark 6.1.4 — Really, the *strange parts* of T_1, T_2, T_3 come from mimicking the fact that abc is a square in the example.

Lemma 6.1.5

Let K and E be as in Theorem 6.1.3. Let $P = (x, y) \in E(K)$. If all $x - e_i \in K^2$ for $i = 1, 2, 3$, then there exists $Q \in E(K)$ such that $P = 2Q$.

Proof. Write $x - e_i = t_i^2$ for $i = 1, 2, 3$ for $t_i \in K$. Then

$$y^2 = t_1^2 t_2^2 t_3^2 = (t_1 t_2 t_3)^2.$$

Assume, after possibly changing the sign of some t_i, that $y = t_1 t_2 t_3$. Set

$$Q = (x_0, y_0), \quad x_0 = x + t_1 t_2 + t_2 t_3 + t_3 t_1, \quad y_0 = (t_1 + t_2)(t_2 + t_3)(t_3 + t_1).$$

We claim that $Q \in E(K)$, and that $2Q = P$. Then

$$x_0 - e_1 = (x - e_1) + t_1 t_2 + t_2 t_3 + t_3 t_1 = t_1^2 + t_1 t_2 + t_2 t_3 + t_3 t_1 = (t_1 + t_2)(t_1 + t_3).$$

Similarly

$$x_0 - e_2 = (t_2 + t_3)(t_2 + t_1), \quad x_0 - e_3 = (t_3 + t_1)(t_3 + t_2).$$

Therefore,

$$(x_0 - e_1)(x_0 - e_2)(x_0 - e_3) = (t_1 + t_2)^2 (t_2 + t_3)^2 (t_3 + t_1)^2 = y_0^2,$$

that is, $Q \in E(K)$. To compute $2Q$, we need to find the tangent line $y = mx + c$. Write

$$f(x) = (x - e_1)(x - e_2)(x - e_3),$$

then

$$\begin{aligned} m = \frac{f'(x_0)}{2y_0} &= \frac{(x_0 - e_2)(x_0 - e_3) + (x_0 - e_3)(x_0 - e_1) + (x_0 - e_1)(x_0 - e_2)}{2y_0} \\ &= \frac{(t_1 + t_2)(t_2 + t_3)(t_3 + t_1)((t_1 + t_2) + (t_2 + t_3) + (t_3 + t_1))}{2y_0} = t_1 + t_2 + t_3. \end{aligned}$$

Then

$$m(x_0 - x) = (t_1 + t_2 + t_3)(t_1 t_2 + t_2 t_3 + t_3 t_1) = (t_1 + t_2)(t_2 + t_3)(t_3 + t_1) + t_1 t_2 t_3 = y_0 + y.$$

Thus, the point $(x, -y)$ also lies on the tangent line to E at (x_0, y_0). So $-2Q = (x, -y) = -P$; that is, $2Q = (x, y) = P$. $\square$

Proof of Theorem 6.1.3. Once we know that δ is a group homomorphism, then $2E(K) \subset \ker \delta$, because $\delta(2P) = \delta(P)^2 = 1$. Lemma 6.1.5 implies that $\ker \delta \subset 2E(K)$, since if $\delta(P) = 1$, then $x - e_i$ are squares, so $P = 2Q$, for some $Q \in E(K)$. Thus, $P \in 2E(K)$. Therefore, we have $\ker \delta = 2E(K)$. We need to prove that if $P_1 + P_2 + P_3 = \mathcal{O}$, then

$$\delta(P_1)\delta(P_2)\delta(P_3) = 1.$$

Note that by definition, $\delta(P) = \delta(-P)$. If $P_1 = P_2 = P_3 = \mathcal{O}$, then obvious. If $P_1 = \mathcal{O}$ and $P_2 \neq \mathcal{O}$, then $P_2 + P_3 = \mathcal{O}$, and

$$\delta(P_1)\delta(P_2)\delta(P_3) = \delta(P_2)\delta(P_3) = \delta(P_2)\delta(-P_2) = \delta(P_2)^2 = 1.$$

By symmetry, assume none of P_1, P_2, P_3 is $\mathcal{O}$. Thus, we have that P_1, P_2, P_3 all lie on some line $Y = mX + c$. Write $P_i = (x_i, y_i)$.

- Assume firstly that $2P_i \neq \mathcal{O}$ for $i = 1, 2, 3$, that is, $x_i \neq e_1, e_2, e_3$. Now, the polynomial $(x - e_1)(x - e_2)(x - e_3) - (mx + c)^2$ has leading coefficient 1 and has roots x_1, x_2, x_3. Hence,

$$(x - x_1)(x - x_2)(x - x_3) = (x - e_1)(x - e_2)(x - e_3) - (mx + c)^2.$$

  Substituting $x = e_1$, $(e_1 - x_1)(e_1 - x_2)(e_1 - x_3) = -(me_1 + c)^2$, that is

$$(x_1 - e_1)(x_2 - e_1)(x_3 - e_1) = (me_1 + c)^2 \in (K^\times)^2.$$

  Similarly, $(x_1 - e_2)(x_2 - e_2)(x_3 - e_2) \in (K^\times)^2$ and $(x_1 - e_3)(x_2 - e_3)(x_3 - e_3) \in (K^\times)^2$. It follows that $\delta(P_1)\delta(P_2)\delta(P_3) = 1$.

- Now assume that one of $2P_1, 2P_2, 2P_3$ is $\mathcal{O}$. Without loss of generality $P_1 = T_1 = (e_1, 0)$. Assume that $2P_2, 2P_3 \neq \mathcal{O}$. The calculation above shows that $\delta(P_1)\delta(P_2)\delta(P_3) \in K^\times/(K^\times)^2 \times 1 \times 1$. By definition,

$$\operatorname{Im} \delta \subset \{(a, b, c) \in (K^\times/(K^\times)^2)^3 \colon abc = 1\}.$$

  Thus, $\delta(P_1)\delta(P_2)\delta(P_3) = 1$ in this case.

- Now assume that at least two of P_1, P_2, P_3 are equal to T_1, T_2, T_3. The only possibility with $P_1 + P_2 + P_3 = \mathcal{O}$ but $P_i \neq \mathcal{O}$ is $\{P_1, P_2, P_3\} = \{T_1, T_2, T_3\}$. Then $\delta(P_1)\delta(P_2)\delta(P_3) = \delta(T_1)\delta(T_2)\delta(T_3) = 1$.

Thus, true. □

Remark 6.1.6 — We note that the assumption $e_i \in K$ is important.

We obtained a group homomorphism

$$\delta \colon E(K)/2E(K) \xrightarrow{\sim} \operatorname{Im}(\delta) \subset \{(a, b, c) \in (K^\times/(K^\times)^2)^3 \colon abc = 1\} \subset (K^\times/(K^\times)^2)^3.$$

$\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is not a finite group, since for each prime $p \in \mathbb{N}$, we have $p \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, and these are linearly independent as elements of $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, which is an $\mathbb{F}_2$-vector space.

Assumption: We assume that E can be written as

$$y^2 = (x - e_1)(x - e_2)(x - e_3), \quad e_1, e_2, e_3 \in \mathbb{Z},$$

that is, we are assuming that all the 2-torsion points are defined over $\mathbb{Q}$ (or in fact, we assume that they have integer co-ordinates since we can always apply admissible change of co-ordinates). These are the points $\mathcal{O}, (e_1, 0), (e_2, 0), (e_3, 0)$.

Proposition 6.1.7

$\operatorname{Im} \delta$ is finite, and, more precisely,

$$\operatorname{Im} \delta \subset \mathbb{Q}(S, 2)^3, \quad S = \{p \text{ prime} : p \mid (e_1 - e_2)(e_2 - e_3)(e_3 - e_1)\},$$

where $\mathbb{Q}(S, 2) \subset \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is the subgroup generated by the elements of S, that is, the coset representatives generated by square-free integers divisible only by primes in S.

Proof. Let p be prime, and consider a point $(x, y) \in E(\mathbb{Q})$. Then

$$y^2 = (x - e_1)(x - e_2)(x - e_3),$$

so $|y|_p^2 = |x - e_1|_p |x - e_2|_p |x - e_3|_p$. Either

1. $|x - e_1|_p, |x - e_2|_p, |x - e_3|_p \in p^{2\mathbb{Z}}$
2. exactly one of $|x - e_1|_p, |x - e_2|_p, |x - e_3|_p \in p^{2\mathbb{Z}}$.

In case 1., then all of these are squares, so there's nothing to show. We need to show that in case 2, we have $p \in S$. So suppose that case 2 holds. If $|x|_p > 1$, then $|x - e_i|_p = |x|_p$ for $i = 1, 2, 3$, because $|e_i|_p \le 1 < |x|_p$. But this contradicts the assumptions of case 2. So $|x|_p \le 1$, and $|x - e_i|_p \le 1$ for $i = 1, 2, 3$. Suppose that $|x - e_1|_p, |x - e_2|_p \notin p^{2\mathbb{Z}}$. Then $|x - e_1|_p, |x - e_2|_p < 1$, so $|e_1 - e_2|_p \le \max\{|x - e_1|_p, |x - e_2|_p\} < 1$, that is, $p \mid (e_1 - e_2)$. We are done by symmetry. □

Remark 6.1.8 — Note that $\mathbb{Q}(S, 2)$ is a finite group. For example, for $S = \{2\}$, $\mathbb{Q}(S, 2) = \{\pm 1, \pm 2\}$. For $S = \{2, 11\}$, $\mathbb{Q}(S, 2) = \{\pm 1, \pm 2, \pm 11, \pm 22\}$. In general, $\#\mathbb{Q}(S, 2) = 2^{1+s}$, where $s = \#S$. Note also that we always have $2 \in S$, since $(e_1 - e_2)(e_2 - e_3)(e_3 - e_1)$ is always even.

Theorem 6.1.9 (Weak Mordell–Weil theorem)
$E(\mathbb{Q})/2E(\mathbb{Q})$ is finite, for $E \colon y^2 = (x - e_1)(x - e_2)(x - e_3)$, $e_1, e_2, e_3 \in \mathbb{Z}$.

Proof. $\delta \colon E(\mathbb{Q})/2E(\mathbb{Q}) \hookrightarrow \mathbb{Q}(S, 2)^3$, and $\mathbb{Q}(S, 2)$ is finite. □

Remark 6.1.10 — In fact $\#(\operatorname{Im}\delta) \ge 4$ for any E, because we have four points coming from $\mathcal{O}$ and $(e_i, 0)$ for $i = 1, 2, 3$. If $s = \#S$, so $s \ge 1$, because $2 \in S$, then $\#\mathbb{Q}(S, 2) = 2^{s+1}$. So $\#\operatorname{Im}\delta \le (2^{s+1})^3 = 8^{s+1}$. By definition, if $\delta(P) = (a, b, c)$, then $abc = 1$. So $\#\operatorname{Im}\delta \le (2^{s+1})^2 = 4^{s+1}$. Considering the signs of $x - e_1$, $x - e_2$, $x - e_3$, and the condition that their product is a square, we get

$$\#(\operatorname{Im}\delta) \le 2^{1+2s}.$$

Corollary 6.1.11
$\#\operatorname{Im}\delta \le 2^{1+2\#S}$.

A Geometric Interpretation

Let $a, b, c \in \mathbb{Z}$ represent an element of $\operatorname{Im}\delta$, and $d \in \mathbb{Z}$ such that $abc = d^2$. Consider the algebraic curve $E_{(a,b,c)}$ determined by the three equations, in affine coordinates (x, y_1, y_2, y_3) in $\mathbb{A}^4$,

$$ay_1^2 = x - e_1, \quad by_2^2 = x - e_2, \quad cy_3^2 = x - e_3.$$

The projective curve can be described as the intersection of quadric surfaces

$$ay_1^2 - by_2^2 = (e_2 - e_1)t^2, \quad ay_1^2 - cy_3^2 = (e_3 - e_1)t^2$$

in $\mathbb{P}^3$ with co-ordinates $[y_1 : y_2 : y_3 : t]$. There is a (4-to-1) covering map

$$\pi_{(a,b,c)} : E_{(a,b,c)}(\mathbb{Q}) \to E(\mathbb{Q})$$

given by $(x, y_1, y_2, y_3) \mapsto (x, dy_1y_2y_3)$. In fact, $\pi_{(a,b,c)}(E_{(a,b,c)}(\mathbb{Q}))$ is a coset of $2E(\mathbb{Q})$ in $E(\mathbb{Q})$ and we have a finite union

$$E(\mathbb{Q}) = \bigcup_{(a,b,c)\in \operatorname{Im}\delta} \pi_{(a,b,c)}\left(E_{(a,b,c)}(\mathbb{Q})\right).$$

It can be shown (but we won't) that $E_{(a,b,c)}$ is in fact isomorphic to E. Note that $\pi_{(a,b,c)}$ is not such an isomorphism since it is 4-to-1. In fact, $\pi_{(a,b,c)}$ does not even necessarily have $\mathcal{O}$ in its image. However, by construction $\pi_{(a,b,c)}$ hits the point P such that $\delta(P)$ is represented by (a, b, c). Hence, we can compose $\pi_{(a,b,c)}$ with the translation map sending a point $Q \in E(\mathbb{Q})$ to $Q - P \in E(\mathbb{Q})$ to get an isogeny from $E_{(a,b,c)}$ to E, which becomes the multiplication by 2 map under the identification of $E_{(a,b,c)}$ with E.

§6.1.iii Mordell-Weil Theorem over General Number Fields

We will give a proof of Mordell–Weil theorem in general here assuming some basic theorems from Algebraic Number Theory.

> **Remark 6.1.12 —** This part of the proof is non-examinable.

What happens if 2-torsion points are not defined over $\mathbb{Q}$? Then, we let K be a finite extension of $\mathbb{Q}$ such that

$$y^2 = (x - e_1)(x - e_2)(x - e_3), \qquad e_1, e_2, e_3 \in \mathcal{O}_K,$$

where $\mathcal{O}_K$ is the ring of algebraic integers of K. We still have the homomorphism

$$\delta : E(K) \to (K^\times/(K^\times)^2)^3.$$

However, to prove that its image is finite requires a little background in algebraic number theory. We will sketch this proof, although we do not assume this knowledge in this module. Here is what we need.

Theorem 6.1.13

Let K be a finite extension of $\mathbb{Q}$ with ring of integers $\mathcal{O}_K$. Then, there exists a ring R with $\mathcal{O}_K \subset R \subset K$ such that R is a principal ideal domain, hence a unique factorization domain and the group of units in R is finitely generated.

(Concretely, let $I_1, \ldots, I_h$ be representatives of finitely many ideal classes of $\mathcal{O}_K$ and let $u_i \in I_i$ for $i = 1, \ldots, h$ be non-zero elements. Then putting $u = u_1u_2\cdots u_h$, we can obtain R as $\mathcal{O}_K[1/u]$, that is the localization of $\mathcal{O}_K$ using the multiplicative set $\{1, u, u^2, \ldots\}$.)

Example 6.1.14

Consider $y^2 = x^3 + x$. The splitting field is $K = \mathbb{Q}(i)$, with $i^2 = -1$. Then $\mathcal{O}_K = \mathbb{Z}[i]$ is a unique factorization domain. Hence, we can take $R = \mathcal{O}_K$. Its group of units is $\{\pm 1, \pm i\} \cong \mathbb{Z}/4\mathbb{Z}$.

Example 6.1.15

Consider $y^2 = x^3 + 5x$. The splitting field is $K = \mathbb{Q}(\sqrt{-5})$. Then $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ is not a unique factorization domain. The only non-trivial ideal class is generated by $(2, 1+\sqrt{-5})$. Therefore, we can take $R = \frac{1}{2}\mathbb{Z}[\sqrt{-5}]$, which is a UFD. Its group of units is generated by $\mathcal{O}_K^\times = \{\pm 1\}$ and 2.

Fix such a ring R. Let S be the set of primes of R dividing $(e_1 - e_2)(e_2 - e_3)(e_1 - e_3)$ in R. Then, by adapting the proof of Proposition 6.1.7, one shows that the image of δ is contained in the group generated by S and the units of R. Therefore, the image of δ is a finitely generated abelian group of exponent 2, hence is finite.

This shows that $E(K)/2E(K)$ is finite. Now, the weak Mordell-Weil theorem over $\mathbb{Q}$ follows from the following proposition.

Proposition 6.1.16

Let $E\colon y^2 = (x-e_1)(x-e_2)(x-e_3)$, be an elliptic curve over $\mathbb{Q}$, let K be a splitting field of $(x-e_1)(x-e_2)(x-e_3)$ (thus, we can assume that $e_1, e_2, e_3 \in \mathcal{O}_K$). Then the canonical homomorphism

$$E(\mathbb{Q})/2E(\mathbb{Q}) \to E(K)/2E(K)$$

has finite kernel (has $\leq 2^{2[K:\mathbb{Q}]}$ elements). Hence, if $E(K)/2E(K)$ is finite, then $E(\mathbb{Q})/2E(\mathbb{Q})$ is also finite.

Proof. A point $P \in E(\mathbb{Q})$ goes to 0 if and only if $P \in E(\mathbb{Q}) \cap 2E(K)$. For each such point P, pick $Q_P \in E(K)$ such that $2Q_P = P$, and define a function (not a homomorphism, in general)

$$\lambda_P\colon \mathrm{Gal}\,(K/\mathbb{Q}) \to E[2](K)$$

by sending $\sigma \in \mathrm{Gal}\,(K/\mathbb{Q})$ to $Q_P - \sigma(Q_P)$. (Note that if $(x, y) \in E(K)$, then $(\sigma(x), \sigma(y)) \in E(K)$ since E is defined over $\mathbb{Q}$.)

- $Q_P - \sigma(Q_P) \in E[2](K)$ because $2(Q_P - \sigma(Q_P)) = 2Q_P - \sigma(2Q_P) = P - \sigma(P) = 0$.
- If $Q_P - \sigma(Q_P) = Q_{P'} - \sigma(Q_{P'})$ for all σ, then $\sigma(Q_P - Q_{P'}) = Q_P - Q_{P'}$ for all σ. But, since K is a normal extension, this implies $Q_P - Q_{P'} \in E(\mathbb{Q})$. Thus, $P - P' = 2(Q_P - Q_{P'}) \in 2E(\mathbb{Q})$.

Thus, we conclude that for P in the kernel, the assignment $P \mapsto \lambda_P$ is injective, but there are only finitely many maps between the finite sets $\mathrm{Gal}\,(K/\mathbb{Q})$ and $E[2](K)$ of sizes $[K : \mathbb{Q}]$ and 4, respectively. Hence, the kernel must be finite. $\square$

§6.2 The Rank of $E(\mathbb{Q})$

We now assume that we know the Mordell-Weil theorem, so that $E(\mathbb{Q}) \cong T \oplus \mathbb{Z}^r$, where T is the finite torsion subgroup. Then $E(\mathbb{Q})/2E(\mathbb{Q}) \cong T/2T \oplus (\mathbb{Z}/2\mathbb{Z})^r$. Thus,

$$\#(\mathrm{Im}\,\delta) = \#(E(\mathbb{Q})/2E(\mathbb{Q})) = \#(T/2T) \times 2^r.$$

Lemma 6.2.1

For any finite abelian group T, let $T[2] = \{x \in T : 2x = 0\}$. Then the following exact sequence holds:

$$0 \to T[2] \to T \xrightarrow{\times 2} T \to T/2T \to 0.$$

In particular, $\#(T/2T) = \#(T[2])$, where $T[2] = \{x \in T \colon 2x = 0\}$.

Proof. Consider the map

$$[2] : T \to T, \quad x \mapsto 2x.$$

This is a group homomorphism with kernel $T[2]$ and image $2T$. Now computing orders, $\#T/\#(T[2]) = \#(2T)$, that is $\#T/\#2T = \#T[2]$. □

Returning to E, if $y^2 = (x - e_1)(x - e_2)(x - e_3)$ with $e_1, e_2, e_3 \in \mathbb{Z}$, and T is the torsion subgroup, we have

$$\#T/2T = \#T[2] = \#\{P \in E(\mathbb{Q}) \mid 2P = \mathcal{O}\} = \#\{\mathcal{O}, (e_1, 0), (e_2, 0), (e_3, 0)\} = 4.$$

So, we proved

$$\#(\operatorname{Im}\delta) = 2^{r+2}.$$

Example 6.2.2

If E is $y^2 = x^3 - x$, we have seen that $\delta = \{(1, 1, 1), (-1, -1, 1), (2, 1, 2), (-2, -1, 2)\}$. Hence $\#(\operatorname{Im}\delta) = 4$. It follows that $r = 0$, and $E(\mathbb{Q}) = \{\mathcal{O}, (0, 0), (\pm 1, 0)\} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example 6.2.3

Let E be $y^2 = (x - 5)(x - 6)(x + 11)$. Let us compute $E(\mathbb{Q})_{\text{tors}}$ and $E(\mathbb{Q})/2E(\mathbb{Q})$.

Solution: First, $(e_1 - e_2)(e_2 - e_3)(e_3 - e_1) = (5 - 6)(6 - (-11))((-11) - 5) = 2^4 \cdot 17$. This is the discriminant. We begin by calculating the torsion.

- The only primes dividing the discriminant are two and seventeen. Modulo three, $y^2 = x(x + 1)(x - 1) \equiv x^3 - x$. This has solutions $(0, 0), (1, 0), (-1, 0)$ in $\mathbb{F}_3$, so $E(\mathbb{F}_3)$ has four points and the y-coordinates are zero, so $E(\mathbb{F}_3) \cong (\mathbb{Z}/2\mathbb{Z})^2$. Then $E(\mathbb{Q})_{\text{tors}} \hookrightarrow E(\mathbb{F}_3) \cong (\mathbb{Z}/2\mathbb{Z})^2$. In fact, this is an isomorphism, since we have four torsion elements form the original curve. So

$$E(\mathbb{Q})_{\text{tors}} = \{\mathcal{O}, (5, 0), (6, 0), (-11, 0)\}.$$

Let

$$\delta \colon E(\mathbb{Q})/2E(\mathbb{Q}) \to (\mathbb{Q}^\times/(\mathbb{Q}^\times)^2)^3, \qquad (x, y) \mapsto (x - 5, x - 6, x + 11), \quad y \neq 0.$$

Then $\operatorname{Im}\delta \subset \{(a,b,c) \in \{\pm1,\pm2,\pm17,\pm34\}^3 \mid abc = 1\}$. From now on, regard δ as

$$\begin{aligned}\delta\colon E(\mathbb{Q})/2E(\mathbb{Q}) &\to (\mathbb{Q}^\times/(\mathbb{Q}^\times)^2)^2 \\ (x,y) &\mapsto (x-5, x-6, x+11), \qquad y \neq 0 \\ \mathcal{O} &\mapsto (1,1,1) \\ (5,0) &\mapsto (-1,-1,1) \\ (6,0) &\mapsto (1,17,17) \\ (-11,0) &\mapsto (-1,-17,17)\end{aligned}$$

To find $\operatorname{Im}\delta$, we need to solve the equations

$$x-5 = au^2, \quad x-6 = bv^2, \quad x+11 = cw^2, \quad u,v,w \in \mathbb{Q}^\times,$$

where $a,b,c \in \{\pm1,\pm2,\pm17,\pm34\}$ such that $abc = 1$, or equivalently

$$au^2 - bv^2 = 1, \qquad abw^2 - au^2 = 16.$$

The second was originally $cw^2 - au^2 = 16$, but we applied the constraint $abc = 1$ (a square), so this is a simplification. This essentially disregards the condition $abc = 1$. For these to have solutions in $\mathbb{Q}^*$ (or even just $\mathbb{R}$), we need $ab > 0$. Also, just the (a,b) parts of $\operatorname{Im}\delta$, we have

$$\delta((5,0)) = (-1,-1),$$

so if $(a,b) \in \operatorname{Im}\delta$, then $(-a,-b) \in \operatorname{Im}\delta$. If $\delta((x,y)) = (a,b)$, then since δ is a homomorphism:

$$\delta((5,0)+(x,y)) = (-1,-1)(a,b) = (-a,-b).$$

So it suffices to determine which $(a,b) \in \operatorname{Im}\delta$ for $a,b > 0$, and double it.

Thus, we only need to work out when $a,b \in \{1,2,17,34\}$. Let ✓ be $(a,b) \in \operatorname{Im}\delta$ and $\times$ be $(a,b) \notin \operatorname{Im}\delta$.

- $\delta(\mathcal{O}) = (1,1)$ and $\delta((6,0)) = (1,17)$, so $(1,1),(1,17) \in \operatorname{Im}\delta$. Then

$a\backslash b$	1	2	17	34
1	✓		✓	
2				
17				
34				

- If $(a,b) = (1,2)$, then $u^2 - 2v^2 = 1$ and $2w^2 - u^2 = 16$, so we can homogenize this so $U^2 - 2V^2 = Z^2$ and $2W^2 - U^2 = 16Z^2$, where $U,V,W,Z \in \mathbb{Z}$ so $2 \mid U,V,W,Z$, so $U = V = W = Z = 0$, a contradiction, so $(1,2) \notin \operatorname{Im}\delta$. Thus, $(1,34) = (1,2)(1,17) \notin \operatorname{Im}\delta$. Then

$a\backslash b$	1	2	17	34
1	✓	$\times$	✓	$\times$
2				
17				
34				

- This argument only used $2 \mid b$ and $2 \nmid a$, so $(17,2),(17,34) \notin \operatorname{Im}\delta$. Then

$a\backslash b$	1	2	17	34
1	✓	$\times$	✓	$\times$
2				
17		$\times$		$\times$
34				

- If $(a,b)=(2,1)$, then $2u^2-v^2=1$ and $w^2-u^2=8$ has a solution $u=v=1$ and $w=3$, so $(2,1)\in \operatorname{Im}\delta$, so $(2,34)=2\cdot 17=34$ and $9^2=36=6^2$, so $(x,y)=(7,\pm 6)$, so $(2,17)=(2,1)(1,17)\in \operatorname{Im}\delta$ and so $(2,2)=(2,1)(1,2)=(2,34)=(2,17)(1,17)$, $(34,2)=(2,1)(17,2)$, $(34,34)=(2,1)(17,34)\in \operatorname{Im}\delta$. Then

$a\backslash b$	1	2	17	34
1	✓	×	✓	×
2	✓	×	✓	×
17		×		×
34		×		×

- If $(a,b)=(17,1)$, then $17u^2-v^2=1$ and $17w^2-17u^2=16$, so $17U^2-V^2=Z^2$ and $17W^2-17U^2=16Z^2$, so $17\mid U,V,W,Z$, so $U=V=W=Z=0$, a contradiction, so $(17,1)\notin \operatorname{Im}\delta$, so $(17,17)=(1,17)(17,1)\notin \operatorname{Im}\delta$. Then

$a\backslash b$	1	2	17	34
1	✓	×	✓	×
2	✓	×	✓	×
17	×	×	×	×
34		×		×

- This argument only used $17\mid a$ and $17\nmid b$, so $(34,1),(34,17)\notin \operatorname{Im}\delta$. Then

$a\backslash b$	1	2	17	34
1	✓	×	✓	×
2	✓	×	✓	×
17	×	×	×	×
34	×	×	×	×

Thus $\#(\operatorname{Im}\delta)=8$ and E has rank one, since $2^{2+r}=8$, and $(7,\pm 6)\in E(\mathbb{Q})$ is not torsion. Hence, we conclude that

$$E(\mathbb{Q})\cong \mathbb{Z}\times \mathbb{Z}/2\mathbb{Z}\times \mathbb{Z}/2\mathbb{Z}.$$

□

§6.3 The Height Machine

To complete the proof of Mordell-Weil theorem, it remains to construct a norm function on $E(\mathbb{Q})$. We will construct such a norm function using heights. Note that weak Mordell by itself is not enough as the following example shows.

Example 6.3.1

$\mathbb{Q}/2\mathbb{Q}=\{0\}$ but the additive group $(\mathbb{Q},+)$ is not finitely generated. The group $(\mathbb{Q},+)$ has the property that for any $x\in\mathbb{Q}$ and for any $n\in\mathbb{Z}$, there exists $y\in\mathbb{Q}$ such that $ny=x$. Such a property cannot hold in a direct sum of (finitely many) cyclic groups.

Definition 6.3.2. Let $\frac{a}{b}\in\mathbb{Q}$ for $a,b\in\mathbb{Z}$ such that $(a,b)=1$. The **height** of $\frac{a}{b}$ is

$$H(\frac{a}{b}):=\max\{|a|,|b|\}$$

and $h(\frac{a}{b})=\log H(\frac{a}{b})$.

Remark 6.3.3 — $h(\frac{a}{b})$ is closely related to the number of digits required to write the rational number $\frac{a}{b}$.

Definition 6.3.4. If $(x, y) \in E(\mathbb{Q})$, we define the **height** of (x, y) as the height of x:

$$H(x, y) = H(x).$$

Set

$$h(P) = \begin{cases} \log H(x, y) & P = (x, y), \\ 0 & P = \mathcal{O}. \end{cases}$$

Remark 6.3.5 — For all $c \in \mathbb{R}$, note that there are only finitely many $P \in E(\mathbb{Q})$ such that $h(P) < c$.

The definition of height that we have given looks a bit ad hoc, but a modification works more cleanly and has desirable properties.

Definition 6.3.6. We define the **canonical height**, or **Tate height**

$$\hat{h}(P) := \lim_{n\to\infty} \frac{h(2^n P)}{4^n}.$$

The fact that the limit exists and has various nice properties will be proved below.

Theorem 6.3.7 (Canonical Height)
Let E be an elliptic curve over $\mathbb{Q}$. There is a unique function

$$\hat{h}\colon E(\mathbb{Q}) \to [0, \infty)$$

with the following properties:

1. There is a constant c such that $|h(P) - \hat{h}(P)| < c$ for all P.
2. $\hat{h}(mP) = m^2 \hat{h}(P)$.
3. (Parallelogram Law) $\hat{h}(P + Q) + \hat{h}(P - Q) = 2\hat{h}(P) + 2\hat{h}(Q)$.
4. $\hat{h}(P) = 0$ if and only if P is torsion.

Remark 6.3.8 — Later, we define the norm function $|P|$ as $|P| := \sqrt{\hat{h}(P)}$. This will satisfy the axioms of a norm function.

The proof of the existence of canonical height function with the stated properties is frustratingly tedious. We will first need to prove two technical lemmas.

Lemma 6.3.9
There exists a constant $C_0 > 0$, depending only on E, such that for all $P \in E(\mathbb{Q})$,

$$h(2P) \geq 4h(P) - C_0.$$

Proof. Let the elliptic curve E be given by $y^2 = f(x)$ for $f(x) = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$. If $P = (x, y)$, and $2P = (x_2, y_2)$, recall from the doubling formula, we have

$$x_2 = \frac{(f'(x))^2 - 8xf(x)}{4f(x)} = \frac{x^4 - 2Ax^2 - 8Bx + A^2}{4(x^3 + Ax + B)}.$$

Write $x = \frac{s}{t}$, where $(s, t) = 1$, so $H(P) = \max\{|s|, |t|\}$, and $x_2 = \frac{m}{n}$, where

$$m(s,t) = s^4 - 2As^2t^2 - 8Bst^3 + A^2t^4, \qquad n(s,t) = 4(s^3t + Ast^3 + Bt^4).$$

Since the polynomials $m(s,1) = (f'(s))^2 - 8sf(s)$, $n(s,1) = 4f(s) \in \mathbb{Z}[s]$ are relatively prime (by smoothness of the elliptic curve), their resultant is an integer $c \neq 0$: $\operatorname{Res}(m(s,1), n(s,1)) = c$. We can therefore apply the Euclidean algorithm to produce polynomials f_1 and $g_1 \in \mathbb{Z}[s]$ of degree 2 and 3, respectively, such that

$$m(s,1)f_1(s) + n(s,1)g_1(s) = c.$$

We homogenize all these polynomials to find homogeneous polynomials $F_1, G_1 \in \mathbb{Z}[s,t]$ of degree 3 such that

$$F_1m + G_1n = ct^7.$$

Exchanging the roles of s and t, we also obtain homogeneous polynomials $F_2, G_2 \in \mathbb{Z}[s,t]$ of degree 3 such that

$$F_2m + G_2n = cs^7.$$

In particular, $(m,n)|(ct^7, cs^7) = c$. Thus,

$$H(2P) \geq \frac{\max\{|m|, |n|\}}{c}.$$

To show $h(2P) \geq 4h(P) - C_0$, or equivalently $H(2P) \geq H(P)^4e^{-C_0}$, it suffices to prove

$$H(P)^4 \leq KH(2P),$$

for some $K > 0$. Now,

$$|t|^7 < \frac{1}{c}|F_1||m| + |G_1||n| \leq K' \max\{|s|^3, |t|^3\} \max\{|m|, |n|\},$$

for some constant K'. Similarly, $|s|^7 \leq K'' \max\{|s|^3, |t|^3\} \max\{|m|, |n|\}$ for some constant K''. Thus,

$$\max\{|s|^7, |t|^7\} \leq \max(K', K'') \max\{|s|^3, |t|^3\} \max\{|m|, |n|\},$$

implying

$$H(P)^4 = \max\{|s|^4, |t|^4\} \leq \max(K', K'') \max\{|m|, |n|\} \leq KH(2P),$$

for some K. □

Remark 6.3.10 — F_1, G_1, F_2, G_2 can be computed explicitly using the general theory of resultants. Compute

$$\operatorname{Res}(m(s,1), n(s,1)) = \begin{vmatrix} 0 & 0 & -2A & -8B & A^2 & 0 \\ 0 & 1 & 0 & -2A & -8B & A^2 \\ 1 & 0 & 4 & 0 & -2A & -8B \\ 0 & 4 & 0 & 4 & 0 & -2A \\ -2A & 0 & -2A & 0 & 4 & 0 \\ A^2 & -2A & -8B & 4 & 0 & 4B \end{vmatrix} = 2^8(4A^3 + 27B^2)^2$$

and show that

$$F_1 m + G_1 n = 4(4A^3 + 27B^2)t^7 \quad F_2 m + G_2 n = 4(4A^3 + 27B^2)s^7$$

for

$$\begin{aligned} F_1 &= 12s^2t + 16At^3 \\ G_1 &= -s^3 + 54st^2 + 27Bt^3 \\ F_2 &= 4(4A^3 + 27B^2)s^3 - 4A^2Bs^2t + 4A(3A^3 + 22B^3)st^2 + 12B(A^3 + 8B^2)t^3 \\ G_2 &= A^2Bs^3 + A(4A^3 + 32B^2)s^2t + 2B(13A^3 + 96B^2)st^2 - 3A^2(A^3 + 8B^2)t^3 \end{aligned}$$

In the next lemma, we will use the following two elementary facts which we leave as exercises.

Exercise 6.3.11. Let $c_1, c_2, d_1, d_2 \in \mathbb{Z}$ then

$$\max(|c_1|, |d_1|) \cdot \max(|c_2|, |d_2|) \leq 2\max(|c_1c_2|, |c_1d_2 + c_2d_1|, |d_1d_2|).$$

Solution: Without loss of generality assume $|c_1| \leq |d_1|$. Now, consider the case $|c_2| \leq |d_2|$. Then the left side is $|d_1d_2|$ and the right side is at least $2|d_1d_2|$. So, the desired inequality holds. Next, consider the case $|c_2| \geq |d_2|$. Then, the left side is $|c_2d_1|$. If $|c_2| \geq |d_1|$ or $|2d_2| \geq |c_2|$, then we see that

$$|c_2d_1| \leq 2\max(|c_2c_1|, |c_1d_2|).$$

So, assume $2|c_1| \leq |d_1|$ and $|2d_2| \leq |c_2|$, then we have

$$2|c_1d_2 + c_2d_1| \geq |c_2d_1| \geq |c_2d_1|$$

as required. □

Exercise 6.3.12. Let $c_1, c_2, d_1, d_2 \in \mathbb{Z}$, with $(c_i, d_i) = 1$ for $i = 1, 2$. Then

$$(c_1c_2, c_1d_2 + c_2d_1, d_1d_2) = 1.$$

Lemma 6.3.13

There exists a constant $c > 0$ such that

$$|h(P+Q) + h(P-Q) - 2h(P) - 2h(Q)| < c.$$

Proof. We will prove the equivalent statement that there exist constants $c', c'' > 0$ such that

$$h(P+Q) + h(P-Q) < 2h(P) + 2h(Q) + c' \tag{6.1}$$
$$h(P+Q) + h(P-Q) > 2h(P) + 2h(Q) - c'' \tag{6.2}$$

Let us prove the inequality (6.1) first. We really prove this in H, so the equivalent statement to prove is

$$H(P+Q)H(P-Q) \leq CH(P)^2H(Q)^2.$$

Let $x_1 = x(P)$, $x_2 = x(Q)$. The x-coordinates of $P+Q$ and $P-Q$ are the roots of a quadratic equation dependent on x_1, x_2 (we leave this as an exercise). Indeed, from the group law, one can derive:

$$x(P+Q) + x(P-Q) = \frac{2(x_1x_2 + A)(x_1 + x_2) + 4B}{(x_1 - x_2)^2}$$

$$x(P+Q)x(P-Q) = \frac{(x_1x_2 - A)^2 - 4B(x_1 + x_2)}{(x_1 - x_2)^2}$$

Put $P = (\frac{a_1}{b_1}, y_1)$, $Q = (\frac{a_2}{b_2}, y_2)$, $P+Q = (\frac{a_3}{b_3}, y_3)$, $P-Q = (\frac{a_4}{b_4}, y_4)$ with $(a_i, b_i) = 1$. It follows using the formulae above that

$$\frac{a_3}{b_3} + \frac{a_4}{b_4} = \frac{g_1}{g_3}, \qquad \frac{a_3a_4}{b_3b_4} = \frac{g_2}{g_3}$$

where

$$\begin{aligned} g_1 &= 2(a_1a_2 + Ab_1b_2)(a_1b_2 + a_2b_1) + 4Bb_1^2b_2^2 \\ g_2 &= (a_1a_2 - Ab_1b_2)^2 - 4B(a_1b_2 + a_2b_1)b_1b_2 \\ g_3 &= (a_1b_2 - b_1a_2)^2. \end{aligned}$$

Since $(a_3, b_3) = (a_4, b_4) = 1$, we also have $(a_3a_4, a_3b_4 + a_4b_3, b_3b_4) = 1$, hence we can find integers x, y, z such that

$$a_3a_4x + (a_3b_4 + a_4b_3)y + b_3b_4z = 1.$$

Then,

$$\begin{aligned} g_3 &= g_3(a_3a_4)x + g_3(a_3b_4 + a_4b_3)y + g_3(b_3b_4)z \\ &= g_2(b_3b_4)x + g_1(b_3b_4)y + g_3(b_3b_4)z. \end{aligned}$$

Hence $b_3b_4 \mid g_3$ and so $|b_3b_4| \leq |g_3|$. From this and $g_3(a_3b_4 + a_4b_3) = g_1(b_3b_4)$ and $g_3(a_3a_4) = g_2(b_3b_4)$, it follows that $|a_3b_4 + a_4b_3| \leq |g_1|$ and $|a_3a_4| \leq |g_2|$. Therefore, we have

$$\begin{aligned} H(P+Q)H(P-Q) &= \max(|a_3|, |a_4|)\max(|a_4|, |b_4|) \\ &\leq 2\max(|a_3a_4|, |a_3b_4 + a_4b_3|, |b_3b_4|) \\ &\leq 2\max(|g_1|, |g_2|, |g_3|). \end{aligned}$$

Now, let $H_1 = H(P) = \max(|a_1|, |b_1|)$ and $H_2 = H(Q) = \max(|a_2|, |b_2|)$.

$$\begin{aligned} |g_1| &\leq 2(H_1H_2 + |A|H_1H_2)(H_1H_2 + H_1H_2) + 4|B|H_1^2H_2^2 = 4(|A| + 1 + |B|)H_1^2H_2^2 \\ |g_2| &\leq (H_1H_2 + |A|H_1H_2)^2 + 4|B|(H_1H_2 + H_2H_1)H_1H_2 = ((1 + |A|)^2 + 8|B|)H_1^2H_2^2 \\ |g_3| &\leq 4H_1^2H_2^2 \end{aligned}$$

So, there is a constant $C > 0$ such that

$$H(P+Q)H(P-Q) \leq CH(P)^2H(Q)^2.$$

Taking log, this gives (6.1). Next, changing $P \to P+Q$ and $Q \to P-Q$, we get

$$h(2P) + h(2Q) < 2h(P+Q) + 2h(P-Q) + c', \quad c' > 0.$$

Now, by Lemma 6.3.9, we have $h(2P) \geq 4h(P) - C_0$ and $h(2Q) \geq 4h(Q) - C_0$ for some $C_0 > 0$. Hence, we get

$$4h(P) + 4h(Q) - 2C_0 < 2h(P+Q) + 2h(P-Q) + c',$$

Letting $c'' = \frac{c'+2C_0}{2}$, we get

$$2h(P) + 2h(Q) - c'' < h(P+Q) + h(P-Q)$$

Thus the inequality (6.2) is established. □

Exercise 6.3.14. Let E be an elliptic curve given by $y^2 = x^3 + Ax + B$. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be points on E, with $x_i \neq \infty$. Show that $\dfrac{4(x_1 - x_2)^2}{(x_1+x_2)^2 - 4A}$

$$= (x_1^2 - 2x_1x_2 + x_2^2 + A)^2 + 2B(y_1^2 + y_2^2 + x_1^2x_2 + x_2^2x_1 + A^2)$$

as a polynomial in t. Show further that the x-coordinates of $P+Q$ and $P-Q$ are zeros of the polynomial in t obtained on putting $t = x_1 + x_2$. Thus verify the formulae for g_1, g_2, g_3 in the above proof.

Proof (of Theorem 6.3.7). On putting $Q = P$ in Lemma (6.3.13) we get

$$|h(2P) - 4h(P)| < c, \quad \text{for all } P.$$

Now,

$$\lim_{n\to\infty} \frac{1}{4^n} h(2^nP) = h(P) + \sum_{j=1}^{\infty} \frac{1}{4^j}\left(h(2^jP) - 4h(2^{j-1}P)\right),$$

which is just a telescoping sum. We have $|h(2^jP) - 4h(2^{j-1}P)| \leq \frac{c}{4^j}$. Hence, the infinite sum converges. Moreover, $|\hat{h}(P) - h(P)| \leq \frac{c}{3}$ (Note that this implies that there are finitely many P below certain height with respect to h, since that is manifestly true for h).

To prove the parallelogram law, observe that

$$\frac{1}{4^n}|h(2^nP + 2^nQ) + h(2^nP - 2^nQ) - 2h(2^nP) - 2h(2^nQ)| \leq \frac{c}{4^n}.$$

Thus, letting $n \to \infty$, the RHS goes to 0, so we achieve equality. Therefore,

$$\hat{h}(P+Q) + \hat{h}(P-Q) - 2\hat{h}(P) - 2\hat{h}(Q) = 0.$$

We do this by induction. For $m = 0, 1$, this is clearly true. Thus, assume that $\hat{h}(mP) = m^2\hat{h}(P)$ and $\hat{h}((m-1)P) = (m-1)^2\hat{h}(P)$. Then, we prove true for $m+1$.

$$\begin{aligned}\hat{h}((m+1)P) &= -\hat{h}((m-1)P) + 2\hat{h}(mP) + 2\hat{h}(P) \\ &= (-(m-1)^2 + 2m^2 + 2)\hat{h}(P) = (m+1)^2\hat{h}(P)\end{aligned}$$

Therefore $\hat{h}(mP) = m^2\hat{h}(P)$.

If P is torsion, then $mP = \mathcal{O}$, then $m^2\hat{h}(P) = \hat{h}(mP) = \hat{h}(P) = \mathcal{O}$. Conversely, if $\hat{h}(P) = 0$, then $\hat{h}(mP) = 0$ for all m, but there are finitely many points of height 0, hence the set of multiples of P must be finite, that is P is a torsion point. □

§6.3.i The Pairing $\langle P, Q\rangle$ on $E(\mathbb{Q})$

Let us point out a useful application of heights.

Corollary 6.3.15

There exists a unique $\mathbb{Z}$-bilinear form $\langle P, Q\rangle$ on $E(\mathbb{Q})$ such that $\langle P, P\rangle = \hat{h}(P)$ given by the following pairing

$$\langle P, Q\rangle = \frac{1}{2}\left(\hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q)\right)$$

Proof. By symmetry, it suffices to prove that

$$\langle P+R, Q\rangle = \langle P, Q\rangle + \langle R, Q\rangle$$

which in terms of $\hat{h}$ is

$$\hat{h}(P+Q+R) - \hat{h}(P+R) - \hat{h}(P+Q) - \hat{h}(Q+R) + \hat{h}(P) + \hat{h}(Q) + \hat{h}(R) = 0.$$

Four applications of the parallelogram law gives

$$\begin{aligned}
\hat{h}(P+R+Q) + \hat{h}(P+R-Q) - 2\hat{h}(P+R) - 2\hat{h}(Q) &= 0\\
\hat{h}(P-R+Q) + \hat{h}(P+R-Q) - 2\hat{h}(P) - 2\hat{h}(Q-R) &= 0\\
\hat{h}(P-R+Q) + \hat{h}(P+R+Q) - 2\hat{h}(P+Q) - 2\hat{h}(R) &= 0\\
2\hat{h}(R+Q) + 2\hat{h}(R-Q) - 4\hat{h}(R) - 4\hat{h}(Q) &= 0
\end{aligned}$$

The alternating sum of these four equations give twice the equation noted above. Of course, this satisfies the equation then. □

Corollary 6.3.16

Suppose $P_1, P_2, \ldots, P_r$ are points in $E(\mathbb{Q})$, and the $r \times r$ determinant

$$\det(\langle P_i, P_j\rangle) \neq 0$$

then $P_1, P_2, \ldots, P_r$ are independent.

Proof. Suppose $a_1P_1 + a_2P_2 + \ldots a_rP_r = 0$ with $a_i \in \mathbb{Z}$ not all zero. Say $a_r \neq 0$, then the last row of the matrix $\langle P_i, P_j\rangle$ is a linear combination of the first $r-1$ rows, hence the determinant vanishes. □

Example 6.3.17

Consider $E\colon y^2 = x^3 + 73$. Then $P = (2, 9)$ and $Q = (3, 10)$ are on $E(\mathbb{Q})$. We can compute

$$\begin{vmatrix} \langle P, P\rangle & \langle P, Q\rangle \\ \langle Q, P\rangle & \langle Q, Q\rangle \end{vmatrix} = \begin{vmatrix} 0.9239 & -0.9770 \\ -0.9770 & 1.9927 \end{vmatrix} = 0.8865\ldots \neq 0$$

Hence, P and Q are independent.

§6.3.ii Proof of Mordell-Weil

Now, we can complete the proof of Mordell-Weil theorem.

Theorem 6.3.18 (Mordell-Weil)
$E(\mathbb{Q})$ is finitely generated.

Proof. It remains to define a norm function. We let $|P| = \sqrt{\hat{h}(P)}$. We need to show that this satisfies the properties of a norm function. All properties of a norm are clear from the properties of $\hat{h}$, except for the triangle inequality. To see the triangle inequality, consider the pairing from Corollary 6.3.15. We claim that this satisfies the Cauchy-Schwarz inequality:

$$|\langle P, Q\rangle|^2 \leq \hat{h}(P)\hat{h}(Q)$$

Indeed, from the bilinearity, we have

$$0 \leq \hat{h}(mP + nQ) = m^2\hat{h}(P) + 2mn\langle P, Q\rangle + n^2\hat{h}(Q)$$

Hence,

$$\hat{h}(P)t^2 + 2\langle P, Q\rangle t + \hat{h}(Q) \geq 0$$

for all $t \in \mathbb{Q}$. By continuity the inequality extends to all of $t \in \mathbb{R}$. Therefore, this quadratic polynomial has at most one root, hence its discriminant must be ≤ 0, which gives the Cauchy-Schwarz inequality, that is, $|\langle P, Q\rangle| \leq |P||Q|$. Now, the triangle inequality is an easy consequence:

$$\begin{aligned} |P+Q|^2 = \langle P, P\rangle + 2\langle P, Q\rangle + \langle Q, Q\rangle &= |P|^2 + 2\langle P, Q\rangle + |Q|^2 \\ &\leq |P|^2 + 2|P||Q| + |Q|^2 = (|P| + |Q|)^2 \end{aligned}$$

Equivalently, $|P + Q| \leq |P| + |Q|$, as desired. □

Exercise 6.3.19. Show that an elliptic curve cannot have two independent rational points of order 4, i.e. points a, b such that $4a = 4b = \mathcal{O}$, $2a \neq 2b$, $2a \neq \mathcal{O}$, $2b \neq \mathcal{O}$.

Solution: Suppose we did. Then, we have two distinct 2-torsion points, $(r, 0), (s, 0)$, so we have three 2-torsion points, say given by $(t, 0)$.

$$y^2 = (x - t)(x - r)(x - s).$$

Since two of these three come from 4-torsion points, they have preimages under the map δ. In other words, they lie in the kernel of δ. Suppose that $(r, 0)$ and $(s, 0)$ are the two maps which map to zero under δ. Then, we must have that $r - t$ and $r - s$ to be a square and $s - t$ and $s - r$ to be a square. However, $r - s$ and $s - r$ are both squares then, but these are both integers. A priori, these are reals, and the negative and positive of a real cannot both be squares, so this is not possible. □

7 Elliptic Curves over $\mathbb{C}$

The goal of this section to show that elliptic curves over $\mathbb{C}$ correspond to Riemann surface structures on a torus.

§7.1 Elliptic Functions

Definition 7.1.1. Let ω_1, ω_2 be two complex number that are linearly independent over $\mathbb{R}$. Then,

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \{m\omega_1 + n\omega_2 : n, m \in \mathbb{Z}\} \subset \mathbb{C}$$

is a **lattice** i.e. a discrete subgroup of $(\mathbb{C}, +)$.

We have a Riemann surface structure on the torus $\mathbb{C}/\Lambda$ induced by the covering map $\mathbb{C} \to \mathbb{C}/\Lambda$. Moreover, since Λ is a normal subgroup of $(\mathbb{C}, +)$, the factor group $\mathbb{C}/\Lambda$ inherits an abelian group structure.

Definition 7.1.2. An **elliptic function** is a doubly periodic meromorphic function on $\mathbb{C}$.

Explicitly, a (ω_1, ω_2)-periodic function on $\mathbb{C}$ is a map $f : \mathbb{C} \to \mathbb{C}$ such that

$$f(z + \omega_1) = f(z), \qquad f(z + \omega_2) = f(z)$$

Such a map descends to a map $\mathbb{C}/\Lambda \to \mathbb{C}$ and since $\mathbb{C}/\Lambda$ is compact, this implies f is bounded. Therefore, f cannot be holomorphic. Otherwise, f would be entire and bounded. Let us next turn to doubly periodic functions. Recall from complex analysis that a function $f : \mathbb{C} \to \mathbb{C}$ is meromorphic if at every $a \in \mathbb{C}$ it has a Laurent series expansion

$$f(z) = \sum_{i=\infty}^{\infty} c_i (z - a)^i$$

where $i_0 \in \mathbb{Z}$. If $i_0 < 0$ we say f has a **pole** of order $|i_0|$ at a, and if $i_0 > 0$, we say f has a zero of order i_0. We call $c_{-1} = \mathrm{Res}_a f$ the **residue** of f at a.

Definition 7.1.3. A **fundamental parallelogram** Π for the lattice $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ is a subset of $\mathbb{C}$ of the form

$$z_0 + t_1\omega_1 + t_2\omega_2, \quad 0 \le t_i < 1.$$

Clearly, a doubly periodic function on $\mathbb{C}$ is determined by its values on a fundamental parallelogram Π.

Proposition 7.1.4

Let Π be a fundamental parallelogram for Λ, and f is an elliptic function which has no poles on its boundary $\partial\Pi$. Then the sum of the residues of f in Π is 0.

Proof. Let $a_1, \dots, a_m$ be the poles of f inside Π. By the Cauchy integral formula and the periodicity of f, we have

$$2\pi i \sum_{i=1}^{m} \operatorname{Res}_{a_i} f = \oint_{\partial\Pi} f(z)\, dz = 0,$$

since opposing edges cancel. □

Proposition 7.1.5

Let Π be a fundamental parallelogram for Λ, and f is an elliptic function which has no poles on its boundary $\partial\Pi$. Let $\{m_i\}$ be the orders of various zeros of f and $\{n_j\}$ be the orders of various poles. Then $\sum_i m_i = \sum_j n_j$.

Proof. Consider $g(z) = \frac{f'(z)}{f(z)}$. The residue of $g(z)$ at a point z_0 is n if $n > 0$ is the order of zero of f at z_0, 0 if f has no pole or zero at z_0 and $-n$ if $n > 0$ is the order of a pole of f at z_0. Now, applying Proposition 7.1.4 gives the result. □

Definition 7.1.6. The **degree** of an elliptic function f is the number $\sum_i m_i = \sum_j n_j$ from Proposition 7.1.5. We denote it as $\deg(f)$.

> **Exercise 7.1.7.** Let f an elliptic function and $d = \deg(f)$. Let $p_1, p_2, \dots, p_d$ and $q_1, q_2, \dots, q_d$ be the zeros and poles of f, repeated as necessary. Show that $\sum_{i=1}^{d} p_i = \sum_{i=1}^{d} q_i \pmod{\Lambda}$. (Hint: Use $zf'(z)/f(z)$.)

Solution: We use the residue theorem again on $\partial\Pi$. Note that we can assume that $\partial\Pi$ doesn't contain any poles or zeroes, since f admits finitely many such points, so we can just perturb by some $z_0 \in \mathbb{C}$: $z_0 + \partial\Pi$. The key is to integrate $z\frac{f'(z)}{f(z)}$ as stated in the hint. Let γ_1 denote the bottom of the fundamental parallelogram and γ_3 denote the top. Then, we can calculate that

$$\int_{\gamma_3} z\frac{f'(z)}{f(z)} dz = -\int_{\gamma_1} (z+\omega_2)\frac{f'(z+\omega_2)}{f(z+\omega_2)} dz = -\int_{\gamma_1} (z+\omega_2)\frac{f'(z)}{f(z)} dz.$$

In the above, we used doubly-periodicity at the last equality. Then, if we sum the integrals along γ_1 and γ_3, we find the following:

$$-\int_{\gamma_1} \omega_2 \frac{f'(z)}{f(z)} dz = -\int_{z_0}^{z_0+\omega_1} \omega_2 \frac{f'(z)}{f(z)} dz = \omega_2 \left(\log(f(z_0+\omega_1)) - \log(f(z_0))\right) = -\omega_2 \cdot 2\pi i k,$$

where $k \in \mathbb{Z}$. If we do the same for vertical components, we find that

$$\int_{z_0+\partial\Pi} z\frac{f'(z)}{f(z)} dz = 2\pi i(\ell\omega_1 - \omega_2 k) \in 2\pi i \cdot \Lambda.$$

On the other hand, we apply the residue theorem to the LHS. If f has an order m zero at some point p, then $f(z) = (z-p)^m h(z)$, for some holomorphic h such that $h(p) \neq 0$. Thus, locally around p, we have that

$$z\frac{f'(z)}{f'(z)} = m\frac{z}{(z-p)} + \frac{h'(z)}{h(z)} = m \cdot \frac{p}{z-p} + m + \frac{h'(z)}{h(z)}.$$

Therefore, the residue theorem tells us that the LHS is equal to

$$\sum_{x\in\mathbb{C}/\Lambda} \nu_f(x)\cdot x,$$

since m is the order of f at x. Since there are finitely many zeroes/ poles, this expands to write

$$\sum_{i=1}^{d} p_i - \sum_{i=1}^{d} q_i \in \Lambda,$$

by the integral above. □

By Proposition 7.1.4, we see that an elliptic function must have at least two poles (counted with multiplicity) on the torus $\mathbb{C}/\Lambda$. Therefore, the simplest doubly periodic functions must have two poles, either a double pole or two simple poles with opposite residues. We can take a function $1/z^2$ which has a double pole at the origin and make it periodic by summing its translates over the period lattice. This naive attempt leads to a divergent series, but the idea is salvaged by considering

$$\wp(z) = \frac{1}{z^2} + \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} \left(\frac{1}{(z-(m\omega_1+n\omega_2))^2} - \frac{1}{(m\omega_1+n\omega_2)^2}\right)$$

This function is called the **Weierstrass $\wp$ function**.

Lemma 7.1.8

Given a lattice $\Lambda \subset \mathbb{C}$ and $r > 2$, the sum $\sum_{\lambda\in\Lambda\setminus\{(0,0)\}} \frac{1}{|\lambda|^r}$ converges.

Proof. There exists a constant $c > 0$, such that

$$|m\omega_1 + n\omega_2| \geq c(|m| + |n|) \quad \text{for all } (m,n) \in \mathbb{Z}^2$$

It is easy to see this by noticing that the function $(m,n) \mapsto \frac{|m\omega_1+n\omega_2|}{(|m|+|n|)}$ is defined on $\mathbb{R}^2 \setminus \{(0,0)\}$ and is invariant under scaling $(m,n) \mapsto (tm,tn)$, hence its values are determined by its values on the unit circle, which is a compact set.

Next, let us consider the $4k$ elements $m\omega_1 + n\omega_2$ of Λ such that $|m| + |n| = k$. For such (m,n), we have $|m\omega_1 + n\omega_2| \geq ck$, thus we have:

$$\sum_{\lambda\in\Lambda\setminus\{(0,0)\}} \frac{1}{|\lambda|^r} \leq \sum_{k=1}^{\infty} \frac{4k}{(ck)^r} = 4c^{-r}\sum_{k=1}^{\infty} k^{1-r} < \infty$$

□

Corollary 7.1.9

$\wp(z)$ defines a meromorphic function on $\mathbb{C}/\Lambda$ with a double pole at each lattice point and no other pole.

Proof. We show that the sum converges for $|\lambda| > 2|z|$, since there are only finitely many terms for $|\lambda| \leq 2|z|$. We have for $|\lambda| > 2|z|$,

$$\left|\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2}\right| = \left|\frac{(z-2\lambda)z}{(z-\lambda)^2\lambda^2}\right| \leq \frac{(|z|+2|\lambda|)|z|}{|z/2|^2|\lambda|^2} \leq \frac{10|z|}{|\lambda|^3}$$

Hence, by Lemma 7.1.8, the series defining $\wp(z)$ converges for all z.

Note from the form of the construction that $\wp(z) = \wp(-z)$, hence $\wp(z)$ is an even meromorphic function. We can get an odd meromorphic function by differentiation

$$\wp'(z) = -2 \sum_{(m,n)\in\mathbb{Z}^2} \frac{1}{(z-(m\omega_1+n\omega_2))^3}.$$

$\wp'(z)$ is a doubly periodic meromorphic function with a pole of order 3 at 0. Moreover,

$$\wp'(-z) = -\wp'(z)$$

In other words, $\wp'(z)$ is an odd function.

$\wp'(z)$ is also well-defined. It is obvious that $\wp'(z)$ is (ω_1, ω_2)-periodic. To see that $\wp(z)$ is (ω_1, ω_2)-periodic, consider the functions:

$$\wp(z+\omega_1) - \wp(z) \qquad \text{and} \qquad \wp(z+\omega_2) - \wp(z)$$

Differentiating these, by double periodicity of $\wp'(z)$, we see that both of these functions have to be constant. To determine the constant, we evaluate these functions at $z = -\omega_1/2$ and $z = -\omega_2/2$, respectively and use the fact that $\wp(z)$ is even. Thus, the constant is zero, and we have that $\wp(z)$ is an elliptic function. It is also clear then from the formulae that there are double points at each lattice point, since $\wp(z) = \wp(z-\omega_1) = \wp(z-\omega_2)$. □

Remark 7.1.10 — Because it has a unique pole at 0 with multiplicity 2, $\wp(z)$ has degree 2, therefore it must have two zeros, and similarly $\wp'(z)$ must have three zeros. It is easy to find the zeros of $\wp(z)$. Namely,

$$\wp'(\frac{\omega_1}{2}) = \wp'(\frac{\omega_2}{2}) = \wp'(\frac{\omega_1+\omega_2}{2}) = 0$$

This follows from periodicity of $\wp'(z)$ and that it is an odd function. Since $\wp(z)$ is an even function, the zeros of $\wp(z)$ must be given by $\pm z_0$ for some z_0. It is difficult to give a formula for z_0.

Exercise 7.1.11. Let e_1, e_2, e_3 be the values of $\wp(\omega_1/2)$, $\wp(\omega_2/2)$, $\wp((\omega_1+\omega_2)/2)$

1. Show that e_1, e_2, e_3 are all distinct.
2. Show that for any $a \in \mathbb{C} \setminus \{e_1, e_2, e_3\}$, the equation $\wp(z) = a$ has exactly two distinct solutions.

Solution: 1. Let $\omega_3 = \omega_1 + \omega_2$. Consider the polynomials $h_i(z) = \wp(z) - e_i$ for $i = 1, 2, 3$. Then

$$h_i(\omega_j/2) = h_i(\omega_i/2) = 0.$$

Hence $h_i(z)$ vanishes to order 2 at $\omega_i/2$. On the other hand, it has a double pole of order 2 at 0 and no other poles. Since the number of poles equals the number of zeroes for elliptic functions, we find that $h_i(z)$ does not vanish anywhere else. In particular, $h_i(\omega_j/2) \neq 0$ for $i \neq j$. It follows that $h_i(z)$ are 3 distinct polynomials, hence e_1, e_2, e_3 are all distinct.

2. This follows from the fact that $\wp(z)$ has degree 2, and the roots of $\wp'(z)$ are $\omega_1/2, \omega_2/2$ and $(\omega_1+\omega_2)/2$. □

We are now ready to compute the function field of $\mathbb{C}/\Lambda$.

Theorem 7.1.12

The field of meromorphic functions on $\mathbb{C}/\Lambda$, $K(\mathbb{C}/\Lambda)$, is generated by $\wp(z)$ and $\wp'(z)$ over $\mathbb{C}$. In other words, every elliptic function is a rational function of $\wp(z)$ and $\wp'(z)$.

Proof. Suppose f is an elliptic function. We can write it as a sum of even and odd functions:

$$f(z) = \frac{f(z)+f(-z)}{2} \quad + \quad \frac{f(z)-f(-z)}{2}.$$

Furthermore, if f is odd then the product $f\wp'$ is even, so it suffices to prove that every even function is a rational function of $\wp$. Thus, let us assume that f is an even elliptic function. f can be viewed as a holomorphic function from $\mathbb{C}/\Lambda \to \mathbb{P}^1$. Let $p_1, \dots, p_n$ and $q_1, \dots, q_n$ be the zeros and poles of f, repeated as necessary. Note that their numbers is the same n since this is the degree of f. Given a zero a and a pole b of f, we define:

$$g(z) = \begin{cases} \frac{f(z)(\wp(z)-\wp(b))}{\wp(z)-\wp(a)}, & a \notin \Lambda,\ b \notin \Lambda, \\ f(z)(\wp(z)-\wp(b)), & a \in \Lambda,\ b \notin \Lambda, \\ \frac{f(z)}{\wp(z)-\wp(a)}, & a \notin \Lambda,\ b \in \Lambda. \end{cases}$$

Now, $g(z)$ is even and doubly periodic meromorphic function, and g is either constant or else

$$\#\{g^{-1}(0)\} = \#\{f^{-1}(0)\} - 2$$

where $\#$ denotes the count of zeros in the quotient $\mathbb{C}/\Lambda$. This equality can be verified by using the fact that for $c \notin \Lambda$, the function

$$\wp(z) - \wp(c)$$

has exactly two zeros at $\pm c$, and has exactly two poles at 0. Hence, after repeating this construction finitely many times, we obtain that $f(z)$ is a rational function of $\wp(z)$. $\square$

Remark 7.1.13 — This argument of canceling poles and zeroes, then reaching that it's a holomorphic map on a compact surface, so constant, is often used.

Corollary 7.1.14

The field $K(\mathbb{C}/\Lambda)$ of meromorphic functions on $\mathbb{C}/\Lambda$ is isomorphic to the field extension $\mathbb{C}(\wp, \wp')$.

Now, since $(\wp'(z))^2$ is an even elliptic function, it should be expressible as a rational function of $\wp(z)$. In fact, we have:

Theorem 7.1.15

Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ be a lattice. Let $e_1 = \wp(\omega_1/2), e_2 = \wp(\omega_2/2)$ and $e_3 = \wp((\omega_1+\omega_2)/2)$. Then,

$$(\wp'(z))^2 = 4\wp(z)^3 - g_2\wp(z) - g_3 = 4(\wp(z)-e_1)(\wp(z)-e_2)(\wp(z)-e_3)$$

where $g_2 = 60\sum_{\lambda\in\Lambda\setminus\{0,0\}} \lambda^{-4}$ and $g_3 = 140\sum_{\lambda\in\Lambda\setminus\{0,0\}} \lambda^{-6}$.

Proof. The even function $\wp$ has Laurent series expansion:

$$\wp(z) = \frac{1}{z^2} + 0 + az^2 + bz^4 + \dots$$

Then,

$$\wp'(z) = -2\frac{1}{z^3} + 2az + 4bz^3 + \dots$$

Consider the function

$$k(z) = \wp'(z)^2 - 4\wp(z)^3 + g_2\wp(z) + g_3$$

where $g_2 = 20a$ and $g_3 = 28b$. Using the above formulae, we can check that $k(z)$ restricts to a holomorphic function near 0 and it vanishes at 0. Furthermore, since $\wp, \wp'$ are doubly periodic, we conclude that $k(z)$ is also doubly periodic, hence a constant and is identically zero. Thus, we conclude that

$$\wp'(z)^2 - 4\wp(z)^3 + g_2\wp(z) + g_3 = 0$$

The required series expansions of g_2 and g_3 can be obtained by computing the series expansion of

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda\in\Lambda\setminus\{0,0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2}\right)$$

at $z = 0$. Namely,

$$\begin{aligned}
\wp(z) &= \frac{1}{z^2} + \sum_{\lambda\in\Lambda\setminus\{(0,0)\}} \left(\frac{1}{\lambda^2}\Big(1 + \frac{z}{\lambda} + \left(\frac{z}{\lambda}\right)^2 + \dots\Big)^2 - \frac{1}{\lambda^2}\right) \\
&= \frac{1}{z^2} + \sum_{\lambda\in\Lambda\setminus\{(0,0)\}} \sum_{m=1}^{\infty} (m+1)\left(\frac{z}{\lambda}\right)^m \frac{1}{\lambda^2} \\
&= \frac{1}{z^2} + \sum_{m=1}^{\infty} c_m z^m
\end{aligned}$$

where

$$c_m = \sum_{\lambda\in\Lambda\setminus\{(0,0)\}} \frac{m+1}{\lambda^{m+2}}.$$

Finally, note that $\wp'(\frac{\omega_1}{2}) = \wp'(\frac{\omega_2}{2}) = \wp'(\frac{\omega_1+\omega_2}{2}) = 0$ for $i = 1, 2, 3$ as $\wp'(z)$ is an odd function. The polynomial $4x^3 - g_2x - g_3$ has roots e_1, e_2, e_3. $\square$

As e_1, e_2, e_3 are distinct, it follows that $g_2^3 - 27g_3^2 \neq 0$, hence the cubic

$$E\colon y^2 = 4x^3 - g_2x - g_3 = 4(x-e_1)(x-e_2)(x-e_3)$$

defines an elliptic curve E over $\mathbb{C}$.

We see that there is a bijection

$$\Phi\colon \mathbb{C}/\Lambda \to E(\mathbb{C}) \quad z \mapsto (\wp(z), \wp'(z)) \quad 0 \mapsto \mathcal{O}$$

where $E(\mathbb{C})$ is the elliptic curve.

Recall that $\mathbb{C}/\Lambda$ has the structure of an abelian group. We next show that the bijection Φ is actually an isomorphism of abelian groups. We need a lemma first.

Lemma 7.1.16

Suppose $\psi\colon \mathbb{C}/\Lambda \times \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda$ is holomorphic in each variable and continuous, then there exists $a, b, c \in \mathbb{C}$ such that $\psi(z_1, z_2) = az_1 + bz_2 + c \pmod{\Lambda}$ for all $z_1, z_2 \in \mathbb{C}$.

Proof. Using the covering maps $\mathbb{C} \to \mathbb{C}/\Lambda$, we can lift ψ to a function $\Psi\colon \mathbb{C} \times \mathbb{C} \to \mathbb{C}$ which is continuous and holomorphic. This satisfies

$$\Psi(z_1 + m\omega_1 + n\omega_2,\, z_2) = \Psi(z_1, z_2) + m_1\omega_1 + n_1\omega_2$$

with $m_1, n_1 \in \mathbb{Z}$ depending on z_1, z_2, m, n. For fixed m, n, we have that m_1, n_1 is continuous in (z_1, z_2), so constant. Therefore, differentiation gives

$$\frac{\partial \Psi}{\partial z_1}(z_1 + m\omega_1 + n\omega_2, z_2) = \frac{\partial \Psi}{\partial z_1}(z_1, z_2), \qquad \frac{\partial \Psi}{\partial z_2}(z_1 + m\omega_1 + n\omega_2, z_2) = \frac{\partial \Psi}{\partial z_2}(z_1, z_2)$$

Thus, $\frac{\partial \Psi}{\partial z_1}$ and $\frac{\partial \Psi}{\partial z_2}$ are holomorphic and also periodic in the first variable, and therefore constant in z_1. Similarly, we can show that they are constant in z_2. Hence, $\frac{\partial \Psi}{\partial z_1} = a$ and $\frac{\partial \Psi}{\partial z_2} = b$ for some constants $a, b \in \mathbb{C}$ and $F(z_1, z_2) = az_1 + bz_2 + c$ for some $c \in \mathbb{C}$. □

Theorem 7.1.17

The biholomorphic map $\Phi\colon \mathbb{C}/\Lambda \to E(\mathbb{C})$ sending $z \mapsto (\wp(z), \wp'(z))$ satisfies

$$\Phi(z_1 + z_2) = \Phi(z_1) + \Phi(z_2)$$

Proof. Consider $\psi\colon \mathbb{C}/\Lambda \times \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda$ given by $\psi(z_1, z_2) = \Phi^{-1}(\Phi(z_1) + \Phi(z_2)) \pmod{\Lambda}$. This is holomorphic in each variable and is continuous. Hence, $\psi(z_1, z_2) = az_1 + bz_2 + c \pmod{\Lambda}$ for some $a, b, c \in \mathbb{C}$. Since $\psi(0, 0) = 0$, we have $c = 0$. Since $\psi(z_2, 0) = z_2 \pmod{\Lambda}$, it follows that $a = 1$ and similarly $b = 1$. Thus, $\psi(z_1, z_2) = z_1 + z_2 \pmod{\Lambda}$. Hence, $\Phi(z_1 + z_2) = \Phi(z_1) + \Phi(z_2)$ as required. □

Exercise 7.1.18. The explicit formula for computing with the group law is reflected in the following identities satisfied by $\wp(z)$. Show that

$$\wp(z + w) = \left(\frac{\wp'(z) - \wp'(w)}{2(\wp(z) - \wp(w))}\right)^2 - \wp(z) - \wp(w) \quad \text{if } \wp(z) \neq \wp(w)$$

$$\wp(2z) = \left(\frac{\wp''(z)}{2\wp'(z)}\right)^2 - 2\wp(z) \quad \text{if } \wp'(z) \neq 0$$

Thus, we have seen that any lattice $\Lambda \subset \mathbb{C}$ gives rise to an elliptic curve $E(\Lambda)\colon y^2 = 4x^3 - g_2x - g_3$ in the Weierstrass form. On the other hand, we have seen before that any elliptic curve $E(\mathbb{C})$ can be brought into Weierstrass form $y^2 = 4x^3 - Ax - B$. The natural question is then whether there exists a lattice Λ such that $E(\Lambda) = E(\mathbb{C})$. The answer is yes. This is called the Uniformization Theorem.

Solution: Let $P = (\wp(t), \wp'(t))$, $Q = (\wp(w), \wp'(w))$ be two points on $E(\mathbb{C})$. Then, We apply the addition formula:

$$x_{P+Q} := \frac{1}{4} \cdot \left(\frac{\wp'(w) - \wp'(t)}{\wp(w) - \wp(t)}\right)^2 - \wp(t) - \wp(w).$$

Now, the LHS corresponds to $\wp(t+w)$ by the theorem above. Therefore, the first relation is verified. For the second, we use the doubling formula. We find that

$$\wp(2t) = x_{2P} = \frac{1}{4}\left(\frac{12\wp'(t)^2 - g_2}{2\wp'(t)}\right)^2 - 2\wp(t) = \left(\frac{6\wp'(t)^2 - \frac{g_2}{2}}{2\wp'(t)}\right)^2 - 2\wp(t).$$

Now, we want to attain that $6\wp'(t)^2 - \frac{g_2}{2} = \wp''(t)$. This is true, as we can just different the elliptic curve by z:

$$\frac{d}{dz}\left(\wp'(z)\right)^2 = \frac{d}{dz}\left(4\wp(z)^3 - g_2\wp(z) - g_3\right)$$

This gives

$$2\wp'(z)\wp''(z) = 12\wp(z)^2\wp'(z) - g_2\wp'(z) \implies \wp''(z) = 6\wp'(z) - \frac{g_2}{2}.$$

Therefore, we've verified the results. $\square$

We end this section with the statement of this theorem for completeness.

Theorem 7.1.19 (Uniformization Theorem)
Let $A, B \in \mathbb{C}$ be complex numbers satisfying $A^3 - 27B^2 \neq 0$. Then there exists a unique lattice $\Lambda \subset \mathbb{C}$ satisfying

$$g_2(\Lambda) = A, \qquad g_3(\Lambda) = B.$$

Remark 7.1.20 — The elements of Λ are called the **periods** of E and can be computed using **elliptic integrals**. These are integrals of the form

$$\int \frac{dx}{\sqrt{4x^3 - g_2x - g_3}}$$

where the path of the integration may be any curve which does not pass through a zero of $4x^3 - g_2x - g_3$. Gauss' theory of arithmetic-geometric mean gives a particularly elegant way of computing such integrals.

This concludes the formal part of the course.

8 Final Material

§8.1 Counting points over $\mathbb{F}_p$

Let $E\colon y^2 = f(x)$ be an elliptic curve.

Theorem 8.1.1 (1933, Hasse)

$$|\#E(\mathbb{F}_p) - (p+1)| \leq 2\sqrt{p}.$$

Here is the rough intuition about this. Suppose $p \neq 2$, among the non-zero elements of $\mathbb{F}_p$ half of them are squares and half of them are non-squares. Now, going through the values $x = 0, 1, \ldots, p-1$. If $f(x) = 0$, there is only one solution $y = 0$. If $f(x) \neq 0$, then for half the possible non-zero values of $f(x)$, there are two solutions, and for the other half there are none. So, if $f(x)$ were randomly distributed among squares and non-squares, then the p possible values of x should give approximately p solutions, and then including O we should get $p+1$ solutions.

Define the **error term**

$$a_p = p + 1 - \#E(\mathbb{F}_p),$$

so $\frac{a_p}{\sqrt{p}} \in [-2, 2]$. Thus, we expect a_p to be fairly small compared to p.

§8.1.i Example with $E\colon y^2 = x^3 - x$

Let E be $y^2 = x^3 - x$. Then $E(\mathbb{Q}) = \{O, (0,0), (\pm 1, 0)\}$. If $p > 2$, then E defines an elliptic curve over $\mathbb{F}_p$.

What more can we say about $\#E(\mathbb{F}_p)$?

$$\begin{array}{c} p = 3 \\ \begin{pmatrix} x & 0 & 1 & 2 \\ \#(x,y) & 1 & 1 & 1 \end{pmatrix} \\ \#E(\mathbb{F}_3) = 4 \end{array} \quad \begin{array}{c} p = 5 \\ \begin{pmatrix} x & 0 & 1 & 2 & 3 & 4 \\ \#(x,y) & 1 & 1 & 2 & 2 & 1 \end{pmatrix} \\ \#E(\mathbb{F}_5) = 8 \end{array} \quad \begin{array}{c} p = 7 \\ \begin{pmatrix} x & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \#(x,y) & 1 & 1 & 0 & 0 & 2 & 2 & 1 \end{pmatrix} \\ \#E(\mathbb{F}_7) = 8 \end{array}$$

We can then attain the following values for a_p.

p	3	5	7	11	13	17
a_p	0	-2	0	0	6	2

We can actually formulate a theory for this elliptic curve.

Proposition 8.1.2

If $p \equiv 3 \pmod 4$, then the number of solutions to $y^2 = x^3 - x$ is $p + 1$.

Proof. If $p \equiv 3 \pmod 4$, then -1 is not a quadratic residue modulo p. Then, multiplying the entire equation by -1, then if x_0 is a square, then $-x_0$ is not, and vice versa. $\square$

Proposition 8.1.3

If $p \equiv 1 \pmod 4$, then $a_p = 2a$.

Proof. We won't write the entire argument out, but here is a starting point. If $p \equiv 1 \pmod 4$, then we can write $p = a^2 + b^2$ for some a and b. Fix a and b by demanding $a + bi \equiv 1 \pmod{(1+i)^3}$. (For example, if $p = 5 = 2^2 + 1^2$, then $\{a, b\} = \{\pm 1, \pm 2\}$, so $a = -1$ and $b = 2$.) $\square$

§8.1.ii Complex Multiplication

Let us now go back to the general case. The question now is how are these a_p distributed?

Given an elliptic curve E over a field k, we can consider

$$\mathrm{End}(E) = \mathrm{Hom}(E, E)$$

whose elements are morphisms $\phi\colon E(\bar{k}) \to E(\bar{k})$ sending $\phi(O) = O$. We can define addition and multiplication on such maps by

$$(\phi + \psi)(P) = \phi(P) + \psi(P), \qquad (\phi\psi)(P) = \phi(\psi(P)).$$

Thus $\mathrm{End}(E)$ is a ring.

More generally, given two elliptic curves E_1 and E_2, we can consider $\mathrm{Hom}(E_1, E_2)$ given by morphisms $\phi\colon E_1 \to E_2$ with $\phi(O) = O$. This set is an abelian group by $(\phi + \psi)(P) = \phi(P) + \psi(P)$. It is called the group of **isogenies**.

Multiplication by n map

$$[n]\colon E(k) \to E(k)$$

is a group homomorphism. When k is of characteristic 0, usually this is the whole story of endomorphisms, i.e. $\mathrm{End}(E) \cong \mathbb{Z}$. If there is an endomorphism $\phi\colon E(\bar{k}) \to E(\bar{k})$ that is not a multiplication by n-map, we say that E has a **CM (complex multiplication)**.

Remark 8.1.4 — The curve $E\colon y^2 = x^3 - x$ has the complex multiplication given by $\phi(x, y) = (-x, iy)$. Note that $\phi \circ \phi(P) = -P$. In fact, one can show that $\mathrm{End}(E) \cong \mathbb{Z}[i]$ if k has characteristic 0.

Remark 8.1.5 — The curve $E\colon y^2 = x^3 + 1$ has an endomorphism $\phi(x, y) = (\xi x, -y)$ where $\xi^3 = 1$ is a root of unity. Check that $\phi^3(P) = -P$.

Elliptic curves with CM have many special properties. In some sense, generic curves do not have CM.

Let us consider, the number $\frac{a_p}{\sqrt{p}}$ which lies in the interval $[-2, 2]$ by Hasse's theorem. How are these numbers distributed as p tends to infinity? For example, do we get a uniform description? Let us define an angle $2\theta_p$ between 0 and π by the condition

$$\cos\theta_p = \frac{a_p}{2\sqrt{p}}.$$

We can ask a precise question, by considering $[\alpha,\beta]\subset[0,\pi]$,

$$\lim_{N\to\infty}\frac{\#\{p\le N\colon \theta_p\in[\alpha,\beta]\}}{\#\{p\le N\}}=\frac{\beta-\alpha}{\pi}\quad ?$$

That is, is the quantity on the left is equidistributed? It turns out this is essentially correct for elliptic curves with CM. If we define the probability distribution function

$$P(x)=\lim_{N\to\infty}\frac{\#\{p\le N\colon a_p/2\sqrt{p}=x\}}{\#\{p\le N\}},$$

then a result of of Hecke (1920) shows

$$\lim_{N\to\infty}\frac{\#\{p\le N\colon \theta_p\in[a,b]\}}{\#\{p\le N\}}=P(x)=\frac{\delta_0}{2}+\frac{1}{2\pi}\frac{1}{\sqrt{4-x^2}}.$$

and, letting $x=2\cos\theta$, we deduce that

$$\int_\alpha^\beta P(x)dx=\frac{\delta_{[\alpha,\beta]}}{2}+\int_\alpha^\beta\frac{d\theta}{2\pi}=\frac{1}{2}\delta_{[\alpha,\beta]}+\frac{\beta-\alpha}{\pi},$$

where

$$\delta_{[\alpha,\beta]}=\begin{cases}1 & \text{if } \frac{\pi}{2}\in[\alpha,\beta]\\ 0 & \text{otherwise}\end{cases}.$$

That is, for roughly half of the primes, $a_p=0$, and for the other half of primes, $a_p/\sqrt{p}$ gets equidistributed. This is in agreement with the example we studied. Namely for $E\colon y^2=x^3-x$, we had that $a_p=0$ for primes $p\equiv 3 \pmod 4$, and for the other primes $a_p/\sqrt{p}$ gets equidistributed over the interval $[-2,2]$.

The behavior is very different for generic curves (those that do not have CM).

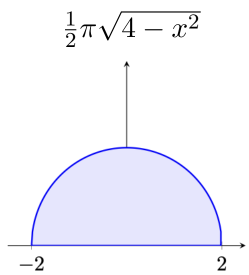
Theorem 8.1.6 (Sato–Tate conjecture 1960, Clozel–Harris–Shepherd–Barron–Taylor 2008)
Over all non-CM elliptic curves,

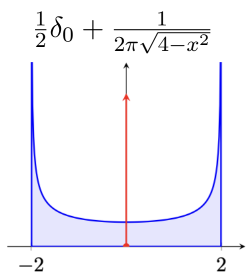
$$P(x)=\frac{1}{2\pi}\sqrt{4-x^2}.$$

In other words,

$$\lim_{N\to\infty}\frac{\#\{p\le N\colon \theta_p\in[\alpha,\beta]\}}{\#\{p\le N\}}=\frac{2}{\pi}\int_\alpha^\beta\sin^2\theta\,d\theta=\frac{\beta-\alpha}{\pi}-\frac{1}{2\pi}(\sin 2\beta-\sin 2\alpha).$$

The distributions look as follows.





As an example of an elliptic curve without CM, we can take E to be $y^2 + y = x^3 - x^2$, the **first elliptic curve in nature**. Then $E(\mathbb{Q}) = \mathbb{Z}/5\mathbb{Z} = \{(0,0)\}$ and the discriminant is -11. If $p \neq 11$, then E defines an elliptic curve over $\mathbb{F}_p$. Computing a_p,

p	2	3	5	7	13	17
a_p	-2	-1	-2	-2	-4	-2

Let $z \in \mathbb{C}$ such that $\mathrm{Im}\,(z) > 0$. Set $q = e^{2\pi i z}$. Consider the infinite product

$$f(z) = q \prod_{n \geq 1} (1 - q^n)^2 (1 - q^{11n})^2 = q - 2q^2 - q^3 + 2q^4 + q^5 + 2q^6 - 2q^7 - 2q^9 \\ - 2q^{10} + q^{11} - 2q^{14} + 4q^{13} + 4q^{14} - q^{15} - 4q^{16} - 2q^{17} + \dots$$

which converges, so $f(z)$ is a holomorphic function. Then $f(z+1) = f(z)$, and $f(-1/z) = (11z)^2 f(z)$, so f is a modular form of level $\Gamma_0(11)$ and weight two.

Theorem 8.1.7 (Modularity of elliptic curves 1957, Wiles 1995 and Breuil-Conrad-Diamond-Taylor 2001)

For each elliptic curve E over $\mathbb{Q}$, there exists a corresponding weight two modular form $f = q + bq^2 + bq^3 + \dots$, such that $b_p = a_p$.

§8.2 L-functions, Ranks of $E(\mathbb{Q})$ and BSD

How large is the torsion subgroup on average? Let E be $y^2 = x^3 + ax + b$. If we choose $a, b \in \mathbb{Z}$ at random, what is the probability that E has any points of order two? That is, what is the probability that $x^3 + ax + b$ has a rational root?

Proposition 8.2.1

0% of elliptic curves over $\mathbb{Q}$ have any points of order two. In fact, it can be shown that 100% of elliptic curves over $\mathbb{Q}$ have trivial torsion subgroup.

Heuristically, if the rank of $E(\mathbb{Q})$ is large, then $\#E(\mathbb{F}_p)$ is also large.

Example 8.2.2

Let $p = 7$. Let E be $y^2 = x^3 - Dx$. If $p \equiv 3 \pmod 4$, then $\#E(\mathbb{F}_p) = p + 1$, so the $a_p = 0$ for all such primes p. But these curves can have rank $0, 1, 2, \dots$

So, the relationship between $E(\mathbb{Q})$ and $\#E(\mathbb{F}_p)$, if there is any, must be more complicated. In fact, there is a beautiful relationship, and we'll now discuss this very briefly.

To any elliptic curve E over $\mathbb{Q}$, there is a **minimal Weierstrass form** E. This is defined to ensure that all the coefficients $a_1, a_2, a_3, a_4, a_6 \in \mathbb{Z}$ and for each prime p the power of p dividing the discriminant of E cannot be decreased by an admissible change of co-ordinates. In other words, the discriminant of E is minimal.

Now, given a minimal Weierstrass form for E, let E_p be the cubic curve obtained by the reduction of E modulo p. Then for each prime p, we get a sequence of numbers

$a_p, \#E_p(\mathbb{F}_{p^n})$, by counting the number of solutions over the finite fields $\mathbb{F}_{p^n}$ for $n = 1, 2, \ldots$. It is convenient to encode this information by defining

$$Z_{E_p}(u) = \exp\left(\sum_{n\geq 1} \frac{\#E_p(\mathbb{F}_{p^n})}{n} u^n\right).$$

Using the exponential generating function, we can also write this as:

$$u\frac{d}{du}\log Z_E(u) = \sum_{n\geq 1} \#E_p(\mathbb{F}_{p^n})u^n.$$

The following proposition shows the motivation behind a slightly complicated form of the generating function that involves the exponential function.

Proposition 8.2.3

$$Z_{E_p}(u) = \begin{cases} \dfrac{1 - a_p u + pu^2}{(1-u)(1-pu)} & \text{if } p \nmid \Delta, \\ \dfrac{1 - a_p u}{(1-u)(1-pu)} & \text{if } p \mid \Delta. \end{cases}$$

The **local zeta function** of E_p is defined to be

$$\zeta_{E_p}(s) = Z_{E_p}(p^{-s}), \quad s \in \mathbb{C}$$

Hasse's theorem implies (and is equivalent to):

1. $\zeta_{E_p}(s) = \zeta_{E_p}(1-s)$
2. If $\zeta_{E_p}(s_0) = 0$, then $\operatorname{Re} s_0 = \frac{1}{2}$.

For this reason, Hasse's theorem is also called the Riemann hypothesis for the local zeta function $\zeta_{E_p}(s)$.

Recall that the **Riemann zeta function** is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

The fact that every positive integer is uniquely expressible as a product of primes (up to rearranging factors) is equivalent to the statement that

$$\zeta(s) = \prod_p \sum_{m=0}^{\infty} p^{-ms} = \prod_p \frac{1}{1-p^{-s}}.$$

It is known that $\zeta(s)$ has a meromorphic continuation to $\mathbb{C}$ which is holomorphic everywhere except at a simple pole at $s = 1$ (with residue 1).

The **Gamma function** is defined by

$$\Gamma(s) = \int_0^{\infty} t^{s-1}e^t\, dt \quad \text{for } \operatorname{Re}(s) > 0.$$

Integration by parts yields the relation $\Gamma(s) = \Gamma(s+1)$, which can be used to extend $\Gamma(s)$ by meromorphic continuation to the complex plane with poles at non-positive integers. (Note also that $\Gamma(n) = (n-1)!$ for positive integers n.)

The Riemann zeta function satisfies a functional equation, which can be stated in terms of the Riemann ξ-function:

$$\xi(s) = \pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\,\zeta(s)$$

in a particularly simple form as

$$\xi(s) = \xi(1-s).$$

Similarly, we can construct a global zeta function for E by taking the product of local functions ζ_{E_p}, but the the denominators of $\zeta_{E_p}(s)$ just lead to $\zeta(s)$ and $\zeta(s-1)$ and give no useful information about E, thus we use the numerators to define the **L-function** which encodes a lot of information about E.

The L-function of $E(\mathbb{Q})$ is

$$L(E,s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} = \prod_{p\nmid\Delta} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \prod_{p\mid\Delta} \frac{1}{1 - a_p p^{-s}},$$

where $a_p = p + 1 - \#E(\mathbb{F}_p)$ are as above. We can equate coefficients to get

$$a_{p^k} = a_{p^{k-1}} a_p - p\, a_{p^{k-2}}$$

(with $a_0 = 1$) and if $n = \prod_j p_j^{e_j}$ then

$$a_n = \prod_j a_{p_j^{e_j}}.$$

Exercise 8.2.4. Factor $1 - a_p T + pT^2 = (1-\alpha_p)(1-\beta_p)$ with $\alpha_p, \beta_p \in \mathbb{C}$. Show that $|\alpha_p| \le 2\sqrt{p}$ implies that $|\alpha_p| = |\beta_p| = \sqrt{p}$. Prove that $a_{p^k} \le (k+1)p^{k/2}$ and $a_n \le d(n)\sqrt{n}$ where $d(n)$ is the number of distinct divisors of n. Use this to conclude that $L(E,s)$ converges for $\operatorname{Re} s > \frac{3}{2}$.

Theorem 8.2.5 (Hasse–Weil conjecture)

$L(E,s)$ has an analytic continuation to all of $\mathbb{C}$, and satisfies a functional equation.

This was conjectured in the 1940s and follows from modularity, which was known for CM elliptic curves $y^2 = x^3 - Dx$ in the 1960s, which were the ones that Birch and Swinnerton-Dyer looked at.

Moreover, there is an integer N_E called the **conductor** of E defined as the product of prime divisors of the discriminant Δ_E when E is in the minimal Weierstrass form, and the function

$$\xi(E,s) = N_E^{\frac{s}{2}}(2\pi)^{-s}\Gamma(s)L(E,s)$$

satisfies the functional equation

$$\xi(E,s) = \pm\xi(E,2-s) \quad \text{for all } s \in \mathbb{C}.$$

The L-function $L(E,s)$ is related to the modular form f_E by the **Mellin transform** and the functional equation is a consequence of modularity of f_E.

If $\phi(t)$ for $t > 0$ is a real function that decays rapidly at ∞ (faster than any polynomial of t) and blows up at most like a polynomial near 0, then the Mellin transform is given by

$$\mathcal{M}(\phi)(s) = \int_0^\infty \phi(t)\, t^{s-1} dt.$$

For example, if $\phi = \sum_{n\geq 1} a_n e^{-nt}$ where a_n are growing polynomially, then

$$\begin{aligned}\mathcal{M}(\phi)(s) &= \int_0^\infty \sum_{n\geq 1} a_n e^{-nt} t^{s-1} dt \\ &= \int_0^\infty e^{-t} t^{s-1} dt \sum_{n\geq 1} a_n n^{-s} \\ &= \Gamma(s) \sum_{n\geq 1} a_n n^{-s}.\end{aligned}$$

The point is that if the integrand is the modular form, then the LHS is the L-function.

Conjecture 8.2.6 (Birch, Swinnerton-Dyer 1965)

The order of vanishing of $L(E,s)$ at $s=1$ is the rank of $E(\mathbb{Q})$.

In other words, around $s = 1$, we have a Taylor series expansion of the form

$$L(E,s) = c(s-1)^r + \text{higher order terms}$$

with $c \neq 0$ and $r = \operatorname{rank} E(\mathbb{Q})$.

Theorem 8.2.7 (Gross-Zagier 1986 and Kolyvagin 1989)

The conjecture holds if $E(\mathbb{Q})$ has rank at most one.

The conjecture is still open if the rank is greater than one and comes with a $1m prize. How large is the rank on average?

Conjecture 8.2.8

100% of elliptic curves over $\mathbb{Q}$ have rank at most one. In fact, 50% have rank zero and 50% have rank one.

Computing the rank of E basically comes down to computing $E(\mathbb{Q})/2E(\mathbb{Q})$, and that comes down to computing $\operatorname{Im}\delta$ for

$$\delta\colon E(\mathbb{Q})/2E(\mathbb{Q}) \hookrightarrow (\mathbb{Q}^\times/(\mathbb{Q}^\times)^2)^3.$$

Define the **2-Selmer group** S_2 to be the same as $\operatorname{im}\delta$, except that you only demand that the equations you write have solutions in $\mathbb{Q}_p$ for all p, and not necessarily in $\mathbb{Q}$. Then compute $\operatorname{im}\delta$ for

$$\delta\colon E(\mathbb{Q})/2E(\mathbb{Q}) \hookrightarrow S_2.$$

The following is the best-known result.

Theorem 8.2.9 (Bhargava–Shankar 2011)
The average rank is less than one.

How large can the rank of E be?

Theorem 8.2.10 (Elkies–Klagsbrun 2024)
There exists $E(\mathbb{Q})$ with rank at least 29.

Elkies-Klagsbrun proved that the following curve has at least 29 independent points:

$$\begin{aligned} y^2 + xy = x^3 &- 27006183241630922218434652145297453784768054621836357954373 85x \\ &+ 55258058551342376475736699591118191821521067302535079608372044779149413277716173\ldots \end{aligned}$$

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